



**HOST SECURITY MODULE 8000
CONSOLE REFERENCE MANUAL**

I270A349 Issue 8

HOST SECURITY MODULE 8000

Console Reference Manual

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Host Security Module 8000 Console Reference Manual

Revision Status

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I270A349-004	HSM 8000 v2.2	Updated to include new commands and features provided in base software v2.2.	November 2005
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I270A349-006.1	HSM 8000 v2.4	Updated to correct errors and omissions.	March 2007
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This manual describes the functionality within the 3.1 base release of HSM 8000 software. For all other versions please refer to the appropriate manual and associated HSM software specifications.

Contact Information

THALES e-SECURITY

Europe, Middle East, Africa

Meadow View House
Crendon Ind. Estate
Long Crendon
Aylesbury
Buckinghamshire HP18 9EQ
UK

Telephone: +44 1844 201800
Fax: +44 1844 208550

Support

Telephone: +44 1844 202566

Fax: +44 1844 208356

emea.support@thales-esecurity.com

Americas

Suite 200
2200 North Commerce Parkway
Weston, FL 33326
USA

Telephone: 1-888-744-4976 (in US)
+1 954-888-6200 (outside US)
Fax: +1 954-888-6211

Support

Telephone: 1-800-521-6261 (in U.S.)
+1 954-888-6277 (outside U.S.)

Fax: +1 954-888-6233

support@thalessec.com

Asia Pacific

Units 2205-06, 22/F
Vicwood Plaza
199 Des Voeux Road, Central
Hong Kong

Telephone: +852 2815 8633
Fax: +852 2815 8141

Support

Telephone: +852 2815 8633

Fax: +852 2815 8141

asia.support@thales-esecurity.com

<http://www.thalesgroup.com/esecurity>

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Chapter I - Introduction

General

The HSM 8000 (Host Security Module) series of equipment provides cryptographic commands to support network and point-to-point data security. Acting as a peripheral to a Host computer, the HSM provides the cryptographic facilities required to implement key management, message authentication and Personal Identification Number (PIN) encryption in real time online environments. The HSM is made physically secure by locks, electronic switches and tamper-detection circuits.



Figure I - HSM 8000: Front View showing Normal Operation

The HSM supports a number of standard commands and can be customised to perform client-specific cryptographic commands. Standard commands include:

- Generating and verifying Personal Identification Numbers (PINs) such as those used with bank accounts and credit cards.
- Generating encrypted card values such as Card Verification Values (CVVs) for the payment card industry.
- PIN solicitation, to obtain a new PIN from a card holder (against a reference number).
- Generating keys for use in Electronic Funds Transfer Point of Sale (EFTPOS) systems.
- Key management in non-EFTPOS systems.
- Generating and verifying Message Authorisation Codes (MACs) for messages transferred via telecommunications networks.

The HSM is normally online to the Host and does not require operator monitoring or intervention. The HSM performs cryptographic processing in response to commands from the Host. The Host sends command messages, which consist of command codes and other fields that are required by the HSM in order to process the commands. The HSM processes the command messages and generates response messages, which also contain a variable number of fields (depending on the message type). Some commands, mainly involving plain text data, are entered by the user via the associated HSM console.

About this Manual

This manual is a reference document containing details of all commands that can be used on the HSM console. For other HSM information, see the following manuals:

- HSM 8000 Security Operations Manual
- HSM 8000 Installation Manual
- HSM 8000 Image Loader User Manual
- HSM 8000 Host Programmers Manual
- HSM 8000 Host Command Reference Manual

Physical Description

An HSM can stand alone or can be part of a system consisting of a number of HSM units in a standard 19-inch cabinet. The front panel of each unit is accessible from the front of the cabinet. The rear of the cabinet has a lockable door which gives access to the rear panel of each unit. The units are supported on telescopic runners so that they can slide out via the front of the cabinet.

Front Panel

On the front panel (see Figure 1) are two cam locks, which secure the HSM in the rack. The locks have different keys, so the HSM can be removed from the rack only when the two authorised key holders are present. After installation, it is only necessary for both keyholders to be present when secure operations are to be carried out on the HSM such as changing the Local Master Keys (LMKs).

There are five coloured LED indicators on the front panel. These are the POWER, ERROR, ACTIVITY, ALARM and SECURE indicators.

The smartcard reader is an ISO card compliant type with automatic card ejection. The card is ejected at standard points in HSM operation:

- On completion of a smartcard related console command.
- Following premature console command termination when the user presses the CTRL-C key combination.
- When the HSM is reset by the RESET button on its front panel.
- During diagnostic testing.
- When an EJECT console command is input by the user.

Console Port

This is a 9-way D-type female socket for connection to a standard console terminal. It is a replica of the console connector on the rear panel; a console terminal must not be connected to the front panel connector while a console terminal is connected to the rear panel connector. The console port is configured as a Data Communications Equipment (DCE). Almost any asynchronous ASCII terminal is suitable for use with the HSM as a console terminal.

RESET Button

The red RESET button is recessed in the front panel to prevent accidental activation. It is used to take the HSM out of offline mode and back into normal operation. The reset also ejects any smartcard that is in the smartcard reader. If the RESET button is pressed while the HSM is online, the HSM performs a reboot and the rebooting message is displayed on the console terminal (if connected).

The RESET button must be pressed for **two seconds** for it to take effect.

Power LED

The green POWER indicator is illuminated while the HSM has power applied to it. If the POWER LED goes out, this means that either the power has been disconnected or a fuse has blown.

Error LED

The red ERROR indicator is normally extinguished. It is illuminated if the HSM's continuous automatic self checks have detected a fault. The ERROR LED also indicates the state of the error log. If a new error has occurred since the error log was last checked then the LED flashes. Once

the error log has been investigated it illuminates continuously until the log is erased. See ERRORLOG command on page 88.

Secure LED

The yellow SECURE indicator flashes during normal operation. This shows that the alarm circuitry (movement and temperature alarms) is active and the LMKs are loaded. The mesh circuitry, which detects an attempted intrusion into the crypto-processor, is always active and is not shown by the secure LED. The conditions for the SECURE indicator are:

Off	LMKs not loaded
On steady	LMKs loaded, alarms disabled
Flashing	LMKs loaded, alarms enabled (either temperature alarm or movement alarm or both)

Alarm LED

The red ALARM indicator illuminates when the HSM is triggered into an alarmed state by a security compromise. All secure data stored in the HSM is erased. To extinguish the ALARM LED, the HSM must be rebooted by powering off and powering on again. Following an alarm condition, the LMK(s) will need to be reloaded into the HSM. If the alarm condition is still present after rebooting, the ALARM LED remains illuminated; in this case the HSM must be returned to THALES for investigation and repair.

Activity LED

The green ACTIVITY indicator shows host activity on the HSM. It illuminates when data is either received or transmitted. It flashes when a small amount of data is being sent and illuminates steadily when the HSM is busy. If no data is being received from the host it remains extinguished.

Locks

The locks on the front panel of the HSM provide security in two ways:

- When in the locked position, the HSM cannot be removed from the rack. Both locks need to be unlocked for the unit to be removed; this can only be achieved by the presence of two authorised key holders, as each lock requires a different key. The mechanical locking of the unit into the rack thus provides low level resistance to a direct attack. Note that the unit itself cannot be opened.
- Micro-switches are attached to the locks, wired to inputs on the internal circuit board, allowing the security state of the HSM to be changed. Three states are supported, online (both locked), offline (one locked and one unlocked) and secure (both unlocked). See table below.

State	Left hand lock	Right hand lock
Normal (online)	Locked (activated)	Locked (activated)
Offline	Locked	Unlocked
Offline	Unlocked	Locked
Secure	Unlocked	Unlocked

Rear Panel

The power input (via an IEC connector) and fuses (20 mm type) are housed in a module on the right (see Figure 2).

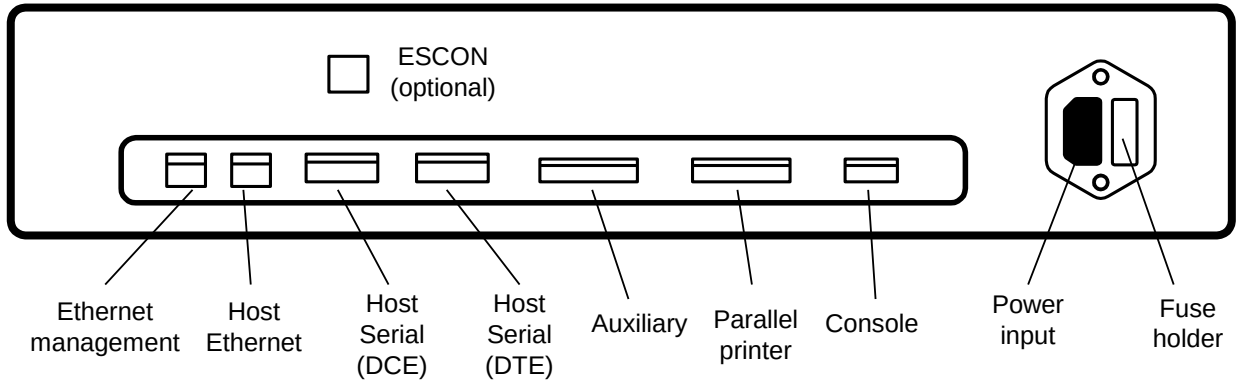


Figure 2 – HSM 8000 Rear View

The host ethernet and ethernet management ports are RJ45 sockets.

The two host Serial ports are 26-way, D-type female sockets.

The auxiliary port is a 25-way, D-type male socket and the parallel printer port is a 25-way D-type female socket.

The console port is a 9-way D-type female socket, which is replicated on the front panel.

ESCON versions have an additional MT-RJ connector for the optical fibre. This is only fitted on the HSM 8000 ESCON; other variants have a blanking plate fitted instead.

See the HSM 8000 Installation Manual for details of the host ports and configuration procedures.

Important Note: In order to ensure that an HSM only processes commands on behalf of the legitimate host computer, it is strongly recommended that a private Ethernet network segment is used. The only devices on this network should be the host and its associated HSM(s).

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Chapter 2 - Configuration Commands

This chapter describes the commands used to configure the HSM to work with the host system. It also includes those commands that provide information to assist with the installation and configuration of the HSM.

The HSM provides the following console commands to support configuration operations:

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Configure Commands

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **ConfigCmds**

Function: To enable or disable host and console commands. All available commands are enabled by default.

Commands are enabled or disabled using the following syntax:

[+ or -] [C or H] [<Command Code>]

+ indicates that the specified command should be enabled.

- indicates that the specified command should be disabled.

C indicates that <Command Code> is a Console command.

H indicates that <Command Code> is a Host command.

<Command Code> is the command code to be enabled or disabled, and may contain the wildcard character '*'. If the first character is '*', then the second character is absent, and this matches all command codes of the specified type. If the second character is '*', then this matches all command codes of the specified type starting with the given first character.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- List of host commands to enable.
- List of console commands to enable.
- List of host commands to disable.
- List of console commands to disable.

Outputs:

- Complete list of enabled host commands.
- Complete list of enabled console commands.

Errors:

- Invalid entry

Notes:

- When a disabled host command is invoked, error code 68 is returned.
- When a disabled console command is invoked, the message "Function not defined or not allowed" is displayed.

Example 1: This example demonstrates the use of the **ConfigCmds** console command to enable one console command (DE) and disable one host command (A4).

```
Secure> ConfigCmds <Return>
List of enabled Console commands: GC, GS, EC, FK
List of enabled Host commands: A0, A4, GG, GY
Enter command code (e.g. +CDE) or Q to Quit: +CDE <Return>
List of enabled Console commands: GC, GS, EC, FK, DE
List of enabled Host commands: A0, A4, GG, GY
Enter command code (e.g. +CDE) or Q to Quit: -HA4 <Return>
List of enabled Console commands: GC, GS, EC, FK, DE
List of enabled Host commands: A0, GG, GY
Enter command code (e.g. +CDE) or Q to Quit: Q <Return>
Secure>
```

Example 2: *This example demonstrates the use of the **ConfigCmds** console command using the wildcard character '*' to disable all non-core host commands, and then enable just those host commands beginning with 'A'. See Appendix B for a list of core commands.*

```
Secure> ConfigCmds <Return>
List of enabled Console commands: GC, GS, EC, FK
List of enabled Host commands: A0, A4, GG, GY
Enter command code (e.g. +CDE) or Q to Quit: -H* <Return>
List of enabled Console commands: GC, GS, EC, FK
List of enabled Host commands:
Enter command code (e.g. +CDE) or Q to Quit: +HA* <Return>
List of enabled Console commands: GC, GS, EC, FK
List of enabled Host commands: SA, SC, SE, SG
Enter command code (e.g. +CDE) or Q to Quit: Q <Return>
Secure>
```

Configure PIN Block Formats

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **ConfigPB**

Function: To enable or disable PIN block formats. The default settings for the available PIN blocks are listed in Appendix C.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- PIN block format identifier.

Outputs:

- List of enabled PIN block formats.

Errors:

- Invalid entry

Example: *This example enables the HSM PIN Block format 03.*

```
Secure> ConfigPB <Return>
List of enabled PIN Block formats:
01 - ISO 9564-1 & ANSI X9.8 format 0
05 - ISO 9564-1 format 1
35 - MasterCard Pay Now & Pay Later format
41 - Visa/Amex new PIN only format
42 - Visa/Amex new & old PIN format
47 - ISO 9564-1 & ANSI X9.8 format 3

Enter + or - followed by PIN Block format or Q to Quit: +03 <Return>

List of enabled PIN Block formats:
01 - ISO 9564-1 & ANSI X9.8 format 0
03 - Diebold & IBM ATM format
05 - ISO 9564-1 format 1
35 - MasterCard Pay Now & Pay Later format
41 - Visa/Amex new PIN only format
42 - Visa/Amex new & old PIN format
47 - ISO 9564-1 & ANSI X9.8 format 3
Enter + or - followed by PIN Block format or Q to Quit: Q <Return>
Secure>
```

Configure Security

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CS**

Function: To set the security configuration of the HSM and some processing parameters. CS converts all lower-case alpha values to upper case for display purposes, except for the Card issuer Password. Operation is menu-driven, as shown in the examples. The security settings can optionally be saved to a smartcard. All the settings, along with their default values, are listed in Appendix I.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- PIN length [4-12]: a one or two-digit number in the range 4 to 12.
- Echo [oN/off]: N or F
- Atalla ZMK variant support [oN/off]: N or F
- Transaction key scheme: Racal, Australian or None? [R/A/N]: R or A or N
- User storage key length [S/D/T]: S, D or T
- Erase LMKs? [Y/N]: confirm Y or N
- Select clear PINs? [Y/N]: Y or N
- Enable ZMK translate command? [Y/N]: Y or N
- Enable X9.17 for import? [Y/N]: Y or N
- Enable X9.17 for export? [Y/N]: Y or N
- Solicitation batch size [1-1024]: a one to four-digit number, range 1 to 1024
- Enable Single-DES [Y/N]: Y or N
- Prevent Single-DES keys masquerading as double or triple-length key? [Y/N]: Y or N
- Single/double length ZMKs [S/D]: S or D (Single or Double)
- Decimalisation table Encrypted/Plaintext [E/P]: E
- Enable decimalisation table checks? [Y/N]: Y or N
- PIN encryption algorithm: A or B (Visa method or Racal Method)
- Card/password authorisation [C/P]: C or P (Card or Password)
- Card issuer password [ENTER = no change]: 8 alphanumeric printable characters
- Authorised State required when importing DES key under RSA key? [Y/N]: Y or N
- Minimum HMAC verification length in bytes [5-64]: number, range 5-64
- Enable PKCS#11 import and export for HMAC keys? [Y/N]: Y or N
- Enable ANSI X9.17 import and export for HMAC keys? [Y/N]: Y or N
- Enable ZEK encryption of all printable ASCII chars? [Y/N]: Y or N
- Enable ZEK encryption of "Base94" ASCII chars? [Y/N]: Y or N
- Enable ZEK encryption of "Base64" ASCII chars? [Y/N]: Y or N
- Enable ZEK encryption of "Hex-only" ASCII chars? [Y/N]: Y or N
- Restrict Key Check Values to 6 hex chars? [Y/N]: Y or N
- Enable multiple authorised activities? [Y/N]: Y or N
- Enable persistent authorised activities? [Y/N]: Y or N
- Enable variable length PIN offset? [Y/N]: Y or N
- Enable weak PIN checking? [Y/N]: Y or N
- Enable PIN Block format 34 as output format for PIN translations to ZPK? [Y/N]: Y or N
- Default LMK identifier? [00-99]: a 2 digit number, range 00 to 99
- Management LMK identifier? [00-99]: a 2 digit number, range 00 to 99

- Use HSM clock for date/time validation? [Y/N]: Y or N
- Additional padding to disguise key length? [Y/N] : Y or N
- Key export and import in trusted format only? [Y/N] : Y or N
- Secure Link password [ENTER = no change]: 8 alphanumeric printable characters

- Save SECURITY settings to smartcard? [Y/N]: Y or N

Outputs:

- Prompts according to the settings chosen (see examples below).

Errors:

- Invalid Entry
- Card not formatted to save/retrieve HSM settings.
Attempt with another card? [Y/N]

Example 1:

```
Secure> CS <Return>
PIN Length [4-12]: 4 <Return>
Echo [oN/ofF]: N <Return>
Atalla ZMK variant support [oN/ofF]: F <Return>
Transaction Key Scheme: Racal, Australian or None [R/A/N]: N <Return>
User storage key length [S/D/T]: S <Return>
```

LMKs must be erased before remaining parameters can be set.

```
Erase LMKs? [Y/N]: N <Return>
```

```
Save SECURITY settings to smartcard? [Y/N]: N <Return>
Secure>
```

Example 2:

```
Secure> CS <Return>
PIN length [4-12]: 4 <Return>
Echo [oN/ofF]: F <Return>
Atalla ZMK variant support [oN/ofF]: F <Return>
Transaction key scheme Racal or Australian or None? [R/A/N]: R <Return>
User storage key length [S/D/T]: T <Return>
```

LMKs must be erased before remaining parameters can be set.

```
Erase LMKs? [Y/N]: Y <Return>
```

```
Select clear PINs? [Y/N]: N <Return>
Enable ZMK translate command? [Y/N]: N <Return>
Enable X9.17 for import? [Y/N]: N <Return>
Enable X9.17 for export? [Y/N]: N <Return>
Solicitation batch size [1-1024]: 1024 <Return>
```

```
Enable Single DES? [Y/N]: Y <Return>
Prevent single-DES keys masquerading as double or triple-length key? [Y/N]
Y Y <Return>
Making default length for ZMKs: [S/D]: D <Return>
Decimalisation table Encrypted/Plaintext [E/P]: E <Return>
Enable Decimalisation Table Checks? [Y/N]: Y <Return>
```

```
PIN encryption algorithm [A/B]: A <Return>
Card/password authorisation (local) [C/P]: C <Return>
Card issuer password (local) [Enter = no change]: <Return>
Authorised state required when importing DES key under RSA key [Y/N]: Y
<Return>
Minimum HMAC verification length in bytes? [5-64]: 10 <Return>
Enable PKCS#11 import and export for HMAC keys? [Y/N]: N <Return>
Enable ANSI X9.17 import and export for HMAC keys? [Y/N]: N <Return>
Enable ZEK encryption of all printable ASCII chars? [Y/N]: N <Return>
Enable ZEK encryption of "Base94" ASCII chars? [Y/N]: N <Return>
Enable ZEK encryption of "Base64" ASCII chars? [Y/N]: Y <Return>
```

```
Restrict Key Check Value to 6 hex chars? [Y/N]: Y <Return>
Enable multiple authorised activities? [Y/N]: Y <Return>
Enable persistent authorised activities? [Y/N]: Y <Return>
Enable variable length PIN offset? [Y/N]: N <Return>
Enable weak PIN checking? [Y/N]: N <Return>
Enable PIN Block format 34 as output format for PIN translations to ZPK?
[Y/N]: N <Return>

Default LMK identifier? [00-99]: 00 <Return>
Management LMK identifier? [00-99]: 00 <Return>
Use HSM clock for date/time validation? [Y/N]: N <Return>
Additional padding to disguise key length? [Y/N]: N <Return>
Key export and import in trusted format only? [Y/N]: Y <Return>
Secure Link password (local) [Enter = no change]: <Return>

Save SECURITY settings to smartcard? [Y/N]: Y <Return>
Insert card and press ENTER: <Return>
SECURITY settings saved to the smartcard.
Secure>
```

View Security Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QS**

Function: Reports the security configuration of the HSM and some processing parameters, plus the LMK check value.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- PIN length
- Encrypted PIN length
- Echo: OFF or ON
- Atalla ZMK variant support: OFF or ON
- Transaction key support: Racal, Australian or None
- User storage key length: Single, Double or Triple
- Select clear PINs: Y or N
- Enable ZMK translate command: YES or NO
- Enable X9.17 for import: YES or NO
- Enable X9.17 for export: YES or NO
- Solicitation batch size: 1..1024
- Single-DES: Enabled or Disabled
- Prevent single-DES keys masquerading as double or triple-length key: YES or NO
- ZMK length: S or D.
- Decimalisation table Encrypted/plaintext: E or P
- Decimalisation table checks enabled: YES or NO
- PIN encryption algorithm: A or B
- Card/password authorisation (local): C or P
- Authorised State required when Importing DES key under RSA key: YES or NO
- Minimum HMAC verification length in bytes: 5..20
- Enable PKCS#11 import and export for HMAC keys: YES or NO
- Enable ANSI X9.17 import and export for HMAC keys: YES or NO
- Enable ZEK encryption of all printable ASCII chars: YES or NO¹
- Enable ZEK encryption of "Base94" ASCII chars: YES or NO
- Enable ZEK encryption of "Base64" ASCII chars: YES or NO
- Enable ZEK encryption of "Hex-only" ASCII chars: YES or NO
- Restrict Key Check Values to 6 hex chars: YES or NO
- Enable multiple authorised activities: YES or NO
- Enable persistent authorised activities: YES or NO
- Enable variable length PIN offset: YES or NO
- Enable weak PIN checking: YES or NO
- Enable PIN Block format 34 as output format for PIN translations to ZPK: YES or NO
- Default LMK identifier: 00..99
- Management LMK identifier: 00..99
- Use HSM clock for date/time validation: YES or NO

¹ Refer to Appendix H for details of these Reduced ASCII character sets.

- Additional padding to disguise key length: YES or NO
- Key export and import in trusted format only: YES or NO

Errors: None.

Example: Online> QS <Return>
PIN length: 04
Encrypted PIN length: 05
Echo: OFF
Atalla ZMK variant support: OFF
Transaction key support: NONE
User storage key length: TRIPLE

Select clear PINs: NO
Enable ZMK translate command: NO
Enable X9.17 for import: NO
Enable X9.17 for export: NO
Solicitation batch size: 1024
Single-DES: DISABLED
Prevent Single-DES keys masquerading as double or triple-length keys: YES
ZMK length: DOUBLE
Decimalisation tables: ENCRYPTED
Decimalisation table checks: ENABLED
PIN encryption algorithm: A
Card/password authorisation (local): C

Authorised State required when importing DES key under RSA key: YES
Minimum HMAC verification length in bytes: 10
Enable PKCS#11 import and export for HMAC keys: NO
Enable ANSI X9.17 import and export for HMAC keys: NO
Enable ZEK encryption of all printable ASCII chars: NO
Enable ZEK encryption of "Base94" ASCII chars: NO
Enable ZEK encryption of "Base64" ASCII chars: YES
Enable ZEK encryption of "Hex-only" ASCII chars: YES

Restrict Key Check Value to 6 Hex Chars: YES
Enable multiple authorised activities: YES
Allow persistent authorised activities: NO
Enable variable length PIN offset: NO
Enable weak PIN checking: NO
Enable PIN Block format 34 as output format for PIN translations to ZPK: NO

Default LMK identifier: 00
Management LMK identifier: 00
Use HSM clock for date/time validation: NO
Additional padding to disguise key length: NO
Key export and import in trusted format only: YES

Online>

Configure Console Port

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CC**

Function: To set the baud rate and word format for the console port.

Authorisation: This command does not require any authorisation.

Inputs:

- Console baud rate (see example below).
- Console word format (see example below).

Outputs: None.

Errors: None.

Example: Offline> **CC** <Return>

BAUD RATES	WORD FORMATS
1. 300	1. 7 bits, no parity, 1 stop
2. 600	2. 7 bits, odd parity, 1 stop
3. 1200	3. 7 bits, even parity, 1 stop
4. 2400	4. 8 bits, no parity, 1 stop
5. 4800	5. 8 bits, odd parity, 1 stop
6. 9600	6. 8 bits, even parity, 1 stop
7. 19200	
8. 38400	

In response to the prompts, enter the number of the desired option or <RETURN> for no change.
Console baud rate: (current value = 1) 6 <Return>
Console word format: (current value = 2) 4 <Return>
Offline>

Notes: After completing the procedure, reset the HSM by turning the keylocks to the online position and pressing the RESET button on the front panel. The new values do not take effect until this has been done.

The new settings remain operational (even after software updates) until the user changes them, or there is an alarm (in which case they return to default values).

Default Settings (Console Port)

The default settings for the console port are:

- 9600 baud
- 8 data bits
- 1 stop bit
- No parity

View Console Port Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QC**

Function: To display details of the console port configuration of the HSM.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- The console baud rate.
- The console word format.
- Flow control is fixed as XON/XOFF.

Errors: None.

Example:

```
Online> qc <Return>
Baud: 9600
Word format: 8 bits, no parity, 1 stop
Flow control: XON/XOFF
Online>
```

Configure Host Port

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CH**

Function: To configure the Host port to emulate a type of data communications equipment and control equipment, which can be one of the following:

- Standard asynchronous emulation.
- Transparent asynchronous emulation.
- SNA/SDLC synchronous emulation.
- Ethernet.
- ESCON (HSM8-Exxxxx variants only).

The Host port setting can optionally be saved to a smartcard.

After completing the procedure, restart the HSM by pressing the RESET button on the front panel. The new values do not take effect until this has been done.

Authorisation: The HSM must be in the Offline or Secure state to run this command.

Inputs: The options are menu driven and the inputs vary depending on the communication mode selected. See examples below.

Outputs: None.

Errors: None.

Example 1: *In this example, standard asynchronous communications is enabled, the message header length is 4 characters and the terminating characters are set to 0300 hexadecimal (ETX). The Host baud is changed from 300 to 19200 bps; the word format to 8 data bits, no parity and 1 stop bit.*

```
Offline> CH <Return>
Message header length (1-255): 4 <Return>
[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: A <Return>
Transparent mode (Y/N): N <Return>
Terminating characters (4 hex): 0300 <Return>
Interface: DTE or DCE [T/C]: C <Return>
  BAUD RATES                WORD FORMATS
  1.      300                1.    7 bits, no parity, 1 stop
  2.      600                2.    7 bits, odd parity, 1 stop
  3.     1200                3.    7 bits, even parity, 1 stop
  4.     2400                4.    8 bits, no parity, 1 stop
  5.     4800                5.    8 bits, odd parity, 1 stop
  6.     9600                6.    8 bits, even parity, 1 stop
  7.    19200
  8.    38400
In response to the prompts, enter the number of the
desired option or <RETURN> for no change.
Host baud rate: (current value = 1) 7 <Return>
Host word format: (current value = 2) 4 <Return>
Save HOST settings to smartcard? [Y/N]: N <Return>
Offline>
```

Example 2: *In this example, transparent asynchronous communications is enabled and the message header length is set to 6 characters. The Host baud is changed to 9600 bps and the word format is set to 8 data bits, no parity and 1 stop bit. No option of message terminating characters is offered because this is fixed (value X'03, "ETX" character).*

```
Offline> CH <Return>
Message header length (1-255): 6 <Return>
[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: A <Return>
Transparent mode (Y/N): Y <Return>
Interface: DTE or DCE [T/C]: C <Return>
  BAUD RATES                                WORD FORMATS
  1.      300                                1.   7 bits, no parity, 1 stop
  2.      600                                2.   7 bits, odd parity, 1 stop
  3.     1200                                3.   7 bits, even parity, 1 stop
  4.     2400                                4.   8 bits, no parity, 1 stop
  5.     4800                                5.   8 bits, odd parity, 1 stop
  6.     9600                                6.   8 bits, even parity, 1 stop
  7.    19200
  8.    38400
In response to the prompts, enter the number of the
desired option or <Return> for no change.
Host baud rate: (current value = 1) 6 <Return>
Host word format: (current value = 2) 4 <Return>
Save HOST settings to smartcard? [Y/N]: Y <Return>
Insert card and press ENTER: <Return>
HOST settings saved to the smartcard.
Offline>
```

Example 3: *In this example, neither IMS nor CICS support is selected. The HSM is configured as a DCE (Data Communications Equipment), i.e. the HSM supplies the Clock signal for the RS-232-C interface. The Host baud is changed to 19200 bps.*

```
Offline> CH <Return>
Message header length (1-100): 4 <Return>
[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: S <Return>
EBCDIC/ASCII [E/A]: E <Return>
Transparent Mode (Y/N): N <Return>
Station Address (2 hex): C1 <Return>
IMS/CICS/General (I/C/G): G <Return>
DTE/DCE (T/C): C <Return>
  BAUD RATES
  1. 9600
  2. 19200
  3. 38400
  4. 56000
  5. 64000
In response to the prompts, enter the number of the desired option or
<RETURN> for no change.
Host baud rate: (current value = 1) 2 <Return>
Save HOST settings to smartcard? [Y/N]: N <Return>
Offline>
```

Example 4: *In this example, IMS support is selected and the HSM is configured for DCE. The Host baud is changed to 56000 bps.*

```
Offline> CH <Return>
Message header length (1-100): 20 <Return>
[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: S <Return>
EBCDIC/ASCII [E/A]: E <Return>
Transparent Mode (Y/N): N <Return>
Station Address (2 hex): C1 <Return>
IMS/CICS/General (I/C/G): I <Return>
IMS message identifier (1-20): PIN,MAC <Return>
Identifier offset: 3 <Return>
DTE/DCE (T/C): C <Return>
```

BAUD RATES

1. 9600
2. 19200
3. 38400
4. 56000
5. 64000

In response to the prompts, enter the number of the desired option or <RETURN> for no change.

Host baud rate: (current value = 2) 4 <Return>

Save HOST settings to smartcard? [Y/N]: N <Return>

Offline>

Example 5:

In this example, CICS support is selected. The HSM is configured as a DTE (Data Terminal Equipment) and depends on the Host for the Clock signal. The Host baud rate cannot be set. No selection for Transparent mode or Station Address is offered, so they default to NO and CI.

Offline> CH <Return>

Message header length (1-100): 4 <Return>

[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: S <Return>

EBCDIC/ASCII [E/A]: E <Return>

Transparent Mode (Y/N): N <Return>

Station Address (2 hex): C1 <Return>

IMS/CICS/General (I/C/G): C <Return>

DTE/DCE (T/C): T <Return>

Save HOST settings to smartcard? [Y/N]: N <Return>

Offline>

Example 6:

In this example, Ethernet communications using UDP is selected.

Offline> CH <Return>

Message header length (1-255): 4 <Return>

[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: E <Return>

EBCDIC/ASCII (E/A): A <Return>

Enter IP address: 128.100.3.1 <Return>

Enter Well-Known-Port address: 1500 <Return>

UDP [Y/N]: Y <Return>

TCP [Y/N]: N <Return>

Enter default gateway address: 128.100.3.5 <Return>

Enter subnet mask: 255.255.255.000 <Return>

Save HOST settings to smartcard? [Y/N]: N <Return>

Offline>

Example 7:

In this example, Ethernet communications using TCP/IP is selected.

Offline> CH <Return>

Message header length (1-255): 4 <Return>

Asynch, Ethernet, or SNA-SDLC 3274? [A/E/S]: E <Return>

EBCDIC/ASCII (E/A): A <Return>

Enter IP address: 128.100.3.1 <Return>

Enter Well-Known-Port address: 1500 <Return>

UDP [Y/N]: N <Return>

TCP [Y/N]: Y <Return>

Number of connections? [1 - 64]: 6 <Return>

Enter TCP Keep_Alive timeout (1-120): 120 <Return>

Enter default gateway address: 128.100.3.5 <Return>

Enter subnet mask: 255.255.255.000 <Return>

Save HOST settings to smartcard? [Y/N]: N <Return>

Offline>

Example 8:

In this example, Ethernet communications using UDP and TCP/IP is selected.

Offline> CH <Return>

Message header length (1-255): 4 <Return>

[A]sync, [E]thernet, [S]NA-SDLC 3274 [A/E/S]: E <Return>

EBCDIC/ASCII (E/A): A <Return>

Enter IP address: 128.100.3.1 <Return>

```
Enter Well-Known-Port address: 1500 <Return>
UDP [Y/N]: Y <Return>
TCP [Y/N]: Y <Return>
Number of connections [1 - 63]: 6 <Return>
Enter TCP Keep Alive timeout (1-120): 120 <Return>
Enter default gateway address: 128.100.3.5 <Return>
Enter subnet mask: 255.255.255.000 <Return>
Save HOST settings to smartcard? [Y/N]: N <Return>
Offline>
```

Example 9: *In this example, ESCON communications is selected.*

```
Offline> CH <Return>
Message header length (1-255): 4 <Return>
[A]sync, [E]thernet, [S]NA-SDLC 3274, ES[C]ON [A/E/S/C]: C <Return>
EBCDIC/ASCII (E/A): E <Return>
Save HOST settings to smartcard? [Y/N]: N <Return>
Offline>
```

View Host Port Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QH**

Function: To display details of the Host port configuration of the HSM.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs: For all systems:

- The message header length. This is the number of characters at the front of each command from the Host to the HSM (after the STX character). The HSM returns the message header in the response.
- The protocol used.
- The character format (EBCDIC or ASCII).

For an asynchronous system:

- The response delay. This is the delay before the HSM responds to the Host. It allows use of half-duplex Host communications that require a defined delay between the transmission of a command and the response from the HSM.

For an SNA - SDLC synchronous system:

- Whether the interface is DCE or DTE.
- The IBM environment. This can be IMS or CICS.
- Message identifier and offset are shown.
- The interface port type. This can be RS-232.
- The SDLC station address.

For an asynchronous system:

- The Host baud rate.
- The Host word format.

For an Ethernet system:

- The IP address. This is the Internet Protocol address of the HSM in the system.
- The Well-Known-Address. This is the publicised TCP Port address of the HSM.
- Transport method: TCP or UDP.
- Default gateway address. This is the IP address of the default gateway.
- Subnet mask. This is the subnet mask of the attached TCP/IP network.

Errors: None.

Example 1: *In this example, the host interface has been configured for transparent asynchronous communications.*

```
Online> QH <Return>
Message header length: 04
Protocol: Transparent Asynchronous
Character format: ASCII
Terminating Sequence: 03 00
Interface: DCE
Response delay (ms): 00
Baud: 9600
Word format: 7 bits, no parity, 1 stop
Online>
```

Example 2: *In this example, the host interface has been configured for SNA-SDLC communications.*

```
Online> QH <Return>
Message header length: 08
Protocol: SNA/SDLC 3274
Character format: EBCDIC
Station Address: C1
Host environment: IMS
IMS message identifier: PIN,MAC
IMS identifier offset: 03
Interface: RS232, DCE
Baud: 64000
Word format: 8 bits, no parity
Online>
```

Example 3: *In this example, the host interface has been configured for Ethernet communication using UDP.*

```
Online> QH <Return>
Message header length: 04
Protocol: Ethernet
Character format: ASCII
IP address: 128.100.003.001
Well-Known-Port address: 01500
Transport: UDP
Default gateway address: 128.100.003.005
Subnet mask: 255.255.255.000
Online>
```

Example 4: *In this example, the host interface has been configured for Ethernet communication using TCP/IP.*

```
Online> QH <Return>
Message header length: 04
Protocol: Ethernet
Character format: ASCII
IP address: 128.100.003.001
Well-Known-Port address: 01500
Transport: TCP, 6 connections
TCP keep_alive value: 120
Default gateway address: 128.100.003.005
Subnet mask: 255.255.255.000
Online>
```

Example 5: *In this example, the host interface has been configured for Ethernet communications using UDP and TCP/IP.*

```
Online> QH <Return>
Message header length: 04
Protocol: Ethernet
Character format: ASCII
IP address: 128.100.003.001
Well-Known-Port address: 01500
Transport: UDP and TCP, 6 connections
TCP keep_alive value: 120
Default gateway address: 128.100.003.005
Subnet mask: 255.255.255.000
Online>
```

Example 6: *In this example, the host interface has been configured for ESCON communications.*

```
Online> QH <Return>
Message header length: 04
Protocol: ESCON
Character format: EBCDIC
Online>
```


Configure Printer Port

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CP**

Function: To select either serial printer (connected to the auxiliary port) or parallel printer (connected to the printer port). If Parallel printer is selected then there are no other parameters. If Serial printer is selected then the settings for the auxiliary port must be selected. Note that the auxiliary port can also be configured using the CA (Configure Auxiliary) command; any previous settings are overwritten.

Authorisation: This command does not require any authorisation.

Inputs:

- Serial or parallel printer: S or P (default P).
- Baud rate (for serial only).
- Word format (for serial only).
- Print test page (parallel printer only): Y or N.

Outputs: • Test page.

Errors: None.

Example 1: *This example demonstrates the configuration of a serial-attached printer.*

```
Online> CP <Return>
Series/Parallel? [S/P]: S <Return>
BAUD RATES                                WORD FORMATS
1.      300                                1.  7 bits, no parity, 1 stop
2.      600                                2.  7 bits, odd parity, 1 stop
3.     1200                                3.  7 bits, even parity, 1 stop
4.     2400                                4.  8 bits, no parity, 1 stop
5.     4800                                5.  8 bits, odd parity, 1 stop
6.     9600                                6.  8 bits, even parity, 1 stop
7.    19200
8.    38400
In response to the prompts, enter the number of the desired option or
<RETURN> for no change.
Auxiliary port baud rate: (current value = 6) <Return>
Auxiliary port word format: (current value = 4) 3 <Return>
Print test page? [Y/N]: <Return>
Online>
The values entered take effect immediately after the command has been completed.
```

Example 2: *This example demonstrates the configuration of a parallel-attached printer.*

```
Online> CP <Return>
Serial/Parallel? [S/P]: P <Return>
```

If a printer is detected that is online, the user is invited to print a test page:

```
Print test page? [Y/N]:
```

Otherwise the following is displayed:

```
Printer is offline
```

Default Printer Settings (Serial)

The default settings for the Auxiliary port when Serial printing is selected are:

- Character Set: ASCII
- Interface: RS 232-C (DTE) serial data not parallel
- Baud rate: 9600 bps
- Stop bits: 1
- Data bits: 8
- Parity: None
- Flow Control: XON, XOFF

Note that the parallel printer port has no configurable parameters.

View Printer Port Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QP**

Function: To display details of the Printer port configuration of the HSM.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- Printer port is either serial or parallel.
- Option to print test page.
- The Auxiliary port baud rate (serial only).
- The Auxiliary port word format (serial only).
- Flow control is fixed as XON/XOFF (serial only).

Errors: None.

Example 1:

```
Online> QP <Return>
Printer port: Parallel
Print test page? [Y/N]:
Online>
```

Example 2:

```
Online> QP <Return>
Printer port: Serial
Baud: 19200
Word format: 8 bits, no parity, 1 stop
Flow control: XON/XOFF
Printer is offline
Online>
```

Configure Auxiliary Port

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CA**

Function: To configure the Auxiliary port for use with a serial printer. Note that the auxiliary port can also be configured using the CP (Configure Printer) command; any previous settings are overwritten.

The variables to be configured are as follows:

- Baud rate: 300 to 38.4k.
- Data bits: 7 or 8 data bits.
- Parity bit: Odd, even or no parity.
- Stop bits: 1 stop bit.

Authorisation: This command does not require any authorisation.

Inputs:

- Baud rate (see example below).
- Word format (see example below).

Outputs: None.

Errors: None.

Example:

```
Online> CA <Return>
BAUD RATES                                WORD FORMATS
1.      300                               1.  7 bits, no parity, 1 stop
2.      600                               2.  7 bits, odd parity, 1 stop
3.     1200                               3.  7 bits, even parity, 1 stop
4.     2400                               4.  8 bits, no parity, 1 stop
5.     4800                               5.  8 bits, odd parity, 1 stop
6.     9600                               6.  8 bits, even parity, 1 stop
7.    19200
8.    38400

In response to the prompts, enter the number of the
desired option or <RETURN> for no change.
Auxiliary port baud rate: (current value = 6) <Return>
Auxiliary port word format: (current value = 4) 3 <Return>
Online>
```

The values entered take effect immediately after the command has been completed.

Default Settings (Auxiliary Port)

The default settings for the Auxiliary port are:

- 9600 baud
- 8 data bits
- 1 stop bit
- No parity

View Auxiliary Port Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QA**

Function: To display details of the Auxiliary port configuration of the HSM.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- The Auxiliary port baud rate.
- The Auxiliary port word format.
- Flow control is fixed as XON/XOFF.

Errors: None.

Example:

```
Online> QA <Return>
Baud: 9600
Word format: 7 bits, even parity, 1 stop
Flow control: XON/XOFF
Online>
```

Configure Management Port

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CM**

Function: To configure the Management port, which is an Ethernet port used only for management of the HSM. If connection to the host is via Ethernet then the Ethernet host port is used for that purpose. The Management Ethernet port is used to update the HSM's internal software, updating licensing information, and for enabling management of the HSM via the Local or Remote HSM Manager.

After completing the procedure, restart the HSM by pressing the RESET button on the front panel. The new values do not take effect until this has been done.

Authorisation: The HSM must be in the Offline or Secure state to run this command.

Inputs:

- IP address.
- Default gateway address.
- Subnet mask.
- Enable (local or remote) HSM Manager connection?

Outputs: None.

Errors: None.

Example:

```
Offline> CM <Return>
Management Ethernet Port:
Enter IP address: 192.168.200.90 <Return>
Enter default gateway address: 192.168.200.1 <Return>
Enter subnet mask: 255.255.255.0 <Return>
Enable HSM Manager connection:
Local Only, Remote Only, Both or Disabled? [L/R/B/D]: D <Return>
Offline>
```

Default settings for the Management port are meaningless and before first use must be changed to values suitable for the local connection. Once set, the values are retained until they are changed again by the CM command.

View Management Port Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QM**

Function: To display details of the Management port parameters.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- IP address.
- Default gateway address.
- Subnet mask.
- Enable (local or remote) HSM Manager connection?

Errors: None.

Example:

```
Online> QM <Return>
Management Ethernet Port:
IP address: 192.168.200.90
Default gateway address: 192.168.200.001
Subnet mask: 255.255.255.0
HSM Manager connection: Disabled
Online>
```

Configure Alarms

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CL**

Function: To enable or disable the motion alarm and temperature alarm. The HSM alarm circuitry typically needs to be turned off if the HSM is to be moved. The alarms should be turned on while the HSM is in service. The alarm settings can optionally be saved to a smartcard.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- Temperature alarm status: On or Off.
- Motion alarm status: On or Off.
- Save settings to smartcard: Yes or No.

Outputs: None.

Errors:

- Card not formatted to save/retrieve HSM settings.
Attempt with another card? [Y/N]

Example:

```
Secure> CL <Return>
LMKs must be erased before proceeding.
Erase LMKs? [Y/N]: Y <Return>
Temperature Alarm [oN/ofF]: F <Return>
Motion Alarm [oN/ofF]: F <Return>
Save ALARM settings to smartcard? [Y/N]: Y <Return>
Insert card and press ENTER: <Return>
ALARM settings saved to the smartcard.
Secure>
```


View Alarm Configuration

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **QL**

Function: To display details of the alarm configuration of the HSM.

Authorisation: This command does not require any authorisation.

Inputs: None.

Outputs:

- The Temperature alarm status.
- The Motion alarm status.

Errors: None.

Example:

```
Online> QL <Return>
Temperature Alarm Enabled
Motion Alarm Disabled
Online>
```

Fraud Detection Commands

The HSM provides the following console commands to support fraud detection operations:

Command	Page
Configure Fraud Detection (A5)	51
Re-enable PIN verification (A7)	52

Configure Fraud Detection

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: audit.console		

Command: **A5**

Function: To set the configuration of the HSM fraud detection function.

Authorisation: The HSM must be in the Offline or Secure state to run this command. The HSM must be either in the Authorised State, or the activity **audit.console** must be authorised, using the Authorising Officer cards of the Management LMK. Refer to Appendix J for details about the HSM's Fraud Detection capabilities.

Inputs:

- Fraud detection enabled: Yes/No.
- Limit on number of PIN verifies performed per minute.
- Limit on number of PIN verifies performed per hour.
- Limit on number of PIN attacks detected.

Outputs: None.

Errors:

- Not Authorised - the HSM is not authorised to perform this operation.
- Not Offline - the HSM must be offline to run this command.
- Invalid Entry - the value entered is invalid.

Example:

```
Offline-AUTH> A5 <Return>
HSM Fraud detection is: ON
The following limits are set:
PIN verifies per minute: 100
PIN verifies per hour: 1000
PIN Attack Limit: 100

Enable HSM fraud detection (Y/N): y <Return>
Enter limit on PIN verifies per minute: 200 <Return>
Enter limit on PIN verifies per hour: 2000 <Return>
Enter PIN Attack Limit: 200 <Return>
Offline-AUTH>
```

Re-enable PIN verification

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: audit.console		

Command: **A7**

Function: To reset the configuration of the HSM fraud detection function.

Authorisation: The HSM must be in the Offline state to run this command. The HSM must be either in the Authorised State, or the activity **audit.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs: None.

Outputs: None.

Errors:

- Not Authorised - the HSM is not authorised to perform this operation.
- Not Offline - the HSM must be offline to run this command.
- PIN Verification is not currently disabled

Example:

```
Offline-AUTH> A7 <Return>
PIN verification has been re-enabled
Offline-AUTH>
```

Diagnostic Commands

The HSM provides the following console commands to support diagnostic operations:

Command	Page
Diagnostic Test (DT)	54
View Software Revision Number (VR)	55
View Licensed Host Commands (NETSTAT)	56
Show Network Statistics (GETCMDS)	57
Test TCP/IP Network (PING)	60
Trace TCP/IP route (TRACERT)	61

Diagnostic Test

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DT**

Function: To perform diagnostic tests.
The DT command tests the following parts of the HSM:

- Working memory areas (RAM).
- Program code.
- The DES cryptographic processor.
- Smartcard reader operation.
- The RSA cryptographic processor.
- Battery voltage level.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs: PASS or FAIL messages.

Errors: None.

Example:

```
Offline> DT <Return>
Memory test...OK
Firmware test...OK
DES test...OK
Check smartcard eject...
Cryptographic test... OK
Battery...OK
Random Number Generator test...
  Monobit...OK
  Poker...OK
  Runs...OK
  Long Run...OK
SHA test...OK
MD5 test...OK
Comms Card Communications Test...OK

Diagnostics complete
Offline>
```

View Software Revision Number

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **VR**

Function: To display details of the software release number, revision number and build number.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs: Software revision numbers.

Errors: None.

Example: Online> **VR** <Return>
Base release: 3.1
Revision: XXXX-XXXX
Build Number: XXXX

Bootstrap Version: 2.5.7
Kernel Version: 3.7.7
ESS FPGA Version: 2.41.1
HSM Core API Version: 2.15.21
HSM Application Version: 5.6.17
HSM COMMs Version: 1.3.6

Build location: 01
Host Configuration: Async, Ethernet, SNA-SDLC 3274, ESCON

Serial Number: A3451200102P
Licence Issue No: 1
Performance: 800 TPS
Base Software: Version 3
Crypto: 3DES, RSA
Ship Counter: 1

TR31: Yes
LMK configuration: 5 LMKs

Online>

View Licensed Host Commands

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **GETCMDS**

Function: To display a list of the host commands included in the HSM licence.

Note: Some of the commands listed may require additional licence options enabled. For example the command EI requires the RSA algorithm to be included in the licence in order to function correctly. Refer to section 3 of the HSM Installation Manual for further information on licence options, and how to identify the commands available under the HSM's individual licence.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs: A list of the host commands included in the HSM licence.

Errors: None.

Example: Online> **GETCMDS** <Return>
Available host commands

```
-----
A0 A2 A4 A6 A8 AA AC AE AG AS
AU AW AY B0 BA BC BE BG BI BQ
BS BU BW BY CA CC CE CG CI CK
CM CO CQ CW CY DA DC DE DG DI
DK DM DO DQ DS DW DY EA EC EE
EG EI EK EM EO EQ ES EU EW EY
FA FC FE FG FI FK FM FO FQ FS
FU GA GC GE GG GI GK GM GY HA
HC IA JA JC JE JG KA KC KQ KS
KU LA LC LE LG LI LK LM MA MC
ME MG MI MK MM MO MQ MS MU MW
NC NE NG NO OA OC OE PA PC PE
PG QA QC QQ QS QU QW RA RC RE
RG RI RK RM RO RQ RS RU RW RY
Online>
```


Show Network Statistics

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **NETSTAT**

Function: The HSM records details about network activity on both its Management and Host Ethernet ports for diagnostic and security purposes. As a diagnostic aid, it can provide useful information when configuring the unit. If reviewed periodically, it can also provide evidence of unexpected network activity, which may require further investigation.

The HSM collects information about each 'endpoint' that communicates with it. The information recorded will depend on the particular protocol that was used to send the packet.

For TCP packets, the HSM displays:

- Local TCP port
- Remote IP address and TCP port
- TCP state (ESTABLISHED or CLOSED)
- Timestamp (time when connection was initiated)

For UDP packets, the HSM displays:

- Local UDP port
- Remote IP address and UDP port
- Timestamp (time of most recent UDP packet received from this address)

[Note: The Host command "NI" provides the same information via the Host connection.]

Alternatively, the Ethernet statistics of the selected (Host or Management) port may be displayed, including:

- Number of bytes sent & received.
- Number of non-broadcast packets sent & received.
- Number of broadcast packets sent & received.
- Number of errors detected during send & receive operations.
- Number of unknown protocols detected during send & receive operations.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: `netstat [-m] [-e | -c]`

- m Use Management Ethernet port instead of Host port.
- e Display Ethernet statistics.
- c Clear 'old' connection details

Outputs: Text messages as shown in examples.

Errors: • Invalid parameter

Example 1: *In this example, the host port's IP connections are shown.*

```
Online> netstat <Return>
Connections to Host port 193.240.101.65:
Protocol  Local Port  Remote Address      TCP State  Time (D:H:M:S)
TCP       1032         193.240.101.1:3434  ESTABLISHED 2:04:12:55
TCP       1034         193.240.101.1:3437  CLOSED      2:03:41:45
TCP       1035         193.240.101.241:2338 ESTABLISHED 0:00:32:41
TCP       1036         193.240.101.1:3439  CLOSED      2:04:01:27
UDP       2043         193.240.101.1:4045  1:43:10:19
Online>
```

Example 2: *In this example, the management port's IP connections are shown.*

```
Online> netstat -m <Return>
Connections to Management port 193.240.101.11:
Protocol  Local Port  Remote Address      TCP State  Time (D:H:M:S)
TCP       1032         193.240.101.1:3434  ESTABLISHED 2:04:12:55
UDP       2043         193.240.100.163:4054 1:52:07:11
Online>
```

Example 3: *In this example, the host port's Ethernet statistics are displayed.*

```
Online> netstat -e <Return>
Bytes                Received      Sent
Unicast packets      68493        77321
Non-unicast packets  8620         83
Discards             0            0
Errors               0            0
Unknown protocols    486
Online>
```

Example 4: *In this example, the details of old host connections are not cleared.*

```
Online> netstat -c <Return>
Connections to Host port 193.240.101.65:
Protocol  Local Port  Remote Address      TCP State  Time (D:H:M:S)
TCP       1032         193.240.101.1:3434  ESTABLISHED 2:04:12:55
TCP       1034         193.240.101.1:3437  CLOSED      2:03:41:45
TCP       1035         193.240.101.241:2338 ESTABLISHED 0:00:32:41
TCP       1036         193.240.101.1:3439  CLOSED      2:04:01:27
UDP       2043         193.240.101.1:4045  1:43:10:19
Press 'C' to confirm deletion of old connection details: <Return>
If 'C' is not entered, then the old connection details are not erased.
Online>
```

Example 5: *In this example, the details of old host connections are cleared.*

```
Online> netstat -c <Return>
Connections to Host port 193.240.101.65:
Protocol    Local Port  Remote Address    TCP State    Time(D:H:M:S)
TCP         1032       193.240.101.1:3434 ESTABLISHED  2:04:12:55
TCP         1034       193.240.101.1:3437 CLOSED       2:03:41:45
TCP         1035       193.240.101.241:2338 ESTABLISHED  0:00:32:41
TCP         1036       193.240.101.1:3439 CLOSED       2:04:01:27
UDP         2043       193.240.101.1:4045          1:43:10:19
Press 'C' to confirm deletion of old connection details: c <Return>
Connections to Host port 193.240.101.65:
Protocol    Local Port  Remote Address    TCP State    Time(D:H:M:S)
TCP         1032       193.240.101.1:3434 ESTABLISHED  2:04:12:55
TCP         1035       193.240.101.241:2338 ESTABLISHED  0:00:32:41
TCP         1036       193.240.101.1:3439 CLOSED       2:04:01:27
Online>
```

Test TCP/IP Network

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **PING**

Function: To test the specified network node, and the route to it.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

```
ping [-m] [-t] [-n count] [-i TTL] [-w timeout] target
```

-m Use Management Ethernet port instead of Host port.

-t Ping the specified Host until a console key is pressed.

-n count Number of echo requests to send.

-i TTL Time To Live.

-w timeout Timeout in milliseconds to wait for each reply.

target IP address of target node, e.g. 193.240.101.54

Outputs: Text messages as shown in examples.

Errors:

- Missing IP address.
- Invalid IP address.
- Invalid parameter.

Example 1: *In this example, the route to the network node 193.240.101.34 is tested.*

```
Online> ping 193.240.101.34 <Return>
Pinging 193.240.101.34 with 32 bytes of data:

Reply from 193.240.101.34: bytes=32 time<1ms TTL=30
Reply from 193.240.101.34: bytes=32 time<1ms TTL=30
Reply from 193.240.101.34: bytes=32 time<1ms TTL=30
Reply from 193.240.101.34: bytes=32 time<1ms TTL=30

Ping statistics for 193.240.101.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
Online>
```

Example 2: *In this example, the route to the non-existent network node 193.240.101.35 is tested.*

```
Online> ping 193.240.101.35 <Return>
Pinging 193.240.101.35 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 193.240.101.35:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
Online>
```

Trace TCP/IP route

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **TRACERT**

Function: To view the path taken from the HSM to the specified address.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

```
tracert [-m] [-h maximum hops] [-w timeout] target
```

-m Use Management Ethernet port instead of Host port.

-h maximum hops Maximum number of hops to search for target.

-w timeout Timeout in milliseconds to wait for each reply.

Target IP address of target node, e.g. 193.240.101.54

Outputs: Text messages as shown in examples.

Errors:

- Missing IP address.
- Invalid IP address.
- Invalid parameter.

Example 1: *In this example, the path taken to target node 216.239.59.99 is shown.*

```
Online> tracert 216.239.59.99 <Return>
Tracing route to 216.239.59.99 over a maximum of 30 hops
  1    1 ms    <1 ms    <1 ms    193.240.101.1
  2    3 ms    3 ms     3 ms    193.240.100.3
  3   35 ms   31 ms    33 ms    216.239.59.99
Trace complete.
Online>
```

Example 2: *In this example, the path taken to non-existent target node 216.239.59.98 is shown.*

```
Online> tracert 216.239.59.98 <Return>
Tracing route to 216.239.59.98 over a maximum of 30 hops
  1    1 ms    <1 ms    <1 ms    193.240.101.1
  2    3 ms    3 ms     3 ms    193.240.100.3
  3    *      *        *      Request timed out.
  4    *      *        *      Request timed out.
  5    *      *        *      Request timed out.
  6    *      *        *      Request timed out.
  7    *      *        *      Request timed out.
  8    *      *        *      Request timed out.
  9    *      *        *      Request timed out.
 10   *      *        *      Request timed out.
Trace complete.
Online>
```

Chapter 3 - Local Master Keys (LMKs)

Types of LMKs

A **Variant LMK** is a set of 40 DES keys, arranged in pairs, with different pairs (and variants of those pairs) being used to encrypt different types of keys. This is the standard LMK format supported in all versions of Racal/Thales HSM firmware.

Note: The term “Variant LMK” refers to the ‘variant’ method of encrypting keys; a Variant LMK is not itself a variant of any other key.

A **Keyblock LMK** is a triple-length DES key, and is used to encrypt keys in a keyblock format. It is not compatible with a Variant LMK, and it can only be used to encrypt keys in keyblock form.

Note: The term “Keyblock LMK” refers to the ‘keyblock’ method of encrypting keys; a Keyblock LMK is not itself stored in keyblock form.

Multiple LMKs

With version 3.0 (and later) software, it is possible to install multiple LMKs within a single HSM 8000. The precise details of the number and type of installed LMKs are controlled via the HSM’s licence file:

Licence	Description
HSM8-LIC001v3 (standard v3.0 licence)	Two concurrent LMKs can be installed; however, one must be a Variant LMK, and the other a Keyblock LMK.
HSM8-LIC012 LMK x 2 (optional licence)	Two concurrent LMKs can be installed; they can be any combination of Variant and Keyblock LMKs.
HSM8-LIC013 LMK x 5 (optional licence)	Five concurrent LMKs can be installed; they can be any combination of Variant and Keyblock LMKs.

LMK Table

LMKs are stored in a table within the secure memory of the HSM, with each LMK occupying a different ‘slot’ within the table. Each slot has the following attributes:

Attribute	Description
LMK ID	A 2-digit number which uniquely indicates the location of each LMK within the table. All references to LMKs are made by specifying the LMK Identifier.
Key Scheme	“Variant” for traditional Racal/Thales LMK – key encryption performed using the <i>variant</i> method. “Keyblock” for v3.0 and later software – key encryption performed using the <i>keyblock</i> method.
Algorithm	“3DES (2key)” is used by Variant LMKs. “3DES (3key)” is used by Keyblock LMKs. Other algorithm types may be supported in future software releases.
Status	“Test” indicates that the LMK is used for testing purposes. “Live” indicates that the LMK is used for live production purposes. When installing LMKs, the HSM will prevent any mixing of Test and Live

	LMKs within the same slot (i.e. LMK Value and Old LMK Value must have the same status).
Comments	User-entered text, which can be used to help identify LMKs.
Authorisation	Indicates the authorisation status of the HSM for this particular LMK – either a flag (for Authorised State) or a list of authorised activities.
LMK Check Value	The check value of the LMK.
Old LMK Check Value	The check value of the 'old' LMK (in Key Change Storage).

Use the console command VT (View LMK Tables) to view the contents of the HSM's LMK table (but not the actual LMK values).

LMK Commands

The HSM provides the following console commands to support LMK operations:

Command	Page
Generate LMK Component (GK)	64
Load LMK (LK)	66
Move 'Old' LMKs into Key Change Storage (LO)	68
Verify LMK Store (V)	70
Duplicate LMK Component Sets (DC)	71
Delete LMK (DM)	72
Delete 'Old' LMK from Key Change Storage (DO)	73
View LMK Tables (VT)	74

Generate LMK Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **GK**

Function: To generate component(s) of an LMK, and store the component(s) on smartcards. This command may be used to generate Variant or Keyblock LMK components.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- LMK Scheme (Variant or Keyblock).
- LMK Status (Test or Live).
- Component Set number.
- Three secret values (A, B, C) – HSM generates random values if no values are input.

Outputs:

- LMK components written to smartcards.
- LMK component check value.

Errors:

- Card not formatted – use the FC command to format the card.
- Not a LMK card – the card is formatted for HSM settings or is a licence card.
- Warning – card not blank. Proceed? [Y/N] – LMK card is not blank.
- Overwrite LMK set? [Y/N] – card already contains an LMK component.
- Smartcard error; command/return: 0003 – invalid PIN is entered.
- Invalid PIN; re-enter – a PIN of less than 4 or greater than 8 is entered.

**Example 1:
(Variant LMK)**

This example generates a variant LMK component set, and writes the components to a smartcard.

```
Secure> GK <Return>
Variant scheme or key block scheme? [V/K]: V <Return>
Key status? [L/T]: L <Return>
LMK component set [1-9]: 1 <Return>
Enter secret value A: AAAA AAAA AAAA AAAA <Return>
Enter secret value B: BBBB BBBB BBBB BBBB <Return>
Enter value C: 18052002 <Return>
Insert blank card and enter PIN: **** <Return>
Writing keys
Checking keys
Device write complete, check: ZZZZZZ
```

Remove the smartcard and store it securely.

```
Make another copy? [Y/N]: N <Return>
1 copies made.
```

Repeat the procedure to generate further component sets.

**Example 2:
(Keyblock LMK)**

This example generates a keyblock LMK component, and writes the component to a smartcard.

```
Secure> GK <Return>
Variant scheme or key block scheme? [V/K]: K <Return>
Key status? [L/T]: L <Return>
LMK component set [1-9]: 1 <Return>
Enter secret value A: <Return>
Enter secret value B: <Return>
Enter secret value C: <Return>
Insert blank card and enter PIN: **** <Return>
Writing key
Checking key
Device write complete, check: ZZZZZZ
```

Remove the smartcard and store it securely.

```
Make another copy? [Y/N]: N <Return>
1 copies made.
```

Repeat the procedure to generate further components.

Load LMK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **LK**

Function: To load LMK components from smartcards.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- LMK Identifier: 2 numeric digits.
- Smartcards with LMK components.
- PINs or passwords for the Smartcards.

Outputs:

- Master key check value.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Load failed check comparison - card is blank.
- Not a LMK card - card formatted for HSM settings or is a licence card.
- Card not formatted - card is not formatted.
- Smartcard error; command/return: 0003 - invalid PIN is entered.
- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered.
- Invalid key - a standard Thales test key cannot be given live status.
- Incompatible key status - the components have different status ("live" or "test").

Example 1:
(Variant LMK)

This example loads a variant LMK from smartcards and installs it in the HSM.

```
Secure> LK <Return>
Enter LMK id: 00 <Return>
Enter comments: Live LMK for ABC Bank <Return>
LMK in selected location must be erased before proceeding
Erase LMK? Y <Return>
Load LMK from components
Insert card and enter PIN: **** <Return>
Check: ZZZZZZ
Load more components? [Y/N]: Y <Return>
```

Remove the smartcard. Insert the second and subsequent smartcards and repeat the loading procedure. When all have been loaded and the HSM displays the check value, record the check value.

```
Check: ZZZZZZ
Load more components? [Y/N]: N <Return>
LMK id: 00
LMK key scheme: Variant
LMK algorithm: 3DES (2key)
LMK status: Live
Comments: Live LMK for ABC Bank
Confirm details? [Y/N]: Y <Return>
Use the LO command to load LMKs into key change storage.
Secure>
```

Use the A command to authorise the HSM in order to check the LMK components and passwords or PINs.

**Example 2:
(Keyblock LMK)**

This example loads a keyblock LMK from smartcards and installs it in the HSM.

```
Secure> LK <Return>
Enter LMK id: 01 <Return>
Enter comments: Live LMK for XYZ Bank <Return>
LMK in selected location must be erased before proceeding
Erase LMK? Y <Return>
Load LMK from components
Insert card and enter PIN: **** <Return>
Check: ZZZZZZ
Load more components? [Y/N]: Y <Return>
```

Remove the smartcard. Insert the second and subsequent smartcards and repeat the loading procedure. When all have been loaded and the HSM displays the check value, record the check value.

```
Check: ZZZZZZ
Load more components? [Y/N]: N <Return>
LMK id: 01
LMK key scheme: Key block
LMK algorithm: 3DES (3key)
LMK status: Live
Comments: Live LMK for XYZ Bank
Confirm details? [Y/N]: Y <Return>
Use the LO command to load LMKs into key change storage.
Secure>
```

Use the A command to authorise the HSM in order to check the LMK components and passwords or PINs.

Move 'Old' LMKs into Key Change Storage

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: admin.console		

Command: **LO**

Function: To load an old LMK component set into Key Change Storage for use in translations from old to new keys. Note that the current LMK must be installed before an "old" LMK can be installed. Also note that it is possible to install a Variant LMK as the "old" LMK, and with a Keyblock LMK as the "new" LMK.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **admin.console** must be authorised, using the Authorising Officer cards of the specified LMK.

Inputs:

- LMK identifier: 2 numeric digits.
- Smartcards with old LMK components.
- PINs or passwords for the Smartcards.

Outputs:

- Master key check value.

Errors:

- No LMK loaded - there is no LMK loaded in main memory.
- Invalid LMK identifier - entered identifier out of range
- Key block LMK not permitted - it is not permitted to load a key block LMK into key change storage if a variant LMK is loaded in main memory.
- Load failed check comparison - card is blank.
- Not a LMK card - card formatted for HSM settings or is a licence card.
- Card not formatted - card is not formatted.
- Smartcard error; command/return: 0003 - invalid PIN is entered.
- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered.
- Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.
- Invalid key - a standard Thales test key cannot be given live status.
- Incompatible cards - the component cards have different formats.
- Incompatible key status - the components have different status ("live" or "test").

Example:
(Variant LMK)

This example loads a variant LMK from smartcards and installs it as 'old' LMK 00.

```
Secure-AUTH> LO <Return>
Enter LMK id: 00 <Return>
Enter comments: Old LMK for ABC Bank <Return>
Load old LMK from components.
Insert card and enter PIN: **** <Return>
Check: ZZZZZZ
Load more components? [Y/N]: Y <Return>
```

Remove the smartcard. Insert the second and subsequent smartcards and repeat the loading procedure until all old component sets have been moved into key change storage. When all have been moved and the HSM displays the check value, press N to terminate the procedure.

```
Check: ZZZZZZ
Load more components? [Y/N]: N <Return>
LMK id: 00
LMK key scheme: Variant
LMK algorithm: 3DES (2key)
LMK status: Live
Comments: Old LMK for ABC Bank
Confirm details? [Y/N]: Y <Return>
Secure-AUTH>
```

Example:
(Keyblock LMK)

This example loads a keyblock LMK from smartcards and installs it as 'old' LMK 01.

```
Secure-AUTH> LO <Return>
Enter LMK id: 01 <Return>
Enter comments: Old LMK for XYZ Bank <Return>
Load old LMK from components.
Insert card and enter PIN: **** <Return>
Check: ZZZZZZ
Load more components? [Y/N]: Y <Return>
```

Remove the smartcard. Insert the second and subsequent smartcards and repeat the loading procedure until all old component sets have been moved into key change storage. When all have been moved and the HSM displays the check value, press N to terminate the procedure.

```
Check: ZZZZZZ
Load more components? [Y/N]: N <Return>
LMK id: 01
LMK key scheme: Key block
LMK algorithm: 3DES (3key)
LMK status: Live
Comments: Old LMK for XYZ Bank
Confirm details? [Y/N]: Y <Return>
Secure-AUTH>
```

Verify LMK Store

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **V**

Function: To confirm that the check value is identical to the value that was recorded when the LMK set was installed.
For Variant LMKs, the length of the displayed check value is determined by the CS (Configure Security) setting "Restrict Key Check Value to 6 hex chars".
For Keyblock LMKs, the length of the displayed check value is always 6 hex digits.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- LMK Identifier: 2 numeric digits.

Outputs:

- Master key check value.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.

Example:

```
Online> v <Return>
Enter LMK id: 03 <Return>
Check: ZZZZZZ
Online>
```

Duplicate LMK Component Sets

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DC**

Function: To copy an LMK component onto another smartcard.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- Smartcard with LMK component.
- PIN for the smartcard.

Outputs:

- LMK check value.

Errors:

- Load failed check comparison - card is blank
- Not a LMK card - card formatted for HSM settings or is a licence card
- Card not formatted - card is not formatted
- Smartcard error; command/return: 0003 - invalid PIN is entered
- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered.
- Warning - card not blank. Proceed? [Y/N] - LMK card is not blank
- Overwrite LMK set? [Y/N] - the smartcard already contains an LMK component. It can be overwritten if desired.

Example:

```
Secure> DC <Return>
Insert card to be duplicated and enter PIN: **** <Return>
Insert blank card and enter PIN: **** <Return>
Device write complete, check: ZZZZ ZZZZ ZZZZ ZZZZ
Make another copy? [Y/N]: N <Return>
Secure>
```

Delete LMK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: admin.console		

Command: **DM**

Function: To delete a selected LMK and (if loaded) the LMK in the corresponding location in key change storage.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **admin.console** must be authorised, using the Authorising Officer cards of the specified LMK.

Inputs: • LMK Identifier: 2 numeric digits.

Outputs: • Display of relevant entry from LMK table and the key change storage table.

Errors: • Invalid LMK identifier - no LMK loaded or entered identifier out of range.
• Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.

Example:

```
Secure-AUTH> DM <Return>
Enter LMK id: 01 <Return>
```

LMK table entry:

LMK table:

ID	Auth	Scheme	Algorithm	Status	Check	Comments
01	No	Key Block	3DES(3key)	Test	999999	Test LMK for XYZ Bank

Key change storage table:

ID	Scheme	Algorithm	Status	Check	Comments
01	Variant	3DES(2key)	Test	876543	Old test LMK for XYZ Bank

Confirm LMK deletion [Y/N]: **Y** <Return>

LMK deleted from main memory and key change storage

Secure>

Delete 'Old' LMK from Key Change Storage

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DO**

Function: To delete a selected LMK from key change storage. This command may only be used if an LMK is loaded in the corresponding location in main LMK memory.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- LMK Identifier: 2 numeric digits.

Outputs:

- Display of relevant entry from the key change storage table.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.

Example:

```
Secure> DO <Return>
Enter LMK id: 01 <Return>

Key change storage table entry:
ID Scheme Algorithm Status Check Comments
01 Variant 3DES(2key) Test 876543 Old test LMK for XYZ Bank

Confirm LMK deletion [Y/N]: Y <Return>
LMK deleted from key change storage

Secure>
```

View LMK Tables

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **VT**

Function: To display the LMK table and the corresponding table for key change storage.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs: Displayed LMK table and key change storage table.

Errors: None.

Example: Online-AUTH> **VT** <Return>

LMK table:

ID	Auth	Scheme	Algorithm	Status	Check	Comments
00	Yes (1)	Variant	3DES (2key)	Live	123456	Default LMK, used for RST Bank
01	No	Key Block	3DES (3key)	Test	999999	Test LMK for XYZ Bank
02	Yes (4)	Key Block	3DES (3key)	Live	396723	Live LMK for PQR Bank
03	No	Key Block	3DES (3key)	Live	223344	Management LMK

Key change storage table:

ID	Scheme	Algorithm	Status	Check	Comments
01	Variant	3DES (2key)	Test	876543	Old test LMK for XYZ Bank
02	Key Block	3DES (3key)	Live	085392	Old LMK for PQR Bank

Online-AUTH>

Chapter 4 - Operational Commands

HSM Authorisation Commands

The HSM need to be authorised for certain commands to be executed - usually those involving clear text data.

There are two methods of authorising the HSM – using:

- a single Authorised State;
- multiple Authorised Activities.

If the HSM needs to be placed in Authorised State using the Authorising Officer cards (or passwords) corresponding to a particular LMK, then the command will only be authorised for that particular LMK identifier. For example, if the “FK” console command (“Form Key from Components”) is authorised using the passwords corresponding to the LMK with identifier “00”, then only keys encrypted using LMK “00” may be formed using the command.

It is possible to authorise the HSM using multiple Authorising Officer cards (or passwords), so that the HSM may be simultaneously authorised for different LMKs.

The HSM provides the following console commands to support the authorisation of the HSM:

Command	Page
Enter the Authorised State (A)	76
Cancel the Authorised State (C)	76
Authorise Activity (A)	78
Cancel Authorised Activity (C)	84
View Authorised Activities (VA)	86

Enter the Authorised State

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **A**

Function: To set the HSM into the Authorised State.
The HSM prompts for either Smartcards or Passwords, as applicable, which must correspond to the LMK being authorised.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- LMK Identifier: 2 numeric digits.
- PIN (if applicable): 4 to 8 alphanumeric characters.
- Password (if applicable): 16 alphanumeric characters.

Outputs: • Text messages as shown in examples.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Card not formatted - card is not formatted.
- Not a LMK card - card formatted for HSM settings or is a licence card.
- Smartcard error; command/return: 0003 - invalid PIN is entered.
- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 digits is entered.
- Data invalid; please re-enter - the password is an invalid length.

Example 1: *This example authorises the HSM using smartcards.*

```
Online> A <Return>
Enter LMK id: 00 <Return>
First Officer:
Insert card and enter PIN: **** <Return>
Second Officer:
Insert card and enter PIN: **** <Return>
AUTHORISED
Online-AUTH>
```

Example 2: *This example authorises the HSM using passwords.*

```
Online> A <Return>
Enter LMK id: 00 <Return>
First Officer:
Password: ***** <Return>
Second Officer:
Password: ***** <Return> ← Password too long
Data invalid; please re-enter: ***** <Return>
AUTHORISED
Online-AUTH>
```

Cancel the Authorised State

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **C**

Function: To cancel the Authorised State.
There is an equivalent command available to the host (Host command 'RA')

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- LMK Identifier: 2 numeric digits.

Outputs:

- Text messages as shown in example.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.

Example 1:

```
Online-AUTH> C <Return>
Enter LMK id: 00 <Return>
NOT AUTHORISED for LMK id 00
Online>
```

Authorise Activity

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **A**

Function: To authorise the HSM to perform certain specified activities.
 In command line mode, the operator specifies which activities are to be authorised.
 In menu mode, the operator is prompted to enter the activities.
 In both cases, the specified activities are authorised by submitting two Security Officer cards or passwords, which must correspond to the LMK being authorised.
 The syntax for authorising activities is described in Appendix G.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- LMK Identifier: 2 numeric digits
- Activities to be authorised.
- Timeout value: Number of minutes before HSM will revoke chosen authorised activity.
- PIN (if applicable): 4 to 8 alphanumeric characters.
- Password (if applicable): 16 alphanumeric characters.
- Use "-h" to display help.

Outputs: • Text messages as shown in examples.

Syntax: **Syntax : A** [*<Activity>*] [*<Activity>*] ...

Activity: *<Category>*.*[<Sub-category>]*.*[<Interface>]**[:<Timeout>]*

Category =

generate|component|genprint|import|export|pin|audit|admin|diag|
misc|command

Sub-category (for 'generate|import|export') = key type code, e.g. 001 for ZPK.

Sub-category (for 'pin') = mailer|clear

Interface = host|console

Timeout = value in minutes

Names may be shortened but must remain unique.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Card not formatted - card is not formatted.
- Not a LMK card - card formatted for HSM settings or is a licence card.
- Smartcard error; command/return: 0003 - invalid PIN is entered.
- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered.
- Data invalid; please re-enter: the password is an invalid length.

Notes:

Activities are described in terms of four fields: *Category*, *Sub-Category*, *Interface* and *Timeout*. If the *Timeout* field is omitted, the activity remains authorised until cancelled either by the console command “C” or the host command “RA”.

Omitting either the *Sub-Category* and/or the *Interface* field is equivalent to authorising multiple activities consisting of all possible combinations of valid values for the missing fields. For clarification:

```
pin.mailer
```

is equivalent to:

```
pin.mailer.host
```

```
pin.mailer.console
```

and:

```
pin
```

is equivalent to:

```
pin.clear.console
```

```
pin.clear.host
```

```
pin.mailer.console
```

```
pin.mailer.host
```

When authorising activities, two (or more) activities may overlap, for example:

```
pin
```

```
pin.mailer
```

There is no requirement to attempt to reduce activities to the minimum set. The list of authorised activities simply consists of all those entered (and authorised) by the user.

There is one case when it will be necessary to overwrite an existing activity: when only the *Timeout* field changes. For example, suppose that the following activity is authorised:

```
export.001.console:11
```

and the user uses the ‘A’ command to authorise the following activity:

```
export.001.console:60
```

then this should overwrite the first one (even if the newer activity has a shorter *Timeout* value).

Note: When omitting the sub-category, but including the interface, there should be two delimiters “.” between them:

Example: `export..host` allows export of any (valid) key using a host command.

A full list of activities categories and sub categories is identified in Appendix G.

Example 1:
(Variant or
Keyblock LMK)

This example authorises a single activity via the menu.

```
Online> A <Return>
Enter LMK id: 00 <Return>
No activities are authorised for LMK id 00.
List of authorisable activities:
generate      genprint      component      import
export        pin           audit          admin
diagnostic    misc          command
Select category: pin <Return>
clear         mailer
Select sub-category, or <RETURN> for all: mailer <Return>
host          console
Select interface, or <RETURN> for all: <Return>
Enter time limit for pin.mailer, or <RETURN> for permanent: <Return>
Enter additional activities to authorise? [y/N]: n <Return>

The following activities are pending authorisation:
pin.mailer

First Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 00:
pin.mailer

Online-AUTH>
```

Example 2:
(Variant or
Keyblock LMK)

This example authorises a single activity via the command line.

```
Online> a pin.mailer <Return>
Enter LMK id: 00 <Return>
The following activities are pending authorisation:
pin.mailer

First Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 00:
pin.mailer

Online-AUTH>
```

Example 3:
(Variant LMK)

This example authorises an additional three activities via the menu.

```
Online-AUTH> A <Return>
Enter LMK id: 00 <Return>
The following activities are authorised for LMK id 00:
pin.mailer
List of authorisable activities:
generate      genprint      component      import
export        pin           audit          admin
diagnostic    misc          command
Select category: generate <Return>
000           100           200           001
002           400           003           006
008           009           109           209
309           409           509           709
00a           00b           rsa
Select sub-category, or <RETURN> for all: 000 <Return>
```



```

host          console
Select interface, or <RETURN> for all: c <Return>
Enter time limit for generate.000.console, or <RETURN> for permanent: 60
<Return>

Enter additional activities to authorise? [y/N]: y <Return>
List of authorisable activities:
generate      genprint      component      import
export        pin            audit          admin
diagnostic    misc            command
Select category: export <Return>
000           100           200           001
002           400           003           006
008           009           109           209
309           409           509           709
00a           00b           rsa
Select sub-category, or <RETURN> for all: 001 <Return>
host          console
Select interface, or <RETURN> for all: h <Return>
Enter time limit for export.001.host, or <RETURN> for permanent:
<Return>

Enter additional activities to authorise? [y/N]: y <Return>
List of authorisable activities:
generate      genprint      component      import
export        pin            audit          admin
diagnostic    misc            command
Select category: admin <Return>
host          console
Select interface, or <RETURN> for all: c <Return>
Enter time limit for admin, or <RETURN> for permanent: 240 <Return>

Enter additional activities to authorise? [y/N]: n <Return>
The following activities will be authorised for LMK id 00.
admin..console:240
export.001.host
generate.000.console:60

First Officer
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 00:
admin:240 (240 mins remaining)
export.001.host
generate.000.console:60 (60 mins remaining)
pin.mailer

Online-AUTH>

```

Example 4:
(Variant LMK)*This example authorises an additional three activities via the command line.*

```
Online-AUTH> a gene.000.con:60 exp.001.host admin:240 <Return>
Enter LMK id: 00 <Return>
The following activities will be authorised for LMK id 00:
admin:240
export.001.host
generate.000.console:60

First Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 00:
admin:240 (240 mins remaining)
export.001.host
generate.000.console:60 (60 mins remaining)
pin.mailer

Online-AUTH>
```

Example 5:
(Variant or
Keyblock LMK)*This example authorises a single activity via the command line.*

```
Online> a pin.clear <Return>
Enter LMK id: 01 <Return>
The following activities are pending authorisation:
pin.clear

First Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 01:
pin.clear

Online-AUTH>
```

Example 6:
(Keyblock LMK)*This example authorises an additional three activities via the menu.*

```
Online-AUTH> A <Return>
Enter LMK id: 01 <Return>
The following activities are authorised for LMK id 01:
pin.clear
List of authorisable activities:
generate      genprint      component      import
export        pin           audit         admin
diagnostic    misc          command
Select category: export <Return>
01             B0             C0             11
12             13            D0             21
22             E0            E1             E2
E3             E4            E5             E6
31             32            K0             51
52             M0            M1             M2
M3             M4            M5             61
62             63            64             65
P0             71            72             73
V0             V1            V2
Select sub-category, or <RETURN> for all: 72 <Return>
host          console
Select interface, or <RETURN> for all: C <Return>
Enter time limit for export.72.console, or <RETURN> for permanent: 60
<Return>
```

```

Enter additional activities to authorise? [y/N]: y <Return>
List of authorisable activities:
generate      genprint      component      import
export        pin          audit          admin
diagnostic    misc          command
Select category: admin <Return>
host          console
Select interface, or <RETURN> for all: <Return>
Enter time limit for admin, or <RETURN> for permanent: 240 <Return>

Enter additional activities to authorise? [y/N]: y <Return>
List of authorisable activities:
generate      genprint      component      import
export        pin          audit          admin
diagnostic    misc          command
Select category: misc <Return>
host          console
Select interface, or <RETURN> for all: c <Return>
Enter time limit for admin, or <RETURN> for permanent: <Return>

Enter additional activities to authorise? [y/N]: n <Return>
The following activities will be authorised for LMK id 01.
misc..console
admin:240
export.72.console:60

First Officer
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 01:
misc..console
admin:240 (240 mins remaining)
export.72.console (60 mins remaining)
pin.clear

Online-AUTH>

```

Example 7:
(Keyblock LMK)

This example authorises an additional three activities via the command line.

```

Online-AUTH> a exp.72.con:60 admin:240 misc..console <Return>
Enter LMK id: 01 <Return>
The following activities will be authorised for LMK id 01:
misc..console
admin:240
export.72.console:60

First Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>
Second Officer:
Insert Card for Security Officer and enter the PIN: **** <Return>

The following activities are authorised for LMK id 01:
misc..console
admin:240 (240 mins remaining)
export.72.console (60 mins remaining)
pin.clear

Online-AUTH>

```

Cancel Authorised Activity

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **C**

Function: To cancel one or more Authorised Activities.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: • LMK Identifier: 2 numeric digits.

Outputs: • Text messages as shown in examples.

Syntax: **c** [**<Activity>**] [**<Activity>**] ...

Activity: **<Category>** [**.<Sub-category>**] [**.<Interface>**] [**:<Timeout>**]

Category =

generate|component|genprint|import|export|pin|audit|admin|diag|
misc|command

Sub-category (for 'generate|import|export') = key name, e.g. TPK, MK-AC, etc.

Sub-category (for 'pin') = mailer|clear

Interface = host|console

Timeout = value in minutes

Names may be shortened but must remain unique.

When cancelling an authorised activity which includes a timeout, the original value of the timeout should be specified.

Note: When omitting the sub-category, but including the interface, there should be two delimiters "." between them:

Example: `export..host` allows export of any (valid) key using a host command.

A full list of activities categories and sub categories is identified in Appendix G.

Errors: • Invalid LMK identifier - no LMK loaded or entered identifier out of range.
• Invalid input.

Example 1: *This example cancels an existing activity via the menu.*

(Variant or
Keyblock LMK)

```
Online-AUTH> C <Return>
Enter LMK id: 00 <Return>
Cancel pin.mailer? [y/N] y <Return>
No activities are authorised for LMK id 00.
Online>
```

Note: This example assumes that the activities in the Authorise Activity command Example 1 (above) are active.

Example 2: *This example cancels an existing activity via the command line.*

(Variant or
Keyblock LMK)

```
Online-AUTH> C pin.mailer <Return>
Enter LMK id: 00 <Return>
No activities are authorised for LMK id 00.
Online>
```

Note: This example assumes that the activities in the Authorise Activity command Example 2 (above) are active.

Example 3: *This example cancels an existing activity via the menu.*
(Variant LMK)

```
Online-AUTH> c <Return>
Enter LMK id: 00 <Return>
Cancel admin:240 (194 mins remaining) ? [y/N] y <Return>
Cancel export.001.host? [y/N] n <Return>
Cancel generate.000.console:60 (14 mins remaining)? [y/N] y <Return>
Cancel pin.mailer? [y/N] n <Return>
The following activities are authorised for LMK id 00:
export.001.host
pin.mailer
Online-AUTH>
```

Note: This example assumes that the activities in the Authorise Activity command Example 3 (above) are active.

Example 4: *This example cancels an existing activity via the command line.*
(Variant LMK)

```
Online-AUTH> c gene.000.c admin <Return>
Enter LMK id: 00 <Return>
The following activities are authorised for LMK id 00.
export.001.host
pin.mailer
Online-AUTH>
```

Note: This example assumes that the activities in the Authorise Activity command Example 4 (above) are active.

Example 5: *This example cancels an existing activity via the command line.*

(Variant or
Keyblock LMK)

```
Online-AUTH> C pin.clear <Return>
Enter LMK id: 01 <Return>
No activities are authorised for LMK id 01.
Online>
```

Note: This example assumes that the activities in the Authorise Activity command Example 5 (above) are active.

View Authorised Activities

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **VA**

Function: To view all active authorised activities.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- LMK identifier: 2 numeric digits.

Outputs:

- List of active authorised activities.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.

Example:

```
Online-AUTH> VA <Return>
Enter LMK id: 00 <Return>
The following activities are authorised for LMK id 00:
admin:240 (228 mins remaining)
export.001.host
generate.000.console:60 (48 mins remaining)
pin.mailer
Online-AUTH>
```

Note: This example assumes the activities in the Authorise Activity command Example 4 (above) were authorised 12 minutes ago.

Logging Commands

An Error Log and an Audit Log are provided, each with a command to display the log and a command to clear the log. There is also a command to enable the user to set their time zone, so that the correct time is displayed in audit log reports.

The Error log stores fault information for use by Thales e-Security support personnel. It contains 100 slots for error codes and sub-codes entries. After 100 entries the HSM will overwrite existing entries i.e. event 101 will overwrite the 1st entry. The error log is used to log unexpected software errors, hardware failures and alarm events. Whenever an error occurs, that error code is stored, along with the time, date and severity level. Additional errors that have the same error code cause the time and date of that code to be updated. In this way, each error type remains in the log (with the most recent time and date) and is not lost. The severity levels are:

- Informative (0) Something abnormal happened, but was not important.
- Recoverable (1) Something abnormal happened, but the unit recovered from it without rebooting or losing data.
- Major (2) Something abnormal happened, but the unit recovered from it but may have lost data/information due to restarting a process or re-initializing hardware. The unit may not function in a full capacity.
- Catastrophic (3) Something abnormal happened, and the unit had to reboot to recover.

Only catastrophic errors cause the HSM to reboot. New errors cause the Fault LED on the front panel to flash.

The Audit journal contains 2000 entries for audit records. Whenever the HSM state is altered through power-up, key-lock changes or console commands, the Audit log is updated with the action and the time and date. The Audit log can also be configured to record execution of almost any console or host command. The Audit log records state changes until it is 100% full and for each subsequent state change the earliest (i.e. oldest) record in the log is deleted to make room for the new record. A number of host commands are provided which allow the host computer to extract and archive (print) audit records from the HSM.

Management of the Audit journal is performed from the console using the command 'AUDITOPTIONS', whilst 'AUDITLOG' is used to retrieve the log and 'CLEARAUDIT' to clear the log. The HSM must be put into the secure-authorized state in order to execute the 'AUDITOPTIONS' and 'CLEARAUDIT' console commands.

Note: Auditing host or console commands may impact HSM performance.

The HSM provides the following console commands to support storage and retrieval of HSM settings:

Command	Page
Display the Error Log (ERRLOG)	88
Clear the Error Log (CLEARERR)	89
Display the Audit Log (AUDITLOG)	90
Clear the Audit Log (CLEARAUDIT)	91
Audit Options (AUDITOPTIONS)	92

Display the Error Log

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **ERRLOG**

Function: To display the entries in the error log.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- A listing of the errors in the error log, or text message: "Error log is empty".

Errors: None.

Example 1: Offline> ERRLOG <Return>
Error log is empty
Offline>

Example 2: Offline> ERRLOG <Return>
Error Log (2 entries)

1: ERROR (1): Invalid queue size (Severity: 2, Code = 00000001, Sub-code = 00000002)
2: ERROR (1): Key3 cannot be specified without key2 (Severity: 0, Code = 00000004, Sub-code = 00000003)
Offline>

Clear the Error Log

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CLEARERR**

Function: To clear the entries in the error log.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs: None.

Outputs:

- A confirmation message.

Errors: None.

Example:

```
Secure> CLEARERR <Return>
Error log Cleared
Secure>
```

Display the Audit Log

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **AUDITLOG**

Function: To display the entries in the audit log.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- A listing of the entries in the audit log (maximum 100).
- The following text messages can be output:
 - Audit Log (in entries)
 - Continue displaying audit log entries? Yes/No/Continuous

Errors: None.

Example 1: Offline> **AUDITLOG** <Return>
Audit log is empty
Offline>

Example 2: Offline> **AUDITLOG** <Return>
Audit Log (5 entries)

Counter	Time	Date	Command/Event
00000100	10:32:21	01/04/2004	Console command CH
00000101	10:32:48	01/04/2004	Keylock turned to Online
00000102	10:34:57	01/04/2004	System restarted
00000103	10:36:03	01/04/2004	Host command CA, response FF
00000104	10:42:32	01/04/2004	Host command CA, response 00

Offline>

After 20 entries are displayed continuously, the following text is displayed:

Continue displaying audit log entries? [Y/N/C]:

Clear the Audit Log

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: audit.console		

Command: **CLEARAUDIT**

Function: To clear the entries in the audit log.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **audit.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs: None.

Outputs:

- One of the following text messages:
 - Audit Log Cleared
 - Audit Log is empty

Errors:

- Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.

Example 1:

```
Secure-AUTH> CLEARAUDIT <Return>
Audit Log Cleared
Secure-AUTH>
```

Example 2:

```
Secure-AUTH> CLEARAUDIT <Return>
Audit Log is empty
Secure-AUTH>
```

Audit Options

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: audit.console		

Command: **AUDITOPTIONS**

Function: To configure the HSM's auditing functionality.

Authorisation: The HSM must be in the Offline or Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **audit.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs:

- Changes to configuration:
 - List of audited console commands
 - List of audited host commands
 - Audit user actions (Y/N)
 - Audit counter value

Outputs:

- Current & new configuration details:
 - List of audited console commands
 - List of audited host commands
 - Audit user actions (Y/N)
 - Audit counter value

Errors:

- Command only allowed from Offline-Authorised - the HSM is not in Offline (or Secure) State, or the HSM is not authorised to perform this operation, or both.
- Invalid Entry - the value entered is invalid.
- Card not formatted to save/retrieve HSM settings - Attempt with another card? [Y/N]

Example: Offline-AUTH> **AUDITOPTIONS** <Return>
List of Audited Console Commands:
GC, GS, EC, FK
List of Audited Host Commands:
A0, A4, GG, GY
Audit User Actions:
Enabled
Audit Counter Value:
000000F3

Modify Audited Command List? [Y/N]: **Y** <Return>
Enter command code (e.g. +CDE) or Q to Quit: **+CDE** <Return>
Console command DE added to list
Enter command code (e.g. +CDE) or Q to Quit: **-HA4** <Return>
Host command A4 removed from list
Enter command code (e.g. +CDE) or Q to Quit: **Q** <Return>

Audit User Actions (Y/N): **N** <Return>

Current Audit Counter value is: 000000F3
Enter new value or <RETURN> for no change: **2AAF** <Return>

List of Audited Console Commands:
GC, GS, EC, FK, DE
List of Audited Host Commands:
A0, GG, GY
Audit User Actions:
Disabled
Audit Counter Value:
00002AAF

Save Audit Settings to smartcard? [Y/N]: **Y** <Return>

Insert Card and press Enter: <Return>
Audit Settings written to the smartcard.
Offline-AUTH>

Time and Date Commands

The SETTIME command is used to set the system time and date used by the HSM for the audit log entries. The user should use this command to adjust the time for the local timezone. The time and date can be queried using the GETTIME command.

The HSM provides the following console commands to support storage and retrieval of HSM settings:

Command	Page
Set the Time and Date (SETTIME)	95
Query the Time and Date (GETTIME)	96

Set the Time and Date

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: admin.console		

Command: **SETTIME**

Function: To set the system time and date used by the HSM.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **admin.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs:

- The time in hours and minutes.
- The date in year, month and day.

Outputs:

- Text messages, as in the example below.

Errors:

- Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.
- Response invalid. Re-enter - an invalid value has been entered.

Example:

```
Secure-AUTH> settime <Return>
Enter hours [HH] (24 hour format): 10 <Return>
Enter minutes [MM]: 08 <Return>
Enter year [YYYY] (2002 or above): 2004 <Return>
Enter month [MM]: 02 <Return>
Enter day [DD]: 12 <Return>
The system time has been modified.
Secure-AUTH>
```

Query the Time and Date

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **GETTIME**

Function: To query the system time and date.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- The year, month and date.
- The time in hours, minutes and seconds.

Errors: None.

Example:

```
Online> gettime <Return>
System date and time: Feb 12 10:08:19 2004
Online>
```


HSM Settings, Storage and Retrieval

Commands are provided to save the HSM Alarm, Host and Security settings to a smartcard and to restore the settings to the HSM. Besides the dedicated command to Save HSM Settings to Smartcard, the individual configuration commands have the option to save settings to smartcard:

- CL (Configure Alarms) to save the Alarm configuration.
- CH (Configure Host) to save the Host port configuration.
- CS (Configure Security) to save the Security configuration.
- AUDITOPTIONS (Audit Options) to save the Audit configuration.

The HSM provides the following console commands to support storage and retrieval of HSM settings:

Command	Page
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Save HSM Settings to a Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: admin.console		

Command: **SS**

Function: To save the alarm, host port, security and audit settings to a smartcard.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **admin.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs: None.

Outputs: • Confirmation messages that Alarm, Host, Security and Audit settings are saved.

Errors:

- Card not formatted to save/retrieve HSM settings.
Attempt with another card? [Y/N] - card is formatted for LMK storage or is a licence card.
- Card not formatted. Attempt with another card? [Y/N] - card is not formatted.
- Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.

Example:

```
Secure-AUTH> SS <Return>
Insert card and press ENTER: <Return>
ALARM settings saved to the smartcard.
HOST settings saved to the smartcard.
SECURITY settings saved to the smartcard.
AUDIT settings saved to the smartcard.
Secure-AUTH>
```

Retrieve HSM Settings from a Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: admin.console		

Command: **RS**

Function: To read the alarm, host port, security and audit settings from a smartcard. The user is then prompted to use these to overwrite the existing HSM settings. If the settings on the smartcard were saved using a configuration command (CL, CH, CS and AUDITOPTIONS), then only those settings are overwritten.

Authorisation: The HSM must be in the Secure state to run this command. Additionally, the HSM must be either in the Authorised State, or the activity **admin.console** must be authorised, using the Authorising Officer cards of the Management LMK.

Inputs:

- Overwrite alarm settings: Y/N.
- Overwrite host settings: Y/N.
- Overwrite security settings: Y/N.
- Overwrite audit settings: Y/N.

Outputs:

- The Alarm, Host, Security and Audit settings stored on the smartcard are listed.

Errors:

- Card not formatted to save/retrieve HSM settings.
Attempt with another card? [Y/N] - card is formatted for LMK storage or is a licence card.
- Card not formatted. Attempt with another card? [Y/N] - card is not formatted.
- Command only allowed from Secure-Authorised - the HSM is not in Secure State, or the HSM is not authorised to perform this operation, or both.

Example:

```
Secure-AUTH> RS <Return>
Insert card and press ENTER: <Return>
Temperature Alarm: ON
Motion Alarm: OFF
Overwrite alarm settings with the settings above? [Y/N]: y <Return>
ALARM settings retrieved from smartcard

Comms Mode: BINARY
Message header length: 4
Protocol: ETHERNET
Character format: ASCII
UDP active: YES
TCP active: YES
Number of TCP connections: 1
Well-Known-Port address: 1500
Overwrite host settings with the settings above? [Y/N]: y <Return>
HOST settings retrieved from smartcard

Enable ZMK translate command: YES
PIN length: 04
Old Encrypted PIN length: 05
Echo: ON
Atalla ZMK variant support: ON
Transaction key support: RACAL
User storage key length: SINGLE
Single DES: DISABLED
Select clear PINs: YES
Enable X9.17 for import: YES
Enable X9.17 for export: YES
Solicitation batch size: 1024
ZMK length: Double
PIN encryption algorithm: A
Card/password authorisation: C
Encrypt/Plaintext Decimalisation table: E
Decimalisation table checks enabled: YES
Authorised State required when importing DES key under RSA key: YES
Minimum HMAC verification length in bytes: 10
Enable PKCS#11 import and export for HMAC keys: NO
Enable ANSI X9.17 import and export for HMAC keys: NO
Enable ZEK encryption of all printable ASCII chars: NO
Enable ZEK encryption of "Base94" ASCII chars: NO
Enable ZEK encryption of "Base64" ASCII chars: YES
Enable ZEK encryption of "Hex-only" ASCII chars: YES
Restrict Key Check Value to 6 Hex Chars: YES
Enable Multiple Authorised Activities: YES
Enable support for variable length PIN offset: NO
Enable weak PIN checking: NO
Enable PIN Block format 34 as output format for PIN translations to ZPK:
NO
Default LMK identifier: 00
Default Management LMK identifier: 00
Use HSM clock for date/time validation: YES
Additional padding to disguise key length: NO
Key export and import in trusted format only: YES
Overwrite security settings with the settings above? [Y/N]: y <Return>
SECURITY settings retrieved from smartcard.

User Action: ENABLED
Audit Counter: 00000000
No Host Commands
No Console Commands
Overwrite audit settings with the settings above? [Y/N]: y <Return>
AUDIT settings retrieved from smartcard.

Secure-AUTH>
```

Key Management Commands

The HSM provides the following host commands to support generic key management operations:

Command	Page
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Generate Key Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.{key}.console		

Command: **GC**

Function: To generate a key component and display it in plain and encrypted forms.

	Variant LMK	Keyblock LMK
Authorisation:	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key type code of the key component being generated.	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key usage code of the key component being generated.
Inputs:	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Type: See Key Type Table. • Key Scheme: See Key Scheme Table.	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Scheme: See Key Scheme Table. • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Component Number: 1-9. • Exportability: See Exportability Table. • Optional Block data.
Outputs:	• Clear text key component. • Key component encrypted under an appropriate variant of the selected LMK. • Component check value.	• Clear text key component. • Keyblock containing the component encrypted under the selected LMK. • Component check value.
Errors:	• Invalid LMK identifier - no LMK loaded or entered identifier out of range. • Invalid key type; re-enter - the key type is invalid. See Key Type Table. • Invalid key scheme for key length - the Key Scheme is inappropriate for Key length. • Invalid key scheme - an invalid key scheme is entered. See Key Scheme Table. • Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department. • Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.	

Example 1:
(Variant LMK)

This example generates a component in plaintext & encrypted form.

```
Online-AUTH> GC <Return>
Enter LMK id: 00 <Return>
Enter Key length [1,2,3]: 2 <Return>
Enter Key Type: 001 <Return>
Enter Key Scheme: U <Return>

Clear Component: XXXX XXXX XXXX XXXX
Encrypted Component: U YYYY YYYY YYYY YYYY
```

```
Key check value: ZZZZZZ  
Online-AUTH>
```

Example 2:
(Keyblock LMK)

This example generates a component in plaintext & encrypted form.

```
Online-AUTH> GC <Return>  
Enter LMK id: 01 <Return>  
Enter key length [1,2,3]: 2 <Return>  
Enter key scheme: S <Return>  
Enter key usage: P0 <Return>  
Enter mode of use: N <Return>  
Enter component number [1-9]: 2 <Return>  
Enter exportability: E <Return>  
Enter optional blocks? [Y/N]: N <Return>  
  
Clear component: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX  
Encrypted component: S YYYYYYYY.....YYYYYY  
Key check value: ZZZZZZ  
Online-AUTH>
```

Generate Key and Write Components to Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: component.{key}.console		

Command: **GS**

Function: Generates a key in 2 to 3 component and write the components to smartcards.

	Variant LMK	Keyblock LMK
Authorisation:	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key type code of the key being generated.	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key usage code of the key being generated.
Inputs:	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Type: See Key Type Table. • Key Scheme: See Key Scheme Table. • Number of components: 2-3. • Smartcard PINs.	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Scheme: See Key Scheme Table. • Number of components: 2-3. • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Key Version Number: 00-99. • Exportability: See Exportability Table. • Optional Block data. • Smartcard PINs.
Outputs:	• Key encrypted under an appropriate variant of the selected LMK. • Key check value.	• Keyblock containing the key encrypted under the selected LMK. • Key check value.
Errors:	• Invalid LMK identifier - no LMK loaded or entered identifier out of range. • Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered. • Smartcard error; command/return: 0003 - invalid PIN is entered. • Warning - card not blank. Proceed? [Y/N] - the smartcard entered is not blank. • Overwrite key component? [Y/N] - the smartcard already contains a key component. It can be overwritten if desired. • Device write failed - the component could not be verified. • Invalid key scheme for key length - the Key scheme is inappropriate for Key length. • Invalid key type; re-enter - the key type is invalid. See Key Type Table. • Invalid key scheme - an invalid key scheme is entered. See Key Scheme Table. • Invalid entry - an invalid number of components has been entered. • Not a LMK card - card formatted for HSM storage or is a licence card. • Card not formatted - card is not formatted.	

- Command only allowed from Authorised – the HSM is not authorised to perform this operation.
- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors – the value entered is invalid, or incompatible with previously entered values.

Example 1:
(Variant LMK)

This example writes components to two smartcards, and encrypts the formed key.

```
Online-AUTH> GS <Return>
Enter LMK id: 00 <Return>
Enter Key length [1,2,3]: 1 <Return>
Enter Key Type: 001 <Return>
Enter Key Scheme: 0 <Return>
Enter number of components [2-3]: 2 <Return>
Insert card 1 and enter PIN: **** <Return>
Make additional copies? [Y/N]: N <Return>
Insert card 2 and enter PIN: **** <Return>
Make additional copies? [Y/N]: N <Return>
Encrypted key: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 2:
(Keyblock LMK)

This example writes components to two smartcards, and encrypts the formed key.

```
Online-AUTH> GS <Return>
Enter LMK id: 01 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key scheme: S <Return>
Enter number of components [2-3]: 2 <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 00 <Return>
Enter exportability: E <Return>
Enter optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 00 <Return>
Enter optional block data: L <Return>
Enter more optional blocks? [Y/N]: N <Return>
Insert card 1 and enter PIN: **** <Return>
Make additional copies? [Y/N]: N <Return>
Insert card 2 and enter PIN: **** <Return>
Make additional copies? [Y/N]: N <Return>
Encrypted key: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Encrypt Clear Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.{key}.console		

Command: **EC**

Function: To encrypt a clear text component and display the result at the console.
If the component does not have odd parity, odd parity will be forced before encryption by the selected LMK.

	Variant LMK	Keyblock LMK
Authorisation:	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key type code of the component being encrypted.	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key usage code of the component being encrypted.
Inputs:	• LMK Identifier: 00-99. • Key Type: See Key Type Table. • Key Scheme: See Key Scheme Table. • Clear Component: 16/32/48 hex digits.	• LMK Identifier: 00-99. • Component Length: 1 (single), 2 (double), 3 (triple). • Key Scheme: See Key Scheme Table. • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Component Number: 1-9. • Exportability: See Exportability Table. • Optional Block data. • Clear Component: 16/32/48 hex digits.
Outputs:	• Component encrypted under an appropriate variant of the selected LMK. • Component check value.	• Keyblock containing the component encrypted under the selected LMK. • Component check value.
Errors:	• <code>Invalid LMK identifier</code> - no LMK loaded or entered identifier out of range. • <code>Data invalid; please re-enter</code> - the input data does not contain 16 or 32 or 48 hexadecimal characters. Re-enter the correct number of hexadecimal characters. • <code>Invalid key type; re-enter</code> - the key type is invalid. See Key Type Table. • <code>Invalid key scheme</code> - an invalid key scheme is entered. See Key Scheme Table. • <code>Command only allowed from Authorised</code> - the HSM is not authorised to perform this operation. • <code>Internal failure 12: function aborted</code> - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department. • <code>Various Keyblock field errors</code> - the value entered is invalid, or incompatible with previously entered values.	

**Example 1:
(Variant LMK)***This example encrypts a plaintext component.*

```

Online-AUTH> EC <Return>
Enter LMK id: 00 <Return>
Enter Key type: 001 <Return>
Enter Key Scheme: U <Return>
Enter component: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted component: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>

```

**Example 2:
(Keyblock LMK)***This example encrypts a plaintext component.*

```

Online-AUTH> EC <Return>
Enter LMK id: 01 <Return>
Enter component length [1,2,3]: 2 <Return>
Enter key scheme: S <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter component number [1-9]: 2 <Return>
Enter exportability: E <Return>
Enter optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 00 <Return>
Enter optional block data: L <Return>
Enter more optional blocks? [Y/N]: N <Return>
Enter component: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted component: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>

```

Form Key from Components

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.{key}.console		

Command: **FK**

Function: To build a key from components. If clear components are used, they will not be checked for parity, but odd parity will be forced on the final key before encryption under the selected LMK.

	Variant LMK	Keyblock LMK
Authorisation:	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key type code of the key being formed.	The HSM must be in the Authorised State, or the activity component.{key}.console must be authorised, where 'key' is the key usage code of the key being formed.
Inputs:	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Type: See Key Type Table. • Key Scheme: See Key Scheme Table. • Component Type: X (xor), H (half), E (encrypted), S (smartcard), T (third). • Number of Components: 1-9. • Clear Components: 16/32/48 hex digits.	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Scheme: See Key Scheme Table. • Component Type: X (xor), H (half), E (encrypted), S (smartcard), T (Third). • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Key Version Number: 00-99. • Exportability: See Exportability Table. • Optional Block data. • Clear Components: 16/32/48 hex digits.
Outputs:	• Key encrypted under an appropriate variant of the selected LMK. • Key Check Value.	• Keyblock containing the component encrypted under the selected LMK. • Key Check Value.
Errors:	• <code>Invalid LMK identifier</code> - no LMK loaded or entered identifier out of range. • <code>Incompatible header values</code> - the field values are incompatible between components. • <code>Incompatible key status optional blocks</code> - there is a mismatch between the values contained in one or more key status optional blocks . • <code>Command only allowed from Authorised</code> - the HSM is not authorised to perform this operation. • <code>Invalid key scheme</code> - an invalid key scheme is entered. See Key Scheme Table. • <code>Invalid key type; re-enter</code> - the key type is invalid. See Key Type Table. • <code>Key all zero</code> - the key is invalid. • <code>Invalid entry</code> - an invalid number of components has been entered. • <code>Data invalid; please re-enter</code> - the amount of input data is incorrect. Re-enter the correct number of hexadecimal characters.	

- Invalid PIN; re-enter - a PIN of less than 4 or greater than 8 is entered.
- Smartcard error; command/return: 0003 - invalid PIN is entered.
- No component card - no key component on the provided smartcard.
- Not a LMK card - card formatted for HSM storage or is a licence card.
- Card not formatted - card is not formatted.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.

Notes: Component type H is not permitted for Triple – DES keys.

Example 1:
(Variant LMK)

This example forms a key from plaintext component.

```
Online-AUTH> FK <Return>
Enter LMK id: 00 <Return>
Enter Key Length[1,2,3]: 2 <Return>
Enter Key type: 002 <Return>
Enter Key Scheme: U <Return>
Component type [X,H,E,S,T]: X <Return>
Enter number of components [1-9]: 2 <Return>
Enter component 1: **** **** **** **** **** **** **** **** <Return>
Enter component 2: **** **** **** **** **** **** **** **** <Return>
Encrypted key: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 2:
(Variant LMK)

This example forms a key from components on a smartcard.

```
Online-AUTH> FK <Return>
Enter LMK id: 00 <Return>
Enter Key Length[1,2,3]: 2 <Return>
Enter Key type: 002 <Return>
Enter Key Scheme: U <Return>
Component type [X,H,E,S,T]: S <Return>
Enter number of components (1-9): 2 <Return>
Insert card 1 and enter PIN: **** <Return>
Insert card 2 and enter PIN: **** <Return>
Encrypted key: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 3:
(Variant LMK)

This example forms a key from encrypted components.

```
Online-AUTH> FK <Return>
Enter LMK id: 00 <Return>
Enter Key Length[1,2,3]: 2 <Return>
Enter Key type: 002 <Return>
Enter Key Scheme: U <Return>
Component type [X,H,E,S,T]: E <Return>
Enter number of components (1-9): 2 <Return>
Enter component 1: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter component 2: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted key: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 4: *This example forms a key from plaintext components.*
(Keyblock LMK)

```
Online-AUTH> FK <Return>
Enter LMK id: 01 <Return>
Enter key length [1,2,3]: 1 <Return>
Enter key scheme: S <Return>
Component type [X,H,E,S,T]: X <Return>
Enter number of components [1-9]: 2 <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 99 <Return>
Enter exportability: E <Return>
Enter optional blocks? [Y/N]: N <Return>
Enter component 1: **** **** *  **** <Return>
Enter component 2: **** **** *  **** <Return>
Encrypted key: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 5: *This example forms a key from components on a smartcard.*
(Keyblock LMK)

```
Online-AUTH> FK <Return>
Enter LMK id: 01 <Return>
Enter Key Length[1,2,3]: 2 <Return>
Enter key scheme: S <Return>
Component type [X,H,E,S,T]: S <Return>
Enter number of components (1-9): 2 <Return>
Insert card 1 and enter PIN: **** <Return>
Insert card 2 and enter PIN: **** <Return>
Encrypted key: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 6: *This example forms a key from encrypted components.*
(Keyblock LMK)

```
Online-AUTH> FK <Return>
Enter LMK id: 01 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key scheme: S <Return>
Component type [X,H,E,S,T]: E <Return>
Enter number of components [1-9]: 3 <Return>
Enter key version number: 00 <Return>
Enter optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 03 <Return>
Enter optional block data: 2005:12:21:00 <Return>
Enter more optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 04 <Return>
Enter optional block data: 2007:12:21:00 <Return>
Enter more optional blocks? [Y/N]: N <Return>
Enter component 1: S XXXXXXXX.....XXXXXX <Return>
Enter component 2: S XXXXXXXX.....XXXXXX <Return>
Enter component 3: S XXXXXXXX.....XXXXXX <Return>
Encrypted key: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Generate Key

		Variant ☒	Keyblock ☒
Online ☒	Offline ☒		Secure ☒
Variant LMK	Authorisation: Determined by KTT(G&E) Activity: generate.{key}.console and export.{key}.console		
Keyblock LMK	Authorisation: If export to non-KB. Activity: export.{key}.console		

Command: **KG**

Function: To generate a random key and return it encrypted under the LMK and optionally under a ZMK (for transmission to another party).

	Variant LMK	Keyblock LMK												
Authorisation:	This command examines the ‘Generate’ flag of the given key type within the Key Type Table to determine the authorisation requirement. If the flag is ‘A’, the HSM must either be in the Authorised State, or the activity generate.{key}.console must be authorised, where 'key' is the key type code of the key being generated. If the generated key is required to be exported under the ZMK, this command also examines the ‘Export’ flag of the given key type within the Key Type Table. If the flag is ‘A’, the HSM must either be in the Authorised State, or the activity export.{key}.console must be authorised, where ‘key’ is the key type code of the key being exported.	The authorisation requirement for this command depends solely on the type of export being requested: <table><tr><th>Exported key scheme</th><th>Authorisation</th></tr><tr><td>No export</td><td>None</td></tr><tr><td>‘S’ (Thales Keyblock)</td><td>None</td></tr><tr><td>‘R’ (ANSI TR-31 Keyblock)</td><td>None</td></tr><tr><td>‘U’, ‘T’ (Variant)</td><td>Required</td></tr><tr><td>‘Z’, ‘X’, ‘Y’ (ANSI X9.17)</td><td>Required</td></tr></table> If authorisation is required, the HSM must either be in the Authorised State, or the activity export.{key}.console must be authorised, where 'key' is the key usage code of the key being exported.	Exported key scheme	Authorisation	No export	None	‘S’ (Thales Keyblock)	None	‘R’ (ANSI TR-31 Keyblock)	None	‘U’, ‘T’ (Variant)	Required	‘Z’, ‘X’, ‘Y’ (ANSI X9.17)	Required
Exported key scheme	Authorisation													
No export	None													
‘S’ (Thales Keyblock)	None													
‘R’ (ANSI TR-31 Keyblock)	None													
‘U’, ‘T’ (Variant)	Required													
‘Z’, ‘X’, ‘Y’ (ANSI X9.17)	Required													
Inputs:	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Type: See Key Type Table. • Key Scheme (LMK): See Key Scheme Table. • Key Scheme (ZMK) (if exporting). • ZMK (if exporting). • Key block values if exporting to TR-31 format	• LMK Identifier: 00-99. • Key Length: 1 (single), 2 (double), 3 (triple). • Key Scheme (LMK): See Key Scheme Table. • Key Scheme (ZMK) (if exporting). • ZMK (if exporting). • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Key Version Number: 00-99. • Exportability: See Exportability Table. • Optional Block data. • Exportability of exported key (if exporting).												
Outputs:	• Key encrypted under an appropriate variant of the selected LMK. • Key/Keyblock encrypted under the ZMK (if exporting). • Key Check Value.	• Keyblock containing the key encrypted under the selected LMK. • Key/Keyblock encrypted under the ZMK (if exporting). • Key Check Value.												

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Must be in Authorised State or Activity not authorised - the key type provided requires the HSM to be in Authorised State. See List Of Authorisable Activities.
- Data invalid; please re-enter - the encrypted ZMK does not contain the correct characters, or the key check value does not contain 6 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- Key parity error; please re-enter - the ZMK does not have odd parity on each byte. Re-enter the encrypted ZMK and check for typographic errors.
- Invalid key scheme for key length - the Key scheme is inappropriate for Key length.
- Invalid key scheme - the key scheme is invalid. See Key Scheme Table.
- Invalid key type; re-enter - the key type is invalid. See Key Type Table.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.

Example 1:
(Variant LMK)

This example generates a new key.

```
Online> KG <Return>
Enter LMK id: 00 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key type: 002 <Return>
Enter key scheme (LMK): U <Return>
Enter key scheme (ZMK): <Return>
Enter ZMK: <Return>
Key under LMK: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key Check value: ZZZZZZ
Online>
```

Example 2:
(Variant LMK)

This example generates a new key, and exports it to X9.17 format.

```
Online-AUTH> KG <Return>
Enter LMK id: 00 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key type: 002 <Return>
Enter key scheme (LMK): U <Return>
Enter key scheme (ZMK): X <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Key under LMK: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key under ZMK: X YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 3:
(Variant LMK)

This example generates a new key, and exports it to TR31 format.

```
Online-AUTH> KG <Return>
Enter LMK id: 00 <Return>
Enter key length [1,2,3]: 3 <Return>
Enter key type: 001 <Return>
Enter key scheme (LMK): U <Return>
Enter key scheme (ZMK): R <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
```



```

Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 44 <Return>
Enter exportability: N <Return>
Enter optional blocks? [Y/N]: N <Return>
Key under LMK: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key under ZMK: R YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>

```

Example 4:
(Keyblock LMK)

This example generates a new key, and exports it to X9.17 format.

```

Online-AUTH> KG <Return>
Enter LMK id: 01 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key scheme (LMK): S <Return>
Enter key scheme (ZMK): X <Return>
Enter ZMK: S XXXXXXXX.....XXXXXX <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 22 <Return>
Enter exportability: N <Return>
Enter optional blocks? [Y/N]: N <Return>
Key under LMK: S YYYYYYYY.....YYYYYY
Key under ZMK: X YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>

```

Example 5:
(Keyblock LMK)

This example generates a new key, and exports it to TR31 format.

```

Online> KG <Return>
Enter LMK id: 01 <Return>
Enter key length [1,2,3]: 2 <Return>
Enter key scheme (LMK): S <Return>
Enter key scheme (ZMK): R <Return>
Enter ZMK: S XXXXXXXX.....XXXXXX <Return>
Enter key usage: 72 <Return>
Enter mode of use: N <Return>
Enter key version number: 33 <Return>
Enter exportability: E <Return>
Enter optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 03 <Return>
Enter optional block data: 2005:12:21:00 <Return>
Enter more optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 04 <Return>
Enter optional block data: 2007:12:21:00 <Return>
Enter more optional blocks? [Y/N]: N <Return>
Enter exportability field for exported key block: <Return>
Key under LMK: S YYYYYYYY.....YYYYYY
Key under ZMK: R YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online>

```

Import Key

		Variant ☒	Keyblock ☒
Online ☒		Offline ☒	Secure ☒
Variant LMK	Authorisation: Determined by KTT(I) Activity: import.{key}.console		
Keyblock LMK	Authorisation: If import from non-KB. Activity: import.{key}.console		

Command: **IK**

Function: To import a key from encryption under a ZMK to encryption under an LMK. If the key imported does not have odd parity a warning will be issued and odd parity will be forced on the key before encryption under the specified LMK.

	Variant LMK	Keyblock LMK										
Authorisation:	This command examines the ‘Import’ flag of the given key type within the Key Type Table to determine whether authorisation is required. If required, the HSM must either be in the Authorised State, or the activity import.{key}.console must be authorised, where 'key' is the key type code of the key being imported.	The authorisation requirement for this command depends on the type of import being requested: <table><tr><th>Imported key scheme</th><th>Authorisation</th></tr><tr><td>‘S’ (<i>Thales Keyblock</i>)</td><td>None</td></tr><tr><td>‘R’ (<i>ANSI TR-31 Keyblock</i>)</td><td>None</td></tr><tr><td>‘U’, ‘T’ (<i>Variant</i>)</td><td>Required</td></tr><tr><td>‘Z’, ‘X’, ‘Y’ (<i>ANSI X9.17</i>)</td><td>Required</td></tr></table> If authorisation is required, the HSM must either be in the Authorised State, or the activity import.{key}.console must be authorised, where 'key' is the key usage code of the key being imported.	Imported key scheme	Authorisation	‘S’ (<i>Thales Keyblock</i>)	None	‘R’ (<i>ANSI TR-31 Keyblock</i>)	None	‘U’, ‘T’ (<i>Variant</i>)	Required	‘Z’, ‘X’, ‘Y’ (<i>ANSI X9.17</i>)	Required
Imported key scheme	Authorisation											
‘S’ (<i>Thales Keyblock</i>)	None											
‘R’ (<i>ANSI TR-31 Keyblock</i>)	None											
‘U’, ‘T’ (<i>Variant</i>)	Required											
‘Z’, ‘X’, ‘Y’ (<i>ANSI X9.17</i>)	Required											
Inputs:	• LMK Identifier: 00-99. • Key Type: See Key Type Table. • Key Scheme (LMK): See Key Scheme Table. • ZMK to be used to decrypt the key. • Key/Keyblock to be imported.	• LMK Identifier: 00-99. • Key Scheme (LMK): See Key Scheme Table. • ZMK to be used to decrypt the key. • Key/Keyblock to be imported. For import from Variant/X9.17: • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Key Version Number: 00-99. • Exportability: See Exportability Table. • Optional Block data. For import from TR-31: • Modified Key Usage • Optional Block data.										
Outputs:	• Key encrypted under an appropriate variant of the selected LMK. • Key Check Value.	• Keyblock containing the key encrypted under the selected LMK. • Key Check Value.										

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Must be in Authorised State or Activity not authorised - the key type provided requires the HSM to be in Authorised State. See List Of Authorisable Activities.
- Data invalid; please re-enter - the encrypted ZMK does not contain the correct characters, or the key check value does not contain 6 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- Key parity error; re-enter key - the parity of the ZMK is not odd.
- Warning: key parity corrected - the parity of the key encrypted under the ZMK is not odd.
- Invalid key scheme - the key scheme is invalid. See Key Scheme Table.
- Invalid key type; re-enter - the key type is invalid. See Key Type Table.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.

Example 1:
(Variant LMK)

This example imports a key from X9.17 format.

```
Online> IK <Return>
Enter LMK id: 00 <Return>
Enter Key type: 002 <Return>
Enter Key Scheme: U <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key: X XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted key: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Example 2:
(Variant LMK)

This example imports a key from TR31 format.

```
Online> IK <Return>
Enter LMK id: 00 <Return>
Enter key type: 009 <Return>
Enter key scheme (LMK): U <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key: R XXXXXXXX.....XXXXXX <Return>
Key under LMK: U YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Example 3:
(Keyblock LMK)

This example imports a key from X9.17 format.

```
Online-AUTH> IK <Return>
Enter LMK id: 01 <Return>
Enter key scheme (LMK): S <Return>
Enter ZMK: S XXXXXXXX.....XXXXXX <Return>
Enter key: X XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 27 <Return>
Enter exportability: N <Return>
Enter optional blocks? [Y/N]: N <Return>
Key under LMK: S YYYYYYYYYY.....YYYYYY
```

**Example 4:
(Keyblock LMK)**

Key check value: ZZZZZZ
Online-AUTH>

This example imports a key from TR31 format. Note that the a new (more restrictive) value for the imported keyblock's Key Usage field is entered during the import process – refer to Appendix K for details.

```
Online> IK <Return>
Enter LMK id: 01 <Return>
Enter key scheme (LMK): S <Return>
Enter ZMK: S XXXXXXXX.....XXXXXX <Return>
Enter key: R XXXXXXXX.....XXXXXX <Return>
Enter modified key usage: 72 <Return>
Enter optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 03 <Return>
Enter optional block data: 2005:12:21:00 <Return>
Enter more optional blocks? [Y/N]: Y <Return>
Enter optional block identifier: 04 <Return>
Enter optional block data: 2007:12:21:00 <Return>
Enter more optional blocks? [Y/N]: N <Return>
Key under LMK: S YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online>
```

Export Key

		Variant ☒	Keyblock ☒
Online ☒		Offline ☒	Secure ☒
Variant LMK	Authorisation: Determined by KTT(E) Activity: export.{key}.console		
Keyblock LMK	Authorisation: If export to non-KB. Activity: export.{key}.console		

Command: **KE**

Function: To translate a key from encryption under the specified LMK to encryption under a ZMK.

	Variant LMK	Keyblock LMK										
Authorisation:	This command examines the ‘Export’ flag of the given key type within the Key Type Table to determine whether authorisation is required. If required, the HSM must either be in the Authorised State, or the activity export.{key}.console must be authorised, where 'key' is the key type code of the key being exported.	The authorisation requirement for this command depends on the type of export being requested: <table><tr><th>Exported key scheme</th><th>Authorisation</th></tr><tr><td>‘S’ (Thales Keyblock)</td><td>None</td></tr><tr><td>‘R’ (ANSI TR-31 Keyblock)</td><td>None</td></tr><tr><td>‘U’, ‘T’ (Variant)</td><td>Required</td></tr><tr><td>‘Z’, ‘X’, ‘Y’ (ANSI X9.17)</td><td>Required</td></tr></table> If authorisation is required, the HSM must either be in the Authorised State, or the activity export.{key}.console must be authorised, where 'key' is the key usage code of the key being exported.	Exported key scheme	Authorisation	‘S’ (Thales Keyblock)	None	‘R’ (ANSI TR-31 Keyblock)	None	‘U’, ‘T’ (Variant)	Required	‘Z’, ‘X’, ‘Y’ (ANSI X9.17)	Required
Exported key scheme	Authorisation											
‘S’ (Thales Keyblock)	None											
‘R’ (ANSI TR-31 Keyblock)	None											
‘U’, ‘T’ (Variant)	Required											
‘Z’, ‘X’, ‘Y’ (ANSI X9.17)	Required											
Inputs:	• LMK Identifier: 00-99. • Key Type: See Key Type Table. • Key Scheme (ZMK): See Key Scheme Table. • ZMK to be used to encrypt the key. • Key to be exported. For export to Thales Keyblock & TR-31: • Key Usage: See Key Usage Table. • Mode of Use: See Mode of Use Table. • Key Version Number: 00-99. • Exportability: See Exportability Table. • Optional Block data. <i>Note export from a Variant LMK to Thales Keyblock is not permitted.</i>	• LMK Identifier: 00-99. • Key Scheme (ZMK): See Key Scheme Table. • ZMK to be used to encrypt the key. • Key to be exported. For export to Thales Keyblock or TR-31: • Exportability of exported key.										
Outputs:	• Key/Keyblock encrypted under the ZMK. • Key Check Value.	• Key/Keyblock encrypted under the ZMK. • Key Check Value.										
Errors:	• Invalid LMK identifier – no LMK loaded or entered identifier out of range.											

- Must be in Authorised State or Activity not authorised - the key type provided requires the HSM to be in Authorised State. See List Of Authorisable Activities.
- Data invalid; please re-enter - the encrypted ZMK or key does not contain 16 or 32 hex or 1 alpha + 32 hex or 1 alpha + 48 hex. Re-enter the correct number of hexadecimal characters.
- Key parity error; re-enter key - the ZMK or key does not have odd parity on each byte. Re-enter the key and check for typographic errors.
- Invalid key scheme - the key scheme is invalid. See Key Scheme Table.
- Invalid key type; re-enter - the key type is invalid. See Key Type Table.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.

Example 1: (Variant LMK)

This example exports a key to X9.17 format.

```
Online-AUTH> KE <Return>
Enter Key type: 002 <Return>
Enter Key Scheme: X <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Key under ZMK: X YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 2: (Variant LMK)

This example exports a key to TR31 format.

```
Online-AUTH> KE <Return>
Enter LMK id: 00 <Return>
Enter key type: 001 <Return>
Enter key scheme (ZMK): R <Return>
Enter ZMK: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter key usage: P0 <Return>
Enter mode of use: N <Return>
Enter key version number: 44 <Return>
Enter exportability: N <Return>
Enter optional blocks? [Y/N]: N <Return>
Key under ZMK: R YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 3: (Keyblock LMK)

This example exports a key to X9.17 format.

```
Online-AUTH> KE <Return>
Enter LMK id: 01 <Return>
Enter key scheme (ZMK): X <Return>
Enter ZMK: S XXXXXXXXXX.....XXXXXX <Return>
Enter key: S XXXXXXXXXX.....XXXXXX <Return>
Key under ZMK: X YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online-AUTH>
```

Example 4: *This example exports a key to TR31 format.*
(Keyblock LMK)

```
Online> KE <Return>
Enter LMK id: 01 <Return>
Enter key scheme (ZMK): R <Return>
Enter ZMK: S XXXXXXXX.....XXXXXX <Return>
Enter key: S XXXXXXXX.....XXXXXX <Return>
Enter exportability field for exported key block: <Return>
Key under ZMK: R YYYYYYYY.....YYYYYY
Key check value: ZZZZZZ
Online>
```

Generate a Check Value

		Variant ☒	Keyblock ☒
Online ☒		Offline ☒	Secure ☒
Variant LMK	Authorisation: Required if ≠ 6 digits Activity: generate.{key}.console		
Keyblock LMK	Authorisation: Not required.		

Command: **CK**

Function: To generate a key check value (KCV) for a key encrypted under a specified LMK.

	Variant LMK	Keyblock LMK
Authorisation:	This command only requires authorisation when calculating either 8 or 16 digit Key Check Values. If required, the HSM must either be in the Authorised State, or the activity generate.{key}.console must be authorised, where 'key' is the key type of the key being used. Regardless of the authorisation requirement, this command examines the 'Generate' flag of the given key type within the Key Type Table to determine whether the check value can be calculated.	The HSM does not require any authorisation to run this command. Note: Key Check Values of Keyblocks are always 6-digits in length.
Inputs:	• LMK Identifier: 00-99. • Key Type: See Key Type Table. • Key Length: 1 (single), 2 (double), 3 (triple). • Key.	• LMK Identifier: 00-99. • Key.
Outputs:	• Key Check Value.	• Key Check Value.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Incompatible LMK schemes - the LMK schemes are different.
- Data invalid; please re-enter - incorrect number of characters.
- Key parity error; re-enter key - the entered key does not have odd parity on each byte. Re-enter the complete line (key and Key-Type code) and check for typographic errors.
- Invalid key type; re-enter - the key type is invalid. See Key Type Table.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with

previously entered values.

Example 1:
(Variant LMK)

This example generates a check value of a key.

```
Online-AUTH> CK <Return>
Enter LMK id: 00 <Return>
Enter key type code: 001 <Return>
Enter key length flag [S/D/T]: D <Return>
Enter encrypted key: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```

Example 2:
(Keyblock LMK)

This example generates a check value of a key.

```
Online> CK <Return>
Enter LMK id: 01 <Return>
Enter key block: S XXXXXXXXXXXXXXXXXXXX.....XXXXXXXXXX <Return>
Key check value: ZZZZZZ
Online>
```

Payment System Commands

The HSM provides the following console commands to support some of the card payment systems host commands.

Command	Page
Generate a Card Verification Value (CV)	123
Generate a VISA PIN Verification Value (PV)	125
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Encrypt Decimalisation Table (ED)	129
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Generate a MAC on an IPB (MI)	131

Generate a Card Verification Value

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: misc.console		

Command: **CV**

Function: To generate a VISA CVV or MasterCard CVC.

Authorisation: The HSM must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when decrypting the supplied CVK(s).
- Encrypted CVK
- Primary account number (PAN) for the card: up to 19 decimal digits.
- Card Expiry date: 4 decimal digits.
- Service code: 3 decimal digits.

Outputs: • Card Verification Value: 3 decimal digits.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Command only allowed from Authorised - the HSM is not authorised to perform this operation.
- Data invalid; please re-enter - possibly incorrect key length. Could also be incorrect PAN, card expiry date, or service code length or non-decimal PAN, card expiry date or service code.
- Key parity error; please re-enter - the parity of the key entered is not odd.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.
- Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.

Example 1:
(Variant LMK)

This example generates a CVV using a CVK pair encrypted in variant format.

```
Online-AUTH> cv <Return>
Enter LMK id: 00 <Return>
Enter key A: XXXX XXXX XXXX XXXX <Return>
Enter key B: XXXX XXXX XXXX XXXX <Return>
Enter PAN: 1234567812345678 <Return>
Enter expiry date: 0694 <Return>
Enter service code: 123 <Return>
CVV: 321
Online-AUTH>
```

Example 2:
(Variant LMK)

This example generates a CVV using a double length CVK in variant format.

```
Online-AUTH> cv <Return>
Enter LMK id: 00 <Return>
Enter key A: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter PAN: 1234567812345678 <Return>
Enter expiry date: 0694 <Return>
```

```
Enter service code: 123 <Return>
CVV: 321
Online-AUTH>
```

Example 3:
(Keyblock LMK)

This example generates a CVV using a CVK in keyblock format.

```
Online-AUTH> cv <Return>
Enter LMK id: 01 <Return>
Enter key block: S xxxxxxxx.....xxxxxx <Return>
Enter PAN: 1234567812345678 <Return>
Enter expiry date: 0694 <Return>
Enter service code: 123 <Return>
CVV: 321
Online-AUTH>
```

Generate a VISA PIN Verification Value

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: misc.console		

Command: **PV**

Function: To generate a VISA PIN Verification Value (PVV).

Authorisation: The HSM must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when decrypting the supplied PVK(s).
- Encrypted PVK.
- The PVV data block comprising:
 - The 11 right-most digits of the account number (excluding check digit): 11 decimal digits.
 - The PIN verification key indicator (PVKI): 1 decimal digit.
 - The 4 left-most digits of the clear PIN: 4 decimal digits.

Outputs: • The PIN Verification Value (PVV): 4 decimal digits.

Errors:

- `Invalid LMK identifier - no LMK loaded or entered identifier out of range.`
- `Command only allowed from Authorised - the HSM is not authorised to perform this operation.`
- `Data invalid; please re-enter - the PVK A, PVK B or the PVV data block field is not 16 characters long. Re-enter the correct number of characters.`
- `Key parity error; please re-enter - the PVK A or PVK B does not have odd parity on each byte. Re-enter the encrypted PVK A or PVK B and check for typographic errors.`
- `Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.`
- `Various Keyblock field errors - the value entered is invalid, or incompatible with previously entered values.`

**Example 1:
(Variant LMK)**

This example generates a PVV using a PVK pair in variant format.

```
Online-AUTH> PV <Return>
Enter LMK id: 00 <Return>
Enter key A: XXXX XXXX XXXX XXXX <Return>
Enter key B: XXXX XXXX XXXX XXXX <Return>
Enter PVV data block: XXXXXXXXXXXX N NNNN <Return>
PVV: NNNN
Online-AUTH>
```

**Example 2:
(Variant LMK)***This example generates a PVV using a double length PVK in variant format.*

```
Online-AUTH> PV <Return>
Enter LMK id: 00 <Return>
Enter key A: U XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter PVV data block: XXXXXXXXXXXX N NNNN <Return>
PVV: NNNN
Online-AUTH>
```

**Example 3:
(Keyblock LMK)***This example generates a PVV using a PVK in keyblock format.*

```
Online-AUTH> PV <Return>
Enter LMK id: 01 <Return>
Enter key block: S XXXXXXXXXXXX.....XXXXXXXX <Return>
Enter PVV data block: XXXXXXXXXXXX N NNNN <Return>
PVV: NNNN
Online-AUTH>
```

Load the Diebold Table

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: misc.console		

Command: **R**

Function: To load the Diebold table into user storage in the HSM.

Authorisation: The HSM must be online and must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when encrypting the supplied values.
- Location in user storage at which to store the Diebold table. This value must be between 0 and 5E0 (hexadecimal). Ensure that the location of the table does not conflict with any other previously-defined storage area: 3 hexadecimal characters.
- Diebold Table: 512 hexadecimal characters (entered as 32 sets of 16 characters).

Outputs:

- The 512-character encrypted table: 16 lines of 32 hexadecimal characters each.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Command only allowed from Online-Authorised - the HSM is not online, or the HSM is not authorised to perform this operation, or both.
- Invalid index - the specified location in user storage is out of range. Enter a valid value.
- Data invalid; please re-enter - the entered index is not 3 hexadecimal characters long, or a table entry is not 16 hexadecimal characters long. Re-enter the correct number of hexadecimal characters.
- Invalid table: duplicate or missing values - some of the data entered is not a valid entry for a Diebold table. Check the table and re-enter the data, checking for typographic errors.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example: Online-AUTH> R <Return>
 Enter LMK id: 00 <Return>
 Enter index (000 - 5FF): xxx <Return>
 Now enter table, 16 hex digits/line
 Line 01: xxxx xxxx xxxx xxxx <Return>
 XXXX XXXX XXXX XXXX OK? [Y/N] y <Return>
 Line 02:
 ...
 ...
 Line 32: xxxx xxxx xxxx xxxx <Return>
 XXXX XXXX XXXX XXXX OK? [Y/N] y <Return>

 XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX
 XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX
 ...
 ...
 XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX
 XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX (16 lines of encrypted table are displayed)
 Online-AUTH>

Note: The result of the “R” command gives no indication as to the LMK scheme or LMK identifier used in the command. When this value is used with other (host) commands, the user must ensure that the correct LMK is specified in the command.

Encrypt Decimalisation Table

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: misc.console		

Command: **ED**

Function: To encrypt a 16 digit decimalisation table for use with host commands using IBM 3624 PIN Generation & Verification.

Authorisation: The HSM must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when encrypting the decimalisation table.
- Decimalisation table. 16 decimal digits that specify the mapping between hexadecimal & decimal numbers.
- The HSM by default checks that the decimalisation table contains at least 8 different digits, with no digit repeated more than 4 times. This feature may be disabled using the Configure Security parameter “Enable decimalisation table check”. Disabling of this feature is not recommended.

Outputs: • Encrypted decimalisation table: 16 Hex characters.

Errors:

- `Invalid LMK identifier` - no LMK loaded or entered identifier out of range.
- `Not Authorised` - the HSM is not authorised to perform this operation.
- `Decimalisation table invalid` - the decimalisation table is not all decimal or does not contain at least 8 different digits with no digit repeated more than 4 times.
- `Master Key Parity Error` - the contents of the HSM storage have been corrupted or erased. Do not continue. Inform the security department.

Example 1: *This example encrypts a decimalisation table.*

(Variant or
Keyblock LMK)

```
Online-AUTH> ED <Return>
Enter LMK id: 00 <Return>
Enter decimalisation table: 0123456789012345 <Return>
Encrypted decimalisation table: XXXX XXXX XXXX XXXX
Online-AUTH>
```

Note: The result of the “ED” command gives no indication as to the LMK scheme or LMK identifier used in the command. When this value is used with other (host) commands, the user must ensure that the correct LMK is specified in the command.

Translate Decimalisation Table

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required		
Activity: misc.console		

Command: **TD**

Function: To translate an encrypted decimalisation table from Encryption under an old LMK to encryption under the corresponding new LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when translating the decimalisation table.
- Encrypted Decimalisation table. This is the result of encrypting a decimalisation table using the ED command.
- The HSM by default checks that the decimalisation table contains at least 8 different digits, with no digit repeated more than 4 times. This feature may be disabled using the Configure Security parameter “Enable decimalisation table check”. Disabling of this feature is not recommended.

Outputs:

- Encrypted decimalisation table (encrypted under the new LMK): 16 Hex characters.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Not Authorised - the HSM is not authorised to perform this operation.
- Decimalisation Table Invalid - decimalisation table not all decimal or does not contain at least 8 different digits with no digit repeated more than 4 times.
- Master Key Parity Error - the contents of the HSM storage have been corrupted or erased. Do not continue. Inform the security department.
- No LMK in Key Change Storage - Key Change storage is empty.

Example 1:
(Variant or
Keyblock LMK)

```
Online-AUTH> TD <Return>
Enter LMK id: 00 <Return>
Enter decimalisation table encrypted under old LMK : XXXXXXXXXXXXXXXXXX <Return>
Decimalisation table encrypted under new LMK      : YYYYYYYYYYYYYYYYYY
Online-AUTH>
```

Note: The result of the “TD” command gives no indication as to the LMK scheme or LMK identifier used in the command. When this value is used with other (host) commands, the user must ensure that the correct LMK is specified in the command.

Generate a MAC on an IPB

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: misc.console		

Command: **MI**

Function: To generate a MAC on the Cryptogram component of a CAP IPB.

Authorisation: The HSM must be either in the Authorised State, or the activity **misc.console** must be authorised, using the Authorising Officer cards of the relevant LMK.

Inputs:

- LMK identifier: indicates the LMK to use when generating the MAC.
- 8 byte IPB represented as 16 hex ASCII characters.

Outputs:

- 4 byte MAC over the plaintext IPB input data.

Errors:

- Invalid LMK identifier - no LMK loaded or entered identifier out of range.
- Command only allowed from Authorised - the HSM is not authorised to perform this operation.
- IPB is not 8 bytes. Please re-enter - the validation of the IPB failed.
- Warning: Less than 16 '1' bits in IPB - the IPB contains less than 16 '1' bits.

Example:

```
Online-AUTH> MI <Return>
Enter LMK id: 00 <Return>
Enter IPB: FFFFFFFF00000000 <Return>
MAC: FB1A 3C1A
Online-AUTH>
```

Note: The result of the "MI" command gives no indication as to the LMK scheme or LMK identifier used in the command. When this value is used with other (host) commands, the user must ensure that the correct LMK is specified in the command.

Smartcard Commands

The HSM provides the following console commands to support the smartcard operations:

Command	Page
Format a Smartcard (FC)	133
Create an Authorising Officer Smartcard (CO)	135
Verify the Contents of a Smartcard (VC)	136
Change a Smartcard PIN (NP)	137
Read Unidentifiable Smartcard Details (RC)	138
Eject a Smartcard (EJECT)	139

NOTE: DO NOT REPEATEDLY ENTER INVALID PINS. A CARD "LOCKS" AFTER SEVEN INVALID PINS HAVE BEEN ENTERED, AND IT CAN BE "UNLOCKED" ONLY BY REFORMATTING, WHICH DELETES THE ENTIRE CONTENTS OF THE CARD.

Format a Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **FC**

Function: To format a smartcard for use by the HSM.
Different formats are used for LMK storage and saving HSM settings.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- (LMK cards): Smartcard PIN: 4 to 8 alphanumeric characters.
- Date: 6 numeric character format DDMMYY.
- Time: 6 numeric characters; format hhmmss.
- Issuer ID: maximum 35 alphanumeric characters.
- User ID: maximum 35 alphanumeric characters.

Outputs:

- Text messages:
 - Insert card and press ENTER.
 - Format card for HSM settings/LMKs? [H/L]
 - Enter new PIN for smartcard.
 - Re-enter new PIN.
 - Enter format code.
 - Enter date.
 - Enter time.
 - Enter Issuer ID.
 - Enter User ID.
 - Format complete.
 - Card already formatted, continue? [Y/N].

Errors:

- Invalid PIN; re-enter - the PIN entered is fewer than 4 or greater than 8.
- PINs did not agree - the new PINs entered for the card did not match each other.
- Invalid input. Entry must be in numeric format - non numeric value is entered for time or date.

Example 1:

```
Online> FC <Return>
Insert card and press ENTER: <Return>
Card already formatted, continue? [Y/N]: Y <Return>
Format card for HSM settings/LMKs? [H/L]: L <Return>
Erasing card
Formatting card . . .
Enter new PIN for Smartcard: ***** <Return>
Re-enter new PIN: ***** <Return>
Enter time [hhmmss]: 153540 <Return>
Enter date [ddmmyy]: 261093 <Return>
Enter User ID: Joe Small <Return>
Enter Issuer ID: Big Bank plc <Return>
Format complete
Online>
```

Example 2: Online> FC <Return>
 Insert card and press ENTER: <Return>
 Card already formatted, continue? [Y/N]: Y <Return>
 Format card for HSM settings/LMKs? [H/L]: H <Return>
 Erasing card
 Formatting card . . .
 Format complete
 Online>

Create an Authorising Officer Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **CO**

Function: To copy the Password for an Authorising Officer to another smartcard so that it can be used to set the HSM into the Authorised State.

Authorisation: The HSM must be in the Offline or Secure state to run this command.

Inputs: • Smartcard PIN: 4 to 8 alphanumeric characters.

Outputs: • Text messages:

- Insert Card for Component Set 1 or 2 and enter the PIN.
- Insert Card for Authorising Officer and enter the PIN.
- Copy Complete.

Errors:

- Card not formatted - card not formatted
- Not a LMK card - card formatted for HSM settings or licence card.
- Smartcard error; command/return: 0003 - an invalid PIN was entered.
- Invalid PIN; re-enter - PIN is fewer than 4 or greater than 8 digits.
- Card not blank - copy failed.

Example:

```
Offline> co <Return>
Insert Card for Component Set 1 or 2 and enter PIN: **** <Return>
Insert Card for Authorising Officer and enter PIN: **** <Return>
Copy complete.
Offline>
```

Verify the Contents of a Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **VC**

Function: To verify the contents of the smartcards held by a Component Holder. The HSM reads the LMK Component Set from the smartcard, computes the check value, compares this with the check value stored on the card and displays the result.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- Smartcard PIN: 4 to 8 alphanumeric characters.

Outputs:

- Component Set check value:
 - For Variant LMKs, the length of the displayed check value is determined by the CS (Configure Security) setting “Restrict Key Check Value to 6 hex chars”.
 - For Keyblock LMKs, the length of the displayed check value is always 6 hex digits.
- Comparison: Pass or Fail.
- Text messages:
 - Check:
 - Compare with card:

Errors:

- Card not formatted - card not formatted
- Not a LMK card - card formatted for HSM settings or licence card.
- Smartcard error; command/return: 0003 - an invalid PIN was entered.
- Invalid PIN; re-enter - PIN is fewer than 4 or greater than 8 digits.

Example:

```
Online> vc <Return>
Insert card and enter PIN: **** <Return>
Check: 012345.
Compare with card: Pass.
Online>
```

If a smartcard is defective or cannot be successfully verified, replace it. Copy a verified smartcard (from the same set of components) onto a replacement.

NOTE: DISPOSE OF THE FAULTY SMARTCARD IN A SECURE MANNER.

Change a Smartcard PIN

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **NP**

Function: To select a new PIN for a smartcard without changing any of the other details stored on the card.
The old PIN must be submitted before a change is effected and the new PIN must be supplied correctly at two consecutive prompts.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- Smartcard PIN: 4 to 8 alphanumeric characters

Outputs:

- Text messages:
 - Insert Card and press ENTER.
 - Enter current PIN.
 - Enter new PIN for smartcard.
 - Re-enter new PIN.
 - PIN change completed.

Errors:

- Card not formatted - card not formatted
- Not a LMK card - card formatted for HSM settings or licence card.
- Smartcard error; command/return: 0003 - an invalid PIN was entered.
- Invalid PIN; re-enter - PIN is fewer than 4 or greater than 8 digits.
- PINs did not agree - the new PINs entered for the smartcard did not match.

Example:

```
Online> NP <Return>
Insert Card and press ENTER: <Return>
Enter current PIN: **** <Return>
Enter new PIN for smartcard: **** <Return>
Re-enter new PIN: **** <Return>
PINs did not agree
Enter new PIN for smartcard: **** <Return>
Re-enter new PIN: **** <Return>
PIN change completed
Online>
```

Read Unidentifiable Smartcard Details

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RC**

Function: To read otherwise unidentifiable cards.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- Text messages:
 - Insert Card and press ENTER when ready.
 - This card is formatted for saving and retrieving HSM settings.
 - Version, as stored on card: decimal integer.
 - Date, as stored on card; format: YY/MM/DD.
 - Time, as stored on card; format: hh:mm:ss.
 - User ID, as stored on card; free format alphanumeric.
 - Issuer ID, as stored on card; free format alphanumeric.
 - Data Zone Size, as stored on card: decimal integer.
 - Max Data Free, as stored on card: decimal integer.

Errors:

- Card not formatted – card not formatted
- Not a LMK card – card formatted for HSM settings or licence card.

Example 1:

```
Online> RC <Return>
Format version: 0001
Issue time: 11:53:00
Issue date: 93/10/25
User ID: Bill Weasel
Issuer ID: Big Bank plc
User-data zone size: 0000
Free: 0392
Online>
```

Example 2:

```
Online> RC <Return>
This card is formatted for saving and retrieving HSM settings.
Online>
```

Eject a Smartcard

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **EJECT**

Function: To eject the smartcard from the smartcard reader.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs: None.

Errors: None.

Example: Online> **EJECT** <Return>
Online>

DES Calculator Commands

The HSM provides the following console commands to support the encryption and decryption of data with a given plaintext single, double or triple-length DES key:

Command	Page
Single-Length Key Calculator (N)	141
Double-Length Key Calculator (\$)	142
Triple-Length Key Calculator (T)	143

Single-Length Key Calculator

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **N**

Function: To encrypt and decrypt the given data block with the given single-length key.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- Key (no parity required): 16 hexadecimal characters.
- Data block: 16 hexadecimal characters.

Outputs:

- The data encrypted with the key.
- The data decrypted with the key.

Errors:

- Data invalid; please re-enter - the entered data does not comprise 16 hexadecimal characters. Re-enter the correct number of hexadecimal characters.

Example:

```
Online> N <Return>
Enter key: XXXX XXXX XXXX XXXX <Return>
Enter data: XXXX XXXX XXXX XXXX <Return>
Encrypted: YYYY YYYY YYYY YYYY
Decrypted: YYYY YYYY YYYY YYYY
Online>
```

Double-Length Key Calculator

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **\$**

Function: To encrypt and decrypt the given data block with the given double-length key.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- The double-length key (odd parity is required): 32 hexadecimal characters.
- Data block: 16 hexadecimal characters.

Outputs:

- The data encrypted with the key.
- The data decrypted with the key.

Errors:

- Data invalid; please re-enter - the entered data does not comprise 32 hexadecimal characters. Re-enter the correct number of hexadecimal characters.

Example:


```
Offline> $ <Return>
Enter key: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter data: XXXX XXXX XXXX XXXX <Return>
Encrypted: YYYY YYYY YYYY YYYY
Decrypted: YYYY YYYY YYYY YYYY
Offline>
```

Triple-Length Key Calculator

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **T**

Function: To encrypt and decrypt the given data block with the given triple-length key.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- The triple-length key (odd parity is required): 48 hexadecimal characters.
- Data block: 16 hexadecimal characters.

Outputs:

- The data encrypted with the key.
- The data decrypted with the key.

Errors:

- Data invalid; please re-enter - Re-enter the correct number of hexadecimal characters.

Example:

```
Offline> T <Return>
Enter key: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX
<Return>
Single, Double, or Triple length data? (S,D,T): S <Return>
Enter data: XXXX XXXX XXXX XXXX <Return>
Encrypted: YYYY YYYY YYYY YYYY
Decrypted: YYYY YYYY YYYY YYYY
Offline>
```

Legacy Commands

The following console commands are redundant, but are retained for backwards compatibility. They have been superseded by newer (usually more generic) commands – refer to the individual commands for details.

Note: The following commands always use the default LMK, which must be a variant LMK.

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Generate a ZMK and Write to Smartcards (GZ)	146
Encrypt a Clear Zone Master Key Component (Z)	147
Form a ZMK From Encrypted Components (D)	148
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Generate a Zone Master Key Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **F** (superseded by GC)

Function: To generate a ZMK component and display it in plain and encrypted forms.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- Clear text ZMK component: 16 or 32 hexadecimal characters.
- ZMK component encrypted under a variant of LMK pair 04-05: 16 or 32 hexadecimal characters.
- Component check value; formed by encrypting 64 binary zeros with the component and returning the left-most 24 bits: 6 hexadecimal characters.

Errors:

- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Notes:

- The F, Z and D console commands create and manipulate ZMK components encrypted under LMK 04-05 variant I. This is to maintain backward compatibility with previous releases of firmware, in which ZMK components were the only types of components supported.
- The recommended method of using ZMK components is to use the GC, EC and FK console commands, which process components of any key type.
- The length of the generated ZMK component will be dictated by the console Configure Security setting "ZMK Length: Single/Double".

Example:

```
Online> F <Return>
Clear ZMK Component: XXXX XXXX XXXX XXXX
Encrypted ZMK Component: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online>
```

Generate a ZMK and Write to Smartcards

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: generate.000.console		

Command: **GZ** (superseded by GS)

Function: To generate a ZMK in 2 to 9 component and write the components to Smartcards.
 Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **generate.000.console** must be authorised.

Inputs: • Number of components, 1 numeric digit.

Outputs: • ZMK encrypted under LMK pair 04-05: 16 or 32 hexadecimal characters.
 • ZMK Check value; formed by encrypting 64 binary zeros with the ZMK; 16 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors: • Command only allowed from Authorised – the HSM is not authorised to perform this operation.
 • Invalid PIN; re-enter – the entered PIN is not 4 – 8 digits.
 • Smartcard error; command/return: 0003 – invalid PIN is entered.
 • Not a LMK card – card formatted for HSM storage or is a licence card.
 • Card not formatted – card is not formatted.
 • Warning – card not blank. Proceed? [Y/N] – the smartcard entered is not blank.
 • Overwrite ZMK component? [Y/N] – a ZMK component already exists on the card.
 • Invalid entry – invalid number of components entered.
 • Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Notes: • The length of the generated ZMK component will be dictated by the console Configure Security setting "ZMK Length: Single/Double".

Example: Online-AUTH> **GZ** <Return>
 Enter number of components [2-3]: **2** <Return>
 Insert card 1 and enter PIN: ******** <Return>
 Make additional copies? [Y/N]: **N** <Return>
 Insert card 2 and enter PIN: ******** <Return>
 Make additional copies? [Y/N] **N** <Return>
 Encrypted ZMK: YYYY YYYY YYYY YYYY
 Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
 Online-AUTH>

Encrypt a Clear Zone Master Key Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.000.console		

Command: **Z** (superseded by EC)

Function: To encrypt a clear text component and display the result at the console.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **generate.000.console** must be authorised.

Inputs: • Clear text ZMK component: 16 or 32 hexadecimal characters.

Outputs: • The ZMK component encrypted under a variant of LMK pair 04-05: 16 or 32 hexadecimal characters.
• Component check value; formed by encrypting 64 binary zeros with the component and returning the left-most 24 bits: 6 hexadecimal characters.

Errors: • Command only allowed from Authorised – the HSM is not authorised to perform this operation.
• Data invalid; please re-enter – the input data does not contain 16 or 32 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
• Component parity error; re-enter component – the entered component does not have odd parity on each byte. Ensure the component has odd parity and re-enter.
• Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Notes: • The F, Z and D console commands create and manipulate ZMK components encrypted under LMK 04-05 variant 1. This is to maintain backward compatibility with previous releases of firmware, in which ZMK components were the only types of components supported.
• The recommended method of using ZMK components is to use the GC, EC and FK console commands, which process components of any key type.
• The length of the generated ZMK component will be dictated by the console Configure Security setting "ZMK Length: Single/Double".

Example: Online-AUTH> **Z** <Return>
Enter ZMK Component: ********* <Return>
Encrypted ZMK Component: **YYYY YYYY YYYY YYYY**
Key check value: **ZZZZ ZZZZ ZZZZ ZZZZ**
Online-AUTH>

Form a ZMK From Encrypted Components

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.000.console		

Command: **D** (superseded by FK)

Function: To form a ZMK from encrypted components. The components may either be entered from the console or read from Smartcards.
The manually entered components must have been encrypted using the Z command, or generated using the F command.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **generate.000.console** must be authorised.

Inputs:

- Type of input, smartcard or keyboard.
- The number of key components to be entered: 2 to 9.
- The ZMK components, each encrypted under a variant of LMK pair 04-05: 16 or 32 hexadecimal characters.

Outputs:

- The ZMK encrypted under LMK 04-05: 16 or 32 hexadecimal characters.
- The key check value, formed by encrypting 64 binary zeros with the ZMK, and returning all 64 bits: 16 or 32 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- `Command only allowed from Authorised` - The HSM is not authorised to perform this operation.
- `Invalid entry` - invalid number of components entered.
- `Data invalid; please re-enter` - the input data does not contain 16 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- `Component parity error; re-enter component` - the entered component does not have odd parity on each byte. Re-enter the encrypted component and check for typographic errors.
- `Invalid PIN; re-enter` - the entered PIN is not 4 to 8 digits or the pin does not match the PIN of the card.
- `Card checksum mismatch` - the components on the cards do not match.
- `Smartcard error; command/return: 0003` - invalid PIN is entered.
- `Not a LMK card` - card formatted for HSM settings or is a licence card.
- `Card not formatted` - card is not formatted.
- `No component card` - there are no ZMK components on the card.
- `Internal failure 12: function aborted` - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Notes:

- The F, Z and D console commands create and manipulate ZMK components encrypted under LMK 04-05 variant 1. This is to maintain backward compatibility with previous releases of firmware, in which ZMK components were the only types of components supported.
- The recommended method of using ZMK components is to use the GC, EC and FK console commands, which process components of any key type.
- The length of the generated ZMK component will be dictated by the console Configure Security setting "ZMK Length: Single/Double".

Example 1:

This example forms a ZMK from plaintext components.

```
Online-AUTH> D <Return>
Input components from smartcards? [Y/N]: N <Return>
Enter number of components (2-9): 2 <Return>
Enter encrypted component 1: ***** <Return>
Enter encrypted component 2: ***** <Return>
Encrypted ZMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```

Example 2:

This example forms a ZMK from components on smartcards.

```
Online-AUTH> D <Return>
Input components from smartcards? [Y/N]: Y <Return>
Enter number of components (2-9): 2 <Return>
Insert card 1 and enter PIN: **** <Return>
Insert card 2 and enter PIN: **** <Return>
Encrypted ZMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```

Form a Key from Components

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.{key}.console		

Command: **BK** (superseded by FK)

Function: To build a key from clear components. The components are not checked for parity, but odd parity is forced on the final key before encryption under the LMK.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **component.{key}.console** must be authorised, where 'key' is the key type code of the key being formed.

Inputs:

- Key Type; 1 numeric digit:
 "0" - Base Derivation Key (BDK)
 "1" - Card Verification Key (CVK)
 "2" - Zone PIN Key (ZPK)
- The number of key components to be entered: 2 to 9.
- The clear key component. Each BDK component must contain 32 hexadecimal characters and each CVK or ZPK component must contain 16 hexadecimal characters.

Outputs:

- The key formed by exclusive-ORing the entered components, forcing odd parity and encrypting under the appropriate LMK pair:
 - Key type "0" - LMK pair 28 - 29, 32 hexadecimal digits.
 - Key type "1" - LMK pair 14 - 15 variant 4, 16 hexadecimal digits.
 - Key type "2" - LMK pair 06 - 07, 16 hexadecimal digits.
- The key check value, formed by encrypting a block of zeros with the key, and returning all 64 bits: 16 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- Command only allowed from Authorised - the HSM is not authorised to perform this operation.
- Invalid entry - invalid number of components has been entered.
- Data invalid; please re-enter - the amount of input data is incorrect or non-hexadecimal characters have been entered. Re-enter the correct number of hexadecimal characters.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example 1: *This example forms a BDK from components.*

```
Online-AUTH> BK <Return>
Enter key type [0=BDK, 1=CVK, 2=ZPK]: 0 <Return>
Enter number of components (2-9): 2 <Return>
Enter component 1: ***** <Return>
Enter component 2: ***** <Return>
```

```
Encrypted key: YYYY YYYY YYYY YYYY YYYY YYYY YYYY  
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ  
Online-AUTH>
```

Example 2: *This example forms a CVK from components.*

```
Online-AUTH> BK <Return>  
Enter key type [0=BDK, 1=CVK, 2=ZPK]: 1 <Return>  
Enter number of components (2-9): 3 <Return>  
Enter component 1: ***** <Return>  
Enter component 2: ***** <Return>  
Enter component 3: ***** <Return>
```

```
Encrypted key: YYYY YYYY YYYY YYYY  
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ  
Online-AUTH>
```

Example 3: *This example forms a ZPK from components.*

```
Online-AUTH> BK <Return>  
Enter key type [0=BDK, 1=CVK, 2=ZPK]: 2 <Return>  
Enter number of components (2-9): 2 <Return>  
Enter component 1: ***** <Return>  
Enter component 2: ***** <Return>
```

```
Encrypted key: YYYY YYYY YYYY YYYY  
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ  
Online-AUTH>
```

Encrypt a Key Under LMK Pair 14-15

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.002.console		

Command: **K** (superseded by FK)

Function: To form and encrypt a TMK, TPK or PVK under LMK 14-15. The TMK, TPK or PVK can be entered as a number of components in the range 1 to 9 inclusive. To ensure that the clear key is not displayed on the screen, enter the “^” character before entering the key component. The entered components need not have odd parity, although the final TMK, TPK or PVK has odd parity.
 Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **generate.002.console** must be authorised.

Inputs:

- The number of key components to be entered: 1 to 9.
- The clear key component. Each key component must contain 16 hexadecimal characters.

Outputs:

- The TMK, TPK or PVK encrypted under LMK 14-15: 16 hexadecimal characters.
- The key check value, formed by encrypting a block of zeros with the TMK, TPK or PVK, and returning the left-most 24 bits: 6 hexadecimal characters.

Errors:

- Command only allowed from Authorised – the HSM is not authorised to perform this operation.
- Invalid entry – the number of components is not in the range 1 to 9. Re-enter the correct number of components.
- Data invalid; please re-enter – the entered key component does not contain 16 hexadecimal characters. Re-enter the key component.
- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online-AUTH> K <Return>
Enter number of components (1-9): 2 <Return>
Enter component 1: ***** <Return>
Enter component 2: ***** <Return>
Encrypted key: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```


Import a CVK or PVK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: import.{key}.console		

Command: **IV** (superseded by IK)

Function: To import VISA PVK or CVK from encryption under ZMK to encryption under LMK.
 Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **import.{key}.console** must be authorised, where 'key' is either '402' (CVK) or '002' (PVK).

Inputs:

- ZMK encrypted under LMK pair 04-05: 16 or 32 hexadecimal characters.
- Key type: C or P (for CVK or PVK respectively).
- Key A and B encrypted under the ZMK: 16 hexadecimal characters.
- ZMK variant: 1 or 2 digit, value 0-99 (or <Enter> to ignore). Used only when interworking with Atalla systems. Refer to the CS command. Note that this input is not requested when the ZMK variant support is set to off.

Outputs:

- Key A and B encrypted under LMK 14-15 or variant: 16 hexadecimal characters.
- Key check value: 16 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- Command only allowed from Authorised - the HSM is not authorised to perform this operation.
- Data invalid; please re-enter - incorrect input data length or invalid ZMK variant.
- Key parity error; re-enter - the ZMK or key entered does not have odd parity.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online-AUTH> IV <Return>
Key type [Pvk/Cvk]: C <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: X <Return>, if enabled by CS command)
Enter key A: XXXXXXXXXXXXXXXXXXXX <Return>
Enter key B: XXXXXXXXXXXXXXXXXXXX <Return>
Key A under LMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Key B under LMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```

Generate a Zone PIN Key

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **B** (superseded by KG)

Function: To generate a random ZPK and return it encrypted under the LMK and under a ZMK (for transmission to another party). The ZPK can be a VISA Acquirer or Issuer Working key.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- The ZMK (VISA Zone Control Master Key, ZCMK) encrypted under LMK pair 04-05 (as generated using the D command): 16 or 32 hexadecimal characters.
- The ZMK key check value (as generated using the D command or by extracting the first 6 digits generated using the CK command): 6 hexadecimal characters.
- The ZMK variant: 1 or 2 digit, value 0-99 (or <Enter> to ignore). Used only when interworking with Atalla systems. Refer to the CS command. Note that this input is not requested when the ZMK variant support is set to off.

Outputs:

- The ZPK encrypted under the ZMK: 16 hexadecimal characters.
- The ZPK encrypted under LMK pair 06-07: 16 hexadecimal characters.
- The ZPK check value, formed by encrypting 64 binary zeros with the ZPK and returning the left-most 48 bits: 12 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- Data invalid; please re-enter - the encrypted ZMK does not contain 16 or 32 hexadecimal characters, or the key check value is not 6 characters or the ZMK variant is invalid. Re-enter the correct number of hexadecimal characters.
- Key parity error; re-enter - the ZMK does not have odd parity on each byte. Re-enter the encrypted ZMK and check for typographic errors.
- Check failed, re-enter check value or abort - invalid 6 character check value has been entered.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> B <Return>
Enter encrypted ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter ZMK check value: XXXXXX <Return>
(Enter ZMK variant: X <Return>, if enabled by CS command)
ZPK encrypted for transmission: YYYY YYYY YYYY YYYY
ZPK encrypted for bank: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online>
```

Translate a Zone PIN Key

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: export.001.console		

Command: **WK** (superseded by KE)

Function: To translate a ZPK from encryption under the LMK to encryption under a ZMK.
 Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **export.001.console** must be authorised.

Inputs:

- ZMK encrypted under LMK pair 04-05: 16 or 32 hexadecimal characters.
- The ZPK encrypted under LMK pair 06-07: 16 hexadecimal characters.
- The ZMK variant: 1 or 2 digit, value 0-99 (or <Enter> to ignore). Used only when interworking with Atalla systems. Refer to the CS command. Note that this input is not requested when the ZMK variant support is set to Off.

Outputs:

- The ZPK encrypted under the ZMK: 16 hexadecimal characters.
- The key check value for the ZPK; generated by encrypting 64 binary zeros with the key: 16 hexadecimal characters, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- Command only allowed from Authorised – the HSM is not authorised to perform this operation.
- Data invalid; please re-enter – the encrypted ZMK does not contain 16 or 32 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- Key parity error; re-enter key – the ZMK does not have odd parity on each byte. Re-enter the key and check for typographic errors.
- Key parity error – the ZPK does not have odd parity on each byte. Re-enter the key and check for typographic errors.
- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online-AUTH> WK <Return>
Enter encrypted ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: X <Return>, if enabled by CS command)
Enter encrypted ZPK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
ZPK encrypted under ZMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ ZZZZ ZZZZ
Online-AUTH>
```

Generate a CVK Pair

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **KA** (superseded by KG)

Function: To generate a CVK pair and output the key encrypted under a variant of LMK pair 14-15.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- CVK A encrypted under a variant of LMK pair 14-15: 16 hexadecimal characters.
- The key check value for CVK A; formed by encrypting 64 binary zeros with the key and returning the left-most 24 bits: 6 hexadecimal characters.
- CVK B encrypted under a variant of LMK pair 14-15: 16 hexadecimal characters.
- The key check value for CVK B; formed by encrypting 64 binary zeros with the key and returning the left-most 24 bits: 6 hexadecimal characters.

Errors:

- **Internal failure 12: function aborted** – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> KA <Return>
Encrypted CVK A: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Encrypted CVK B: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Translate a CVK Pair from LMK to ZMK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **KB** (superseded by KE)

Function: To translate a CVK pair from encryption under a variant of LMK pair 14-15 to encryption under a ZMK.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- CVK A encrypted under a variant of LMK pair 14-15: 16 hexadecimal characters.
- CVK B encrypted under a variant of LMK pair 14-15: 16 hexadecimal characters.
- ZMK encrypted under LMK pair 04-05: 16 or 32 hexadecimal characters.
- The ZMK variant: 1 or 2 digit, value 0-99 (or <Enter> to ignore). Used only when interworking with Atalla systems. Refer to the CS command. Note that this input is not requested when the ZMK variant support is set to off.

Outputs:

- CVK A encrypted under the ZMK.
- The key check value for CVK A, formed by encrypting 64 binary zeros with the key and returning the left-most 24 bits: 6 hexadecimal characters.
- CVK B encrypted under the ZMK.
- The key check value for CVK B, formed by encrypting 64 binary zeros with the key and returning the left-most 24 bits: 6 hexadecimal characters.

Errors:

- `Data invalid; please re-enter` – the encrypted key does not contain the correct number of hexadecimal characters or an invalid ZMK variant was entered.
- `Key parity error` – the key does not have odd parity on each byte. Re-enter the key and check for typographic errors.
- `Internal failure 12: function aborted` – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> KB <Return>
Enter encrypted CVK A: XXXX XXXX XXXX XXXX <Return>
Enter encrypted CVK B: XXXX XXXX XXXX XXXX <Return>
Enter encrypted ZMK: XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: X <Return>, if enabled by CS command)
Encrypted CVK A: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Encrypted CVK B: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Generate a Double-Length ZMK Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DD** (superseded by GC)

Function: To generate a double-length random ZMK component and display the value at the console screen.
The command ignores the S/D (single/double length) parameter set by the CS (Configure Security) command.
Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs: None.

Outputs:

- The clear ZMK component.

Errors: None.

Example:

```
Online> DD <Return>
Clear ZMK component: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Online>
```

Form a ZMK from Clear Components

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Required Activity: component.000.console		

Command: **DE** (superseded by FK)

Function: To enter a ZMK as either two single-length components (halves) or as two to nine double-length components.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM must be either in the Authorised State, or the activity **generate.000.console** must be authorised.

Inputs:

- A half-length or full-length flag.
- The number of components.
- The clear components: each 16 or 32 hexadecimal characters.

Outputs:

- The ZMK encrypted under LMK pair 04-05.
- The key check value (KCV) for the ZMK, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- `Command only allowed from Authorised` – the HSM must be in Authorised State.
- `Data invalid; please re-enter` – the input data does not contain 16 or 32 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- `Internal failure 12: function aborted` – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Notes:

- The DE command differs from the D command as follows:
 - It uses clear components (not encrypted components).
 - It forms the ZMK from two 16-character halves, or from two to nine 32-character components.
- When H/F is set to H, two 16-character halves are used: the user is prompted to enter 16 left characters, then 16 right characters. (The unit concatenates the left and right halves).
- When H/F is set to F, two to nine 32-character components are used: the user is prompted to enter the first component, then the second component, then the third, etc., according to the number of components to be entered. (The unit exclusive-OR combines the 32-character components).
- The parity of the components is not checked, but the resulting ZMK has odd parity forced before encryption.
- If the Echo parameter entered in the CS (Configure Security) command has been set to N (on), the clear components are echoed onto the screen as they are entered. If

this is not required, either:

- Use the CS command to set the Echo parameter to F (off);
- or
- Enter ^ (i.e. press the Shift and 6 keys) before entering each component.

Example 1:

```
Online-AUTH> DE <Return>
Half or full-length components? (H/F): H <Return>
Enter clear left half: XXXX XXXX XXXX XXXX <Return>
Enter clear right half: XXXX XXXX XXXX XXXX <Return>
Encrypted ZMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ
Online-AUTH>
```

Example 2:

```
Online-AUTH> DE <Return>
Half or full-length components? (H/F): F <Return>
Enter number of clear components (2-9): 3 <Return>
Enter component 1: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter component 2: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter component 3: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted ZMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ
Online-AUTH>
```


Import a Base Derivation Key (BDK)

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DF** (superseded by IK)

Function: To import a BDK encrypted under a ZMK and translate it to encryption under LMK pair 28-29.

Notes: A KCV for the BDK is also computed and displayed.

The command also prompts for a variant. If the exporter applied a variant to the ZMK, enter the variant number.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- ZMK encrypted under LMK pair 04-05: 32 hexadecimal characters.
- ZMK variant (or <Return> to ignore). The command ignores the setting of the Atalla ZMK variant support parameter entered in the CS (Configure Security) command.
- BDK encrypted under the ZMK: 32 hexadecimal characters.

Outputs:

- BDK encrypted under LMK pair 28-29.
- BDK, KCV, if restrict KCV is enabled in the CS command the output will be restricted to the 6 most significant digits with padding zeros for the remainder.

Errors:

- Data invalid; please re-enter - the key is not 32 hexadecimal characters or an invalid ZMK variant is entered.
- Key parity error; re-enter key - the entered key does not have odd parity. Re-enter the key and check for typographic errors.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> DF <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter ZMK variant: X <Return>
Enter BDK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Encrypted BDK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ
Online>
```

Generate a Base Derivation Key (BDK)

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DG** (superseded by KG)

Function: To generate a random BDK, displaying it encrypted under the LMK pair and under a ZMK, and a BDK check value. Equivalent to Host BI command

Notes: The command also prompts for a variant. If the recipient requires a variant to the ZMK, enter the appropriate variant number.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- ZMK encrypted under LMK pair 04-05 (generated by the DE command): 32 hexadecimal characters.
- ZMK variant (or <Return> to ignore). (The command ignores the setting of the Atalla ZMK variant support parameter entered in the CS (Configure Security) command).
- ZMK key check value (generated by the DE command) or the value generated by the console CK command or Host BU command.

Outputs:

- BDK encrypted under the ZMK: 32 hexadecimal characters.
- BDK encrypted under LMK pair 28-29: 32 hexadecimal characters.
- BDK check value.

Errors:

- Data invalid; please re-enter - the encrypted ZMK does not contain 32 hexadecimal characters or the key check value does not contain 8 hexadecimal characters. Re-enter the correct number of hexadecimal characters.
- Key parity error; please re-enter - the entered ZMK does not have odd parity on each byte. Re-enter the encrypted ZMK and check for typographic errors.
- Check failed; re-enter check value or abort - the ZMK check key value is not correct. Re-enter the correct check value.
- Internal failure 12: function aborted - the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> DG <Return>
Enter encrypted ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
Enter ZMK variant: X <Return>
Enter ZMK check value: XXXX XXXX <Return>
BDK encrypted for transmission: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
BDK encrypted under LMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZ ZZZZ
Online>
```

Generate and Export a Master Load Key (KML)

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DA** (superseded by KG)

Function: To generate a double-length Master Load Key (KML) and return it encrypted under Variant 2 of LMK pair 04-05, and under a Zone Control Master Key (ZCMK). A check value for the KML is also returned.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- ZCMK, encrypted under LMK pair 04-05: 32 hexadecimal characters.
- (Optional) Atalla Variant – 1 or 2 numeric digit; this value is required only if support for Atalla variants is set using the “CS” console command (see Ref.2)

Outputs:

- KML, encrypted under the ZCMK: 32 hexadecimal characters.
- KML, encrypted under Variant 2 of LMK pair 04-05.
- KML check value, formed by encrypting a block of binary zeros with the key and returning the left-most 24 bits of the result: 6 hexadecimal characters.

Errors:

- `Data invalid; please re-enter` – the entered value does not contain 32 hexadecimal characters or invalid ZMK variant was entered. Re-enter the correct number of characters.
- `Key parity error` – the plaintext key does not have odd parity on each byte. Re-enter the correct value.
- `Internal failure 12: function aborted` – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> DA <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: Y <Return>, if enabled by CS command.)
KML encrypted for transmission: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
KML encrypted under LMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Import a Master Load Key (KML)

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **DB** (superseded by IK)

Function: To translate a double-length Master Load Key (KML) from encryption under a Zone Master Key (ZCMK) to encryption under Variant 2 of LMK pair 04-05. A check value for the KML is also returned.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- ZMK, encrypted under LMK pair 04-05: 32 hexadecimal characters.
- (Optional) Atalla Variant – 1 or 2 numeric digit; this value is required only if support for Atalla variants is set using the “CS” console command.
- KML, encrypted under the ZMK: 32 hexadecimal characters.

Outputs:

- KML, encrypted under Variant 2 of LMK pair 04-05.
- KML check value, formed by encrypting a block of binary zeros with the key and returning the left-most 24 bits of the result: 6 hexadecimal characters.

Errors:

- `Data invalid; please re-enter` – the entered value does not contain 32 hexadecimal characters or invalid ZMK variant was entered. Re-enter the correct number of characters.
- `Key parity error` – the plaintext key does not have odd parity on each byte. Re-enter the correct value.
- `Internal failure 12: function aborted` – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example:

```
Online> DB <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: V <Return>, if enabled by CS command).
Enter KML: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
KML encrypted under LMK: YYY YYY YYY YYY YYY YYY YYY YYY
Key check value: ZZZZZZ
Online>
```

Generate a CSCK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **YA** (superseded by KG)

Function: Generates a new CSCK and displays it encrypted under the LMK.
Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- A CSCK length flag.

Outputs:

- The new CSCK, encrypted under LMK 14-15 variant 4.

Errors:

- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example 1:

```
Online> YA <Return>
Enter CSCK length [S/D]: D <Return>
CSCK encrypted under LMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Online>
```

Example 2:

```
Online> YA <Return>
Enter CSCK length [S/D]: S <Return>
CSCK encrypted under LMK: YYYY YYYY YYYY YYYY
Online>
```

Export a CSCK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **YB** (superseded by KE)

Function: This command accepts a Zone Master Key (ZMK) and a CSCK encrypted under the LMK. It decrypts and checks parity on both keys, and if correct encrypts the CSCK under the ZMK and displays it.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- A flag to indicate the length of the ZMK.
- A ZMK encrypted under LMK 04-05 (generated by the “DE” command), 16/32 hexadecimal characters.
- A ZMK variant (or <Return> to ignore).
- Note: the Atalla variant support parameter (set with the “CS” command) is ignored. CSCK encrypted under LMK 14-15 variant 4, 16/32 hexadecimal characters.

Outputs:

- The CSCK encrypted under the ZMK.
- A Key Check Value (KCV) for the CSCK.

Errors:

- Data invalid; please re-enter – the keys are not 16 or 32 hexadecimal digits in length or invalid ZMK variant was entered.
- Key parity error – the key just entered did not have odd parity; check for typographical errors and re-enter.
- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example 1:

```
Online> YB <Return>
Enter ZMK length [S/D]: D <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: V <Return>, if enabled by CS command.)
Enter CSCK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
CSCK encrypted for transmission: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Example 2:

```
Online> YB <Return>
Enter ZMK length [S/D]: S <Return>
Enter ZMK: XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: V <Return>, if enabled by CS command.)
Enter CSCK: XXXX XXXX XXXX XXXX <Return>
CSCK encrypted for transmission: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Import a CSCK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **YC** (superseded by IK)

Function: This command accepts a Zone Master Key (ZMK) encrypted under the LMK and a CSCK encrypted under the ZMK. It decrypts and checks parity on the ZMK, and if correct uses it to decrypt the CSCK. Incoming CSCK parity is ignored, but is forced odd before encryption under the LMK.

Note: This command will only operate using a variant default LMK.

Authorisation: The HSM does not require any authorisation to run this command.

Inputs:

- A flag to indicate the length of the ZMK.
- A ZMK encrypted under LMK 04-05 (generated by the “DE” command), 16/32 hexadecimal characters.
- A ZMK variant (or <Return> to ignore).
- Note: the Atalla variant support parameter (set with the “CS” command) is ignored.
- A CSCK encrypted under the ZMK, 16/32 hexadecimal characters.

Outputs:

- The CSCK encrypted under LMK 14-15 variant 4.
- A Key Check Value (KCV) for the CSCK.

Errors:

- Data invalid; please re-enter – the keys are not 16 or 32 hexadecimal digits in length or invalid ZMK variant was entered.
- Key parity error – the key just entered did not have odd parity; check for typographical errors and re-enter.
- CSCK all-zero – the clear CSCK is all zeros, and will not be translated.
- Internal failure 12: function aborted – the contents of LMK storage have been corrupted or erased. Do not continue. Inform the Security Department.

Example 1:

```
Online> YC <Return>
Enter ZMK length [S/D]: D <Return>
Enter ZMK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: V <Return>, if enabled by CS command.)
Enter CSCK: XXXX XXXX XXXX XXXX XXXX XXXX XXXX XXXX <Return>
CSCK encrypted under LMK: YYYY YYYY YYYY YYYY YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Example 2:

```
Online> YC <Return>
Enter ZMK length [S/D]: S <Return>
Enter ZMK: XXXX XXXX XXXX XXXX <Return>
(Enter ZMK variant: V <Return>, if enabled by CS command.)
Enter CSCK: XXXX XXXX XXXX XXXX <Return>
CSCK encrypted under LMK: YYYY YYYY YYYY YYYY
Key check value: ZZZZZZ
Online>
```

Glossary

Term	Description
Account Number	See Primary Account Number
Acquirer	An institution (frequently a financial institution) that manages terminals and switches on-line payment transactions to the Card Issuer for authorisation and verification
Acquirer Working Key	Visa terminology for a Zone PIN Key, specifically one used to encrypt a PIN block between an Acquirer and Visa; abbreviated to AWK
Advanced Encryption Standard	Cryptographic algorithm, standardised in FIPS 197, that uses a 128-, 192- or 256-bit key; abbreviated to AES
ANSI X9.17	A highly influential ANSI standard defining how symmetric keys should be managed; now withdrawn and generally regarded as inadequate from a security perspective
Application Authentication Cryptogram	A particular type of Application Cryptogram used in EMV transactions, specifically for declined transactions; abbreviated to AAC
Application Cryptogram	EMV terminology for a Message Authentication Code; often abbreviated to AC
Asymmetric Algorithm	Generic term for a cryptographic algorithm in which the decryption key cannot be feasibly deduced from knowledge of the encryption key, and hence the encryption key can be made public, whereas the decryption key must remain secret; examples include RSA, DSA
Atalla	Part of the HP group of companies and supplier of Hardware Security Modules
Atalla Variant	A mechanism to modify a Transport Key, so that different types of key are encrypted under different variants of the Transport Key; required when exchanging keys with Atalla security modules
Australian Transaction Key Scheme	A specific type of Transaction Key Scheme, standardised in AS2605.6.2; often abbreviated to ATKS
Authentication Parameter	A value used in the Racal and Australian Transaction Key Schemes that “proves” the Issuer’s involvement in the transaction; usually denoted Auth Para (or AP)
Authorisation Request Cryptogram	A Message Authentication Code generated by an EMV card using a key derived from a specific Card Master Key and included in a request message sent to the Card Issuer; abbreviated to ARQC
Authorisation Response Code	A transaction response code generated by a Card Issuer and included in the generation of an Authorisation Response Cryptogram; abbreviated to ARC
Authorisation Response Cryptogram	A Message Authentication Code generated by a Card Issuer in response to a request message from an EMV card; abbreviated to ARPC
Authorised Activity	A mode of operation of the HSM 8000 that permits one or more specified sensitive functions to be performed; requires two passwords to be entered into the HSM, either manually or (more usually) via smart cards

Term	Description
Authorised State	Similar to Authorised Activity, except that Authorised State is a “global” authorisation for all sensitive functions
Base Derivation Key	A cryptographic key used to derive transaction keys used in the DUKPT scheme; abbreviated to BDK
Card Issuer	An institution (usually a financial institution) that issues payment cards, such as debit or credit cards, to its customers; the Card Issuer is usually involved in the on-line authorisation and verification of payment transactions involving these cards
Card Master Keys	A generic term for a family of EMV card specific keys derived from the corresponding Issuer Master Keys
Card Security Code	American Express equivalent of a Card Verification Value; may be 3, 4 or 5 digits in length; abbreviated to CSC
Card Security Code Key	Key used to generate or verify an American Express Card Security Code; abbreviated to CSCK
Card Verification Code	MasterCard equivalent of a Card Verification Value; abbreviated to CVC
Card Verification Key	Generic term for a key used with a Card Verification Value or a Card Verification Code generation or verification algorithm; abbreviated to CVK
Card Verification Value	A non-secret 3-digit number derived as a cryptographic function of a card’s Primary Account Number, Service Code and Expiry Date; used by Visa to verify that these card details have not been modified; abbreviated to CVV
Certificate Authority	A trusted party, whose role is to create Public Key Certificates; usually abbreviated to CA
Chip and PIN	A colloquial name for an EMV transaction; see Europay, MasterCard & Visa
Chip Card	See Smart Card
Cipher	See Cryptographic Algorithm
Cipher Block Chaining	A mode of encryption, whereby blocks of data are “chained” together during the encryption process (each block of ciphertext is combined with the next block of plaintext, prior to encryption of that block); for DES this mechanism is standardised in ANSI X3.106 and for 3-DES it is standardised in ANSI X9.52; usually abbreviated to CBC
Cipher Feedback	A mode of encryption, whereby ciphertext is “fed back” into the encryption process to provide a form of “chaining”; for DES this mechanism is standardised in ANSI X3.106 and for 3-DES it is standardised in ANSI X9.52; usually abbreviated to CFB
Ciphertext	Generic term for unintelligible data that is output from the encryption process of a cryptographic algorithm
Collision Resistance	A property of a Hash Function that makes it infeasible to find two separate messages that hash to the same value
Component	See Key Component

Term	Description
Cryptographic Algorithm	A function that transforms data (plaintext) and a Key to data (ciphertext) in such a way that the procedure cannot be reversed without knowledge of the key; also known as an Encryption Algorithm or a Cipher
Cryptographic Key	Value involved in the operation of a Cryptographic Algorithm; must be kept secret for a symmetric algorithm and the decryption key must be kept secret for an asymmetric algorithm
Cryptographic Zone	A generic term to indicate those entities that share common cryptographic keys; usually only two such entities are involved
Customer PIN	See Personal Identification Number
Data Authentication Code	An authentication value calculated by the smart card during an EMV transaction (MasterCard only) and validated by the Card Issuer; abbreviated to DAC
Data Encryption Algorithm	Cryptographic algorithm, standardised in FIPS 46 and ANSI X3.92, that uses a 56-bit key; more commonly known as the Data Encryption Standard (DES)
Data Encryption Key	Generic HSM 8000 terminology for a key used for data encryption; abbreviated to DEK
Data Encryption Standard	See Data Encryption Algorithm
Decimalisation Table	A function that maps hexadecimal characters to numeric characters; used in many PIN generation and verification algorithms, including the Diebold and IBM 3624 schemes
Decryption	The inverse process to Encryption, so that ciphertext is transformed into plaintext by means of a cryptographic algorithm and a key
Default LMK	A specified LMK (when using Multiple LMKs) to provide a backwards compatible mode of use for the HSM 8000
Derived PIN	A value derived from an Account Number and a key, usually as an intermediate step during the PIN generation process; also known as a Natural PIN
Derived Unique Key Per Transaction Scheme	A specific type of Transaction Key Scheme, widely used and standardised in ANSI X9.24-1; usually abbreviated to DUKPT
Diebold PIN Generation Algorithm	Proprietary PIN generation and verification algorithm, invented by Diebold and supported by the HSM 8000; nowadays this scheme is rarely used
Diebold Table	A 256 byte data structure used with the Diebold PIN Generation Algorithm, and stored in the HSM 8000 User Storage
Digital Signature	A cryptographic checksum, appended to a message and used by the recipient of the message to detect whether the message has been corrupted during transmission (either accidentally or deliberately); calculated using the private key of an asymmetric algorithm and verified using the corresponding public key
Digital Signature Algorithm	Public key algorithm, standardised in FIPS 186, used for the generation and verification of digital signatures; usually abbreviated to DSA

Term	Description
Dynamic Card Verification Value	Similar to a (static) Card Verification Value or Card Verification Code, except that it is calculated using a unique transaction key generated by a smart card using a specific Card Master Key
Dynamic Number	A dynamic authentication value calculated by the smart card during an EMV transaction (MasterCard only) and validated by the Card Issuer; abbreviated to DN
Electronic Codebook	A mode of encryption, whereby blocks of data are independently encrypted; for DES this mechanism is standardised in ANSI X3.106 and for 3-DES it is standardised in ANSI X9.52; usually abbreviated to ECB
Electronic Purse	Generic term for a smart card based scheme that stores an electronic version of cash on the card
Encryption	Transformation of meaningful data (plaintext) into unintelligible data (ciphertext) by means of a cryptographic algorithm and a key
Encryption Algorithm	See Cryptographic Algorithm
Europay, MasterCard and Visa	A framework for terminal-based “Chip & PIN” payment transactions, developed by Europay, MasterCard and Visa; usually abbreviated simply to EMV (see www.emvco.com)
Exclusive-or	A mathematical operation to combine two binary numbers; also known as “modulo 2 addition, without carry”; often abbreviated to XOR
GISKE	A proprietary keyblock format, a forerunner to the TR-31 keyblock format, supported by some VeriFone terminals
Hardware Security Module	Alternative name for Host Security Module
Hash Function	A non-cryptographic technique for compressing arbitrarily long strings of data to a fixed length, such that finding Collisions or Pre-images is computationally infeasible; examples include SHA-1, SHA-2 and MD5; also known as a Message Digest
Host Security Module	Tamper-resistant cryptographic processor that accepts commands from a host application via a defined interface; often abbreviated to HSM
HSM 8000	Family of Host Security Modules, manufactured by Thales; in this manual the abbreviation HSM always refers to the HSM 8000 device
IBM 3624 PIN Generation Algorithm	Widely used proprietary PIN generation and verification algorithm, invented by IBM and supported by the HSM 8000
Initialisation Value	See Initialisation Vector
Initialisation Vector	A non-secret value used to initialise the Cipher Block Chaining and Cipher Feedback modes of encryption; may also be used as a “chaining value” when encrypting a large amount of data; also known as an Initialisation Value and often abbreviated to IV
Integrated Circuit Card	See Smart Card
Interchange	A generic term for a Cryptographic Zone involving two or more host systems
Issuer	See Card Issuer

Term	Description
Issuer Master Keys	A generic term for a family of Card Issuer specific keys used in EMV to derive card unique keys and transaction keys; the terminology between Visa and MasterCard varies considerably and is detailed in the "EMV Chip Card Commands" section of the HSM 8000 Host Command Reference Manual
Issuer Working Key	Visa terminology for a Zone PIN Key, specifically one used to encrypt a PIN block between Visa and the Card Issuer; abbreviated to IWK
Key	See Cryptographic Key
Keyblock	A data structure that comprises an encrypted key, a Header that defines how the key may be used and an authenticator, calculated over the Header and the encrypted key
Keyblock Header	The set of fields in a keyblock that determine how the key may be used; typically, such fields will include key usage, mode of use, algorithm and exportability
Keyblock LMK	A key scheme whereby keys are encrypted in Keyblock format using the HSM 8000 Local Master Key (LMK)
Key Change Storage	Tamper-proof area of memory in the HSM 8000 that stores "old" LMK(s); used to permit translation of keys following an LMK change
Key Check Value	A mechanism used to verify the correctness of a cryptographic key; the most common technique is to encrypt a block of binary zeros with the key and use the leftmost 24 bits (6 hexadecimal characters) as the check value; abbreviated to KCV
Key Component	Part of a cryptographic key, which when combined with other key components (e.g. using the exclusive-or operation) forms the complete key; typically used when generating an LMK or generating and distributing a Key Encrypting Key
Key Component Mailer	Tamper-evident form used to convey a Key Component to another party
Keyed-Hash Message Authentication Code	A type of Message Authentication Code (MAC) calculated using a hash function and a secret key; the best known example is known simply as HMAC ; it uses the SHA-1 hash function and is standardised in FIPS 198
Key Encrypting Key	A generic term for a key (usually a symmetric key) that is used to encrypt keys for distribution to another party; usually abbreviated to KEK
Key Management	General term for the processes involved during the lifecycle of a cryptographic key; involves key generation, storage, use, distribution (import and export), change, deletion, archive, revocation, back-up, and so on; for symmetric algorithms, the principal requirement of key management is that keys remain secret and are only used for their intended purpose; for asymmetric algorithms, the principal requirements are that the integrity of public keys is assured and that private keys remain secret
Key Scheme	Mechanism used by the HSM 8000 to indicate the mode of encryption both for stored keys (i.e. encrypted under the LMK) and for distributed keys (i.e. encrypted under a Transport Key)

Term	Description
Key Type Code	Three character value used with a Variant LMK in the HSM 8000 to identify different key types (and hence to identify the LMK Pair used to encrypt the key)
LMK	See Local Master Key
LMK Identifier	A technique used to identify a particular LMK in the HSM 8000 when Multiple LMKs are used
LMK Pair	For a Variant LMK, different key types are encrypted under a pair of (different) variants of the LMK; for example, a Zone PIN Key is encrypted under LMK pair 06-07
LMK Storage	Tamper-proof area of memory in the HSM 8000 that stores the LMK(s)
Local Master Key	A 3-DES key (or a set of such keys) stored in the tamper-protected memory of the HSM 8000 and used for encryption of other keys so they can be stored securely outside the HSM; abbreviated to LMK
MAC	See Message Authentication Code
MAC Residue	The bits of the Message Authentication Block (MAB) that are not used to form the MAC ; typically, when using DES or 3-DES, the MAC is the leftmost 32 bits of the MAB and the MAC Residue is the remaining 32 bits; used, in particular, in the Racal and Australian Transaction Key Schemes
Management LMK	A specified LMK (when using Multiple LMKs) that is used by the HSM 8000 for purposes that are not linked to a particular LMK; for example, authenticating audit trail records
Master Key	Generic term for a cryptographic key that is used as a Transport Key or used to derive card or transaction specific keys; examples include Zone Master Keys and Master Load Keys
Master Load Key	A cryptographic key used to derive card and transaction specific keys used in the Visa Cash scheme; abbreviated to KML
Master/Session Key Scheme	Generic term for a key management scheme, whereby Session Keys are updated frequently, encrypted under a fixed Master Key; often abbreviated to MK/SK
MD5	A Hash Function (Message Digest) that outputs a 128-bit result; nowadays its security is generally regarded as relatively weak
Message Authentication Block	Final ciphertext block when computing a MAC; may also be used when calculating a MAC over a large amount of data; often abbreviated to MAB
Message Authentication Code	A cryptographic checksum, appended to a message and used by the recipient of the message to detect whether the message has been corrupted during transmission (either accidentally or deliberately); calculated using a symmetric algorithm; mechanisms have been standardised in ANSI X9.9, ANSI X9.19 and ISO 9797 (amongst others); usually abbreviated to MAC
Message Digest	See Hash Function
Multiple LMKs	The HSM 8000 can support a number of independent LMKs within a single device; combinations of Variant and Keyblock LMKs can be supported

Term	Description
Natural PIN	See Derived PIN
Offline	Mode of operation of the HSM 8000 that prevents online communication with the host computer system; achieved using a single physical key with the HSM; usually required when changing configuration parameters
Offset	A non-secret value used in some PIN generation algorithms to link a customer PIN to a Derived PIN
Online	Mode of operation of the HSM 8000 that permits communication with a host computer system via the HSM's host port
Optimal Asymmetric Encryption Padding	A mechanism specified in PKCS#1, version 2.0, for padding data prior to encryption using an RSA public key or signing using an RSA private key; abbreviated to OAEP
Optional Keyblock Header	An optional set of fields in a keyblock that typically are used to refine the usage of the key contained in the keyblock; for example, a start and end date/time field may be included
Personal Identification Number	A number known only to the cardholder and used to authenticate the cardholder during a financial transaction; typically 4 to 6 digits, but may be up to 12 digits in length; abbreviated to PIN
PIN	See Personal Identification Number
PIN Block	A block of data that contains a PIN and is padded to the appropriate size prior to encryption; for example, 64 bits in the case of DES
PIN Block Format	Mechanism used to pad a PIN to form a PIN Block
PIN Generation Algorithm	A technique used to generate (and verify) a PIN; usually involves an Account Number and a Key
PIN Mailer	Tamper-evident form used to convey a PIN from the Card Issuer's PIN generation facility to the customer
PIN Solicitation	A technique used to allow a customer to choose his or her own PIN; now rarely used as customers can generally change their PINs at terminals
PIN Translation	The process of decrypting a PIN block from encryption under a key and then re-encrypting it under a different key; required, for example, when a PIN moves from a Terminal Zone to Interchange; the PIN Block Format may also be changed during the translation process
PIN Verification Algorithm	See PIN Generation Algorithm
PIN Verification Key	Generic term for a key used with a PIN generation or verification algorithm; abbreviated to PVK
PIN Verification Value	A non-secret 4-digit number derived as a cryptographic function of a customer's Account Number and the customer's PIN; used by Visa as a method of verifying the customer's PIN; abbreviated to PVV
Plaintext	Generic term for meaningful data that is input to the encryption process of a cryptographic algorithm

Term	Description
Pre-Image Resistance	A property of a Hash Function that makes it infeasible to find a data string that hashes to a specified value
Primary Account Number	A unique sequence of numbers assigned to a cardholder account that identifies the Card Issuer and type of financial transaction card; usually abbreviated to PAN
Private Key	Common term for the decryption key in a Public Key Algorithm; sometimes called a Secret Key
Public Key	Common term for the encryption key in a Public Key Algorithm
Public Key Algorithm	See Asymmetric Algorithm
Public Key Certificate	A data structure that binds together a user's identity and their Public Key, via a Digital Signature generated by a Certificate Authority
Public Key Modulus	A large number (integer) used in arithmetic calculations involving Public Key Algorithms; in general, the larger the number the more secure the algorithm
Racal PIN Encryption Algorithm	Proprietary technique used by the HSM 8000 to encrypt PINs under the LMK for storage on a host database; an alternative to the Visa PIN Encryption Algorithm and sometimes referred to as Algorithm B
Racal Transaction Key Scheme	A specific type of Transaction Key Scheme, standardised in APACS 40 & APACS 70; often abbreviated to RTKS
Reference Number	An encrypted version of an Account Number, used with PIN Solicitation
RSA Public Key Algorithm	Commonly used Public Key Algorithm, named after its inventors, Rivest, Shamir & Adleman; usually abbreviated simply to RSA
Secret Key	See Private Key; sometimes also used to denote a key used in a Symmetric Algorithm
Secure Hash Algorithm (SHA-1)	A Hash Function whose output is always 160 bits in length; standardised in FIPS 180-1; usually abbreviated to SHA-1
Secure Hash Algorithm (SHA-2)	A family of Hash Functions with different output lengths, standardised in FIPS 180-2; specifically, SHA-224, SHA-256, SHA-384 and SHA-512
Secure State	Mode of operation of the HSM 8000 that is required for certain highly sensitive functions (for example, enabling the HSM's tamper-detection circuitry); achieved using two physical keys with the HSM
Semi-Weak Keys	In DES, a Semi-Weak key pair is a set of two keys such that the encryption process with one of the keys is identical to the decryption process using the other key; DES has eight known pairs of such keys
Session Key	Generic term for a cryptographic key that is used (for example) for data or PIN encryption or MAC calculation for a limited number of transactions (often just a single transaction); also known as a Working Key

Term	Description
Smart Card	A plastic card (e.g. a debit or credit card) that includes a microprocessor; also known as a Chip Card or Integrated Circuit Card (ICC)
Symmetric Algorithm	Generic term for a cryptographic algorithm that uses the same key for both encryption and decryption, and hence all keys must remain secret; examples include DES, 3-DES, AES
Terminal	A generic term for any device from which retail payment transactions originate; for example, a Point of Sale (POS) terminal or an Automated Teller Machine (ATM)
Terminal Authentication Key	HSM 8000 terminology for a key used for MAC calculation on a Terminal Zone; abbreviated to TAK
Terminal Master Key	HSM terminology for a Key Encrypting Key used on a Terminal Zone; abbreviated to TMK
Terminal PIN Key	HSM 8000 terminology for a key used for encryption of a PIN Block on a Terminal Zone; abbreviated to TPK
Terminal Zone	A Cryptographic Zone involving one or more terminals and a host system
Thales Keyblock	A proprietary Keyblock format, based on the TR-31 format, and supported by the HSM 8000
TR-31 Keyblock	An ANSI Technical Report that specifies a particular Keyblock format; expected to be widely adopted by the financial community
Transaction Certificate	A particular type of Application Cryptogram used in EMV transactions, specifically for accepted transactions; abbreviated to TC
Transaction Key Scheme	Generic term for a Terminal Key Management scheme that allows keys to change automatically with each transaction
Transport Key	Another name for a Key Encrypting Key
Triple Data Encryption Algorithm	Cryptographic algorithm that extends the use of the Data Encryption Standard (DES) algorithm to use either two or three independent keys (112 or 168 bits, respectively); often abbreviated to TDES or 3-DES
User Storage	Volatile (and non-secure) area of memory in the HSM 8000 that is used to store encrypted keys and other frequently used data elements (such as Decimalisation Tables); mainly used for performance reasons
Variant	A generic term used to describe a mechanism whereby keys or other data are encrypted using some form of “variant” (or “modifier”) of the “base” encryption key, to provide additional security for the key; such techniques are generally regarded as relatively insecure
Variant LMK	A key scheme used by the HSM 8000 whereby different key types are encrypted under different variants of the LMK, to provide cryptographic separation of key types
Visa Cash	A proprietary electronic purse scheme developed by Visa

Term	Description
Visa PIN Encryption Algorithm	Proprietary technique used by the HSM 8000 to encrypt PINs with the LMK for storage on a host database; sometimes referred to as Algorithm A
WatchWord	A Thales proprietary technique that calculates a cryptographic response to a given “challenge”
WatchWord Key	HSM 8000 terminology for a key used to calculate a WatchWord response; abbreviated to WWK
Weak Key	In DES, a Weak Key is one of four known keys such that the encryption and decryption processes are identical when using these keys; more generally, a Weak Key is any cryptographic key which exhibits “undesirable” properties
Working Key	See Session Key
Zone	See Cryptographic Zone
Zone Authentication Key	HSM 8000 terminology for a key used for MAC calculation on Interchange; abbreviated to ZAK
Zone Control Master Key	Visa terminology for a Zone Master Key; abbreviated to ZCMK
Zone Encryption Key	HSM 8000 terminology for a key used for data encryption on Interchange; abbreviated to ZEK
Zone Master Key	HSM terminology for a Key Encrypting Key used for Interchange; abbreviated to ZMK
Zone PIN Key	HSM 8000 terminology for a key used for encryption of a PIN Block on Interchange; abbreviated to ZPK

General Abbreviations

Abbreviation	Description
3-DES	Triple Data Encryption Standard (algorithm)
AAC	Application Authentication Cryptogram (EMV)
AC	Application Cryptogram (EMV)
AES	Advanced Encryption Standard (algorithm); FIPS 197
ANSI	American National Standards Institute
AP	Authentication Parameter (ATKS and RTKS)
ARQC	Authorisation Request Cryptogram (EMV)
ARC	Authorisation Response Code (EMV)
ARPC	Authorisation Response Cryptogram (EMV)
ATKS	Australian Transaction Key Scheme
ATM	Automated Teller Machine
AWK	Acquirer Working Key (Visa)
BDK	Base Derivation Key (DUKPT)
CA	Certificate Authority
CBC	Cipher Block Chaining
CFB	Cipher Feedback
CSC	Card Security Code (American Express)
CSCK	Card Security Code Key (American Express)
CVC	Card Verification Code (MasterCard)
CVK	Card Verification Key (Visa & MasterCard)
CVV	Card Verification Value (Visa)
DAC	Data Authentication Code (EMV, MasterCard)
dCVV	Dynamic Card Verification Value
DEA	Data Encryption Algorithm (FIPS 46)
DES	Data Encryption Standard (algorithm); FIPS 46
DEK	Data Encryption Key (HSM 8000)
DN	Dynamic Number (EMV, MasterCard)
DSA	Digital Signature Algorithm (FIPS 186)
DUKPT	Derived Unique Key Per Transaction Scheme (ANSI X9.24-1)
ECB	Electronic Codebook
EMV	Europay, MasterCard and Visa
FIPS	Federal Information Processing Standard
HMAC	Keyed-Hash Message Authentication Code (FIPS 198)
HSM	Host Security Module (alternatively, Hardware Security Module)

ICC	Integrated Circuit Card
ISO	International Standards Organisation
IV	Initialisation Vector
IWK	Issuer Working Key (Visa)
KCV	Key Check Value
KEK	Key Encrypting Key
KML	Master Load Key (Visa Cash)
LMK	Local Master Key (HSM 8000)
MAC	Message Authentication Code
MK/SK	Master/Session Key (key management scheme)
OAEP	Optimal Asymmetric Encryption Padding
PAN	Primary Account Number
PIN	Personal Identification Number
POS	Point of Sale (terminal)
PVK	PIN Verification Key (HSM 8000)
PVV	PIN Verification Value (Visa)
RSA	Rivest, Shamir & Adleman (public key algorithm)
RTKS	Racal Transaction Key Scheme
SHA-1	Secure Hash Algorithm (FIPS 180-1)
SHA-2	Secure Hash Algorithm (FIPS 180-2)
TAK	Terminal Authentication Key (HSM 8000)
TC	Transaction Certificate (EMV)
TDES	Triple Data Encryption Standard (algorithm)
TMK	Terminal Master Key (HSM 8000)
TPK	Terminal PIN Key (HSM 8000)
TR-31	TR-31 keyblock (format)
WWK	WatchWord Key (HSM 8000)
XOR	Exclusive-or
ZAK	Zone Authentication Key (HSM 8000)
ZCMK	Zone Control Master Key (Visa)
ZEK	Zone Encryption Key (HSM 8000)
ZMK	Zone Master Key (HSM 8000)
ZPK	Zone PIN Key (HSM 8000)

SNA-SDLC Abbreviations

Abbreviation	Description
AID	Attention Identifier.
BB,EB	Begin Bracket, End Bracket. SNA Brackets are used to delimit transactions. In the HSM environment they can be used to carry-out a change of direction (CD) on the link. Bracketing is only of limited use with HSMs.
BC,EC	Beginning of Chain and End of Chain Bits:
	00 : Middle in Chain : MIC.
	01 : End of Chain : LIC.
	10 : Beginning of Chain : FIC.
	11 : Only in Chain : OIC.
BETB	3274 "Between Bracket" State. After EB and before BB (i.e.,]..here..[).
CD	Change Direction so that the other LU has control.
CSI	Bit in RH defining "Code Selection Indicator".
DFC	Data Flow Control command. SNA command to control the flow of data in LU-LU sessions.
FIC	First in Chain. Position of a request (RU) in an SNA Chain.
FMD	Function Management Data command. SNA Command for Data transfer between LUs (LU-LU).
INB	3274 "In Bracket" State. After BB and before EB (i.e., [..here..]).
IPR	Isolated Pacing Response.
LIC	Last in Chain. Position of a request (RU) in an SNA Chain.
LU	SNA Logical Unit, entity which partakes in SNA sessions.
MIC	Middle in Chain. Position of a request (RU) in an SNA Chain.
OIC	Only in Chain.
PEND.BB	3274 "Pending Begin Bracket" State. A 3270 Terminal user has started to type but has not yet pressed <Enter>.
PLU	SNA Primary Logical Unit; normally the application program in the Host computer.
PU	SNA Physical Unit, can contain several LUs (e.g., a 3274 Cluster Controller).
RCV	Receive state.
RH	Request/Response Header (3 bytes).
RQD	Definite Request/Response. The recipient must reply to this message.

RQE	Exception Request/Response. The sender does not expect a reply unless there is a problem.
+RSP	A positive (affirmative) response to an SNA request.
RU	Request/Response Unit.
SBA	Set Buffer Address.
SDT	Start Data Traffic.
SLU	SNA Secondary Logical Unit; normally a terminal or the HSM.
SSCP	SNA System Services Control Point; normally resides in the Communications Controller (37x5) and responsible for controlling access to the PU and SLU.
TH	Transmission Header (6 bytes).
WCC	Write Control Character (3270).
WSF	Write Structured Field.
X'NNNN	A 4-digit hexadecimal value.

Chapter 5 - Remote Management Commands

This chapter describes the commands used to configure an HSM that is to be remotely managed, using the Thales Remote HSM Manager application. It also includes those commands that provide information to assist with the installation and configuration of the HSM.

Note: HSMs must contain an appropriate licence (HSM8-LIC009) before an HSM can be remotely managed.

Remote Management Commands

The HSM provides the following console commands to support remote management configuration:

Command	Page
Intialise CA (RI)	183
Generate an HSM Certificate (RH)	185
Generate Smart Card Certificates (RR)	187
Generate RMK Component (RG)	189
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Reset Administrator/Operator Card (RX)	194

Intialise CA

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RI**

Function: To configure the CA parameter table and generate a CA RSA key pair and write the results to smart cards.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- CA parameters (if modified from the default)
- Number of CA share cards (parameter “n”, $3 \leq n \leq 9$)
- Number of shares to recover the CA private key (parameter “m”, $3 \leq m \leq n$)

Outputs:

- Prompts, as above
- Key generation message
- Prompt to enter smart card and PIN
- Continuation message
- Message giving number of copies of public key card
- Summary of CA information

Errors:

- Invalid value
- Smart card warning – smart card already contains a CA private key share or a public key certificate

Notes:

- The CA private key is broken into a number of shares to be used by the threshold scheme and each share is written to a separate smart card.
- In addition to the CA private key share, the self-certified CA public key and the CA parameter table will be written to each smart card.
- The length of the RSA modulus and the public exponent are determined by the values held in the CA parameter table.
- The CA information will be summarised at the end of the operation to ensure that no errors were made.

Example 1: *This example demonstrates the use of the **RI** console command to generate a CA consisting of 5 (previously formatted) CA cards, any 3 of which are required to recover (and therefore use) the CA's private key.*

```
Secure> RI <Return>
Issuer name: [default = CertAuth]: <Return>
Signature algorithm [RSA]: <Return>
Hash Algorithm [SHA-1, SHA-256(default)]: <Return>
CA RSA key length [1024-2048 (default)]: <Return>
HSM RSA key length [1024-2048 (default)]: 1536 <Return>
Card RSA key length [1024-2048 (default)]: 1024 <Return>
Public exponent [3, 65537 (default)]: <Return>
Enter number (n) of CA private key shares: [3-9]: 5 <Return>
Enter number (m) of CA private key shares: [3-5]: 3 <Return>
Enter 12 character device serial number : CA0000000001 <Return>
Generating CA key pair ...

Insert first CA private key card and enter PIN: ***** <Return>
Insert second CA private key card and enter PIN: ***** <Return>
Insert third CA private key card and enter PIN: ***** <Return>
Insert fourth CA private key card and enter PIN: ***** <Return>
```

Insert fifth CA private key card and enter PIN: ***** <Return>

CA generation complete as follows:

Issuer name: CertAuth
Signature algorithm: RSA
Hash Algorithm: SHA-256
CA RSA key length: 2048
HSM RSA key length: 1536
Card RSA key length: 1024
Public exponent: 65537

Number of CA private key shares: 5
Number of shares to recover private key: 3

Secure>

Generate an HSM Certificate

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RH**

Function: To generate an HSM public key certificate, and stored it inside the HSM.

Authorisation: The HSM may be in the online, offline or secure state.

Inputs: None.

Outputs:

- Prompt to enter smart card and PIN
- Key generation message
- Confirmation message
- CA Parameter Table (as retrieved from the CA share cards)

Errors:

- `Public key error`
- `Private key error`
- `Invalid serial number`

Notes: The CA private key is recovered from “m” share cards. The self-signed CA public key certificate, the CA parameter table, and the threshold scheme parameters are read from each card.

The processing ensures that all “m” CA share cards contain identical copies of the CA parameter table and the threshold scheme parameters.

After the CA private key is recovered, the CA Parameter Table is displayed to the user to ensure that the information is correct.

The HSM generates an RSA key pair and uses the CA private key to create the HSM’s public key certificate. The length of the RSA modulus and the public exponent for the generated key are determined by the values held in the CA parameter table.

The HSM stores its private key, its public key certificate and the self-signed CA public key certificate in secure memory. If an RMK is installed, secure memory and Flash memory are synchronised.

Example: *This example shows the use of the RH command to generate an HSM's certificate. In this example, 3 shares are required to recover the CA private key.*

Online> **RH** <Return>

Insert CA private key card and enter PIN: ***** <Return>

Insert another CA private key card and enter PIN: ***** <Return>

Insert another CA private key card and enter PIN: ***** <Return>

CA parameters as follows:

Issuer name: CertAuth

Signature algorithm: RSA

Hash Algorithm: SHA-256

CA RSA key length: 2048

HSM RSA key length: 1536

Card RSA key length: 1024

Public exponent: 65537

Continue generating HSM Certificate using the above CA parameters

[Y/N]: **Y** <Return>

Generating HSM key pair ...

NO RMK LOADED - SIGNIFICANT KEY LOSS IS POSSIBLE!!!

HSM certificate generated and stored.

Online>

Generate Smart Card Certificates

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RR**

Function: To generate one or more smart card public key certificates and optionally print the session log.

Authorisation: The HSM may be in the online, offline or secure state.

Inputs:

- Transport PIN (minimum 8 digits, maximum 12 digits).

Outputs:

- Prompt to enter smart card and PIN
- CA Parameter Table (as retrieved from the CA share cards)
- Key generation message
- Continuation message
- Confirmation and continuation message
- Session log
- Prompt to commence printing
- Prompt to repeat printing

Errors:

- Public key error
- Private key error
- Invalid value
- Invalid serial number
- Printer error
- Smart card processing error (xxxx)

Notes: The CA private key is recovered from “m” share cards. The self-signed CA public key certificate, the CA parameter table, and the threshold scheme parameters are read from each card.

The processing ensures that all “m” CA share cards contain identical copies of the CA parameter table and the threshold scheme parameters.

After the CA private key is recovered, the CA Parameter Table is displayed to the user to ensure that the information is correct.

Each target smart card is inserted into the HSM’s card reader and the HSM generates an RSA key pair for the target card and uses the CA private key to create the public key certificate. The length of the RSA modulus and the public exponent for the generated key are determined by the values held in the CA parameter table.

The HSM loads the private key, the public key certificate and the self-signed CA public key certificate onto the smart card.

A Transport PIN is also set for the smart card.

Example: *This example shows the use of the RR command to generate a certificate on three smart cards. In this example, 3 shares are required to recover the CA private key.*

Online> RR <Return>

Insert CA private key card and enter PIN: ***** <Return>

Insert another CA private key card and enter PIN: ***** <Return>

Insert another CA private key card and enter PIN: ***** <Return>

CA parameters as follows:

Issuer name: CertAuth

Signature algorithm: RSA

Hash Algorithm: SHA-256

CA RSA key length: 2048

HSM RSA key length: 1536

Card RSA key length: 1024

Public exponent: 65537

Continue generating HSM Certificate using the above CA parameters
[Y/N]: Y <Return>

Enter transport PIN: XXXXXXXX <Return>

Insert target smart card

Generating smart card key pair ...

Keys loaded; process another smart card? [Y/N]: Y <Return>

Insert target smart card

Generating smart card key pair ...

Keys loaded; process another smart card? [Y/N]: Y <Return>

Insert target smart card

Generating smart card key pair ...

Keys loaded; process another smart card? [Y/N]: N <Return>

Smart Card Certificate Generation Session

Date/Time: March 20 11:19:30 2008

Transport PIN: XXXXXXXX

Administrator Cards:

12340093

Number of Administrator Cards: 1

Operator Cards:

99990046

99990044

Number of Operator Cards: 2

Print session log? [Y/N]: Y <Return>

Printing successful; print another copy? [Y/N]: N <Return>

Online>

Generate RMK Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RG**

Function: To generate an RMK component and write the result to smart cards.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- Component number (1, 2 or 3)
- Number of RMK component back-up cards (minimum = 1)

Outputs:

- Prompts to enter component number & number of back-up cards
- Prompt to enter smart card and PIN
- Confirmation of number of copies made

Errors:

- Invalid value
- Smart card warning - smart card already contains an RMK component

Notes: The command forces the operator to make at least one back-up copy of each smart card.

Example: *This example shows the use of the RG command to generate RMK component number 1, and to make one back-up copy.*

```
Secure> RG <Return>
```

```
Enter RMK component number [1,2,3]: 1 <Return>
```

```
Insert RMK component card and enter PIN: ***** <Return>
```

```
Enter number of back-up cards required: 1 <Return>
```

```
Insert back-up card and enter PIN: ***** <Return>
```

```
Total number of copies made: 2
```

```
Secure>
```

Back-up RMK Component

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RB**

Function: To copy an existing RMK component to a back-up smart card.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- Number of RMK component back-up cards (minimum = 1)

Outputs:

- Prompt to enter number of copies
- Prompt to enter component number
- Prompt to enter smart card and PIN
- Continuation message
- Confirmation of number of copies made

Errors:

- Invalid value
- Smart card error
- Smart card warning

Example: *This example shows the use of the RB command to copy RMK component number 3 to two back-up smart cards.*

```
Secure> RB <Return>
```

```
Insert RMK component card and enter PIN: ***** <Return>
RMK component number 3
Continue processing? [Y/N]: Y <Return>
Enter number of back-up cards required: 2 <Return>
Insert back-up card and enter PIN: ***** <Return>
Insert back-up card and enter PIN: ***** <Return>
Total number of copies made: 2
```

```
Secure>
```

Load RMK

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RL**

Function: To load an RMK from component smart cards and (if applicable) synchronise secure memory and Flash memory.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs: • None

Outputs: • Prompt to enter smart card and PIN
• RMK check value
• Continuation message(s)
• Status message(s)

Errors: • Component parity error
• Component number error – component already loaded
• Error in stored private key – validation of the private key stored in Flash memory failed; RMK deleted from memory
• MAC validation failure – the MAC on the public key certificates stored in Flash memory did not validate; RMK **not** deleted from memory

Notes: This command may be run before the HSM key pair has been generated (in which case the RMK is simply loaded into secure memory) or after the HSM key pair has been generated (in which case secure memory and Flash memory will be synchronised).

If the HSM has generated its key pair and is subsequently tampered before the “RL” command is executed then the RSA private key will be deleted from memory and a new key pair will need to be generated.

Example 1: *This example shows the use of the RL command to load an RMK into an HSM which does not contain a private key. Once a private key is created, all Remote Management keys and certificates will be synchronised between secure memory and Flash memory.*

Secure> **RL** <Return>

```
Insert first RMK component card and enter PIN: ***** <Return>
Insert second RMK component card and enter PIN: ***** <Return>
Insert third RMK component card and enter PIN: ***** <Return>
RMK Check Value: 123ABC
Are you sure you want to install this RMK? [Y/N]: Y <Return>
RMK successfully loaded.
Security group successfully written.
Key synchronisation complete.
```

Secure>

Example 2: *This example shows the use of the RL command to load an RMK into an HSM which already includes an RMK. The existing RMK is replaced with the newly loaded RMK, and all Remote Management keys and certificates are synchronised between secure memory and Flash memory.*

```
Secure> RL <Return>

RMK already exists; check value = DEF789
Are you sure you want to overwrite the currently installed RMK? [Y/N]: Y <Return>
Insert first RMK component card and enter PIN: ***** <Return>
Insert second RMK component card and enter PIN: ***** <Return>
Insert third RMK component card and enter PIN: ***** <Return>
RMK check value: 123ABC
Are you sure you want to install this RMK? [Y/N]: Y <Return>
RMK successfully loaded.
Security group successfully written.
Key synchronisation complete.

Secure>
```


Convert Local LMK Card

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RT**

Function: To copy an existing (local) HSM LMK component smart card to a (remote-usable) Multos “Operator” smart card.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs:

- Local LMK card and PIN
- Multos “Operator” card and PIN

Outputs:

- Prompt to enter smart card and PIN
- LMK check value
- Continuation message(s)
- Status message(s)

Errors:

- Not a local LMK card
- Not a remote Operator card
- Not in Secure state
- Invalid PIN; re-enter
- Warning – card not blank. Process? [Y/N]:
- Overwrite LMK set? [Y/N]:
- Load failed check comparison

Notes: This command handles both Variant and Keyblock LMK key types, stored on either legacy (PC3) or current (TT80) HSM smart cards. The command will operate in a similar manner to the “Duplicate LMK Components Sets” (DC) command.

Example 1: *This example shows the use of the RT command to copy an existing (local) HSM LMK component (component no. 1) smart card to a (remote-usable) Multos “Operator” smart card.*

```
Secure> RT <Return>
```

```
Insert Local LMK card and enter PIN: ***** <Return>
```

```
LMK component set [1-9]: 1 <Return>
```

```
Insert Remote Operator card and enter PIN: ***** <Return>
```

```
SC Serial: SC000000012
```

```
Complete: LMK check value: 123ABC
```

```
Make another copy? [Y/N]: N <Return>
```

```
Secure>
```

Reset Administrator/Operator Card

Variant ☒		Keyblock ☒
Online ☒	Offline ☒	Secure ☒
Authorisation: Not required		

Command: **RX**

Function: To erase the contents of an Administrator or Operator MULTOS smartcard, and reset it to the stage whereupon it can be re-issued using the RR console command.

Authorisation: The HSM must be in the Secure state to run this command.

Inputs: • None

Outputs: • Prompt to enter smart card
• Confirmation message
• Continuation message(s)

Errors: • Not in Secure state
• Invalid PIN; re-enter
• Non-MULTOS card inserted

Notes: Use with caution! Using this command erases all contents from the smartcard!

Example 1: *This example shows the use of the RX command to erase the contents of an Administrator (or Operator) MULTOS smartcard.*

```
Secure> RX <Return>
```

```
Warning: Resetting an Administrator/Operator smartcard to its original
state will erase all key material from the card.
```

```
Are you sure? [Y/N]: Y <Return>
```

```
Reset another MULTOS card? [Y/N]: N <Return>
```

```
Secure>
```

Appendix A ERROR CODES

General

The error log lists each error with a severity, an error code and a sub-code. The error log only contains the numerical part of the error code and sub-code. Sub-codes relate to a specific error code.

Severity

Severity	Meaning
INFORMATIVE	Something abnormal happened, but was not important.
RECOVERABLE	Something abnormal happened, but the unit recovered from it without rebooting or losing data.
MAJOR	Something abnormal happened; the unit recovered from it but may have lost data or information due to restarting a process or re-initialising hardware. The unit may not function at full capacity.
CATASTROPHIC	Something abnormal happened and the unit had to reboot to recover.

Error Codes

Number	Code	Meaning
1	SYSTEM	System Errors
2	OS	Operating System Errors
3	HARDWARE	Hardware Errors
4	NETWORK	Network Errors
5	ASSERT	Software Assertion Failure
6	CONFIG	Configuration Errors
7	EXECUTIVE	Software Executive Failure

Sub-Codes

System Errors (Code 1)

Number	Sub-code	Meaning
1	DIAG_POWERUP	Power up diagnostics failed
2	DIAG_EXEC	A diagnostic failed
3	CONSOLE	Console failed
4	STACK_GONE	Task stack depleted
5	NO_MEMORY	System heap depleted
6	BAD_MEM_PTR	Memory pointer corrupted
7	NO_MBLK	No memory blocks
8	MBLK_CORRUPT	Memory block corrupted
9	BUFF_OVERFLOW	Buffer overflow
10	STREAM	Stream error

11	DISK_CORUPT	Disk is corrupt
12	DISK_FULL	Disk is full
13	DISK_FAIL	Disk failed
14	ILLEGAL_INT	Illegal interrupt
15	NOT_HEAP_PTR	Memory pointer is not a heap pointer
16	SW_ERROR	Generic software error
17	SWT	Software watchdog timer
18	INT_UNHANDLED	Interrupt not handled

OS Errors (Code 2)

The sub-codes are the actual error codes returned by the OS call.

Hardware Errors (Code 3)

Number	Sub-code	Meaning
1	RTC	Failed to access the real time clock
2	FPGA_LOAD	Failed to load FPGA
3	FPGA_ACCESS	Failed to access FPGA
4	FPGA_OPERATION	FPGA operation failed
5	FPGA_WRONG_IMAGE	Attempt to load incorrect FPGA image

Network Errors (Code 4)

Number	Sub-code	Meaning
1	DHCP	DHCP failed
2	BOOTP	Bootp failed
3	RARP	RARP failed
4	IP_ADDR	No IP address assigned

Assertion Errors (Code 5)

There are currently no sub-codes.

Configuration Errors (Code 6)

Number	Sub-code	Meaning
1	HW	Illegal hardware configuration
2	VER_NEWER	The SRAM configuration version is newer than the current version.
3	CREATE_FAIL	Couldn't create the configuration database.
4	UPDATE_FAIL	Couldn't upgrade the configuration database.

Executive Errors (Code 7)

Number	Sub-code	Meaning
0	SUCCESS	Operation completed successfully
1	GEN_ERROR	An indeterminate error occurred
2	NO_MEM	Not enough memory

3	NO_FILE	File does not exist
4	BAD_FILE	The file is not an executable
5	BAD_VERS	The file version is not supported
6	BAD_CHKSUM	The file checksum does not match

Appendix B CORE HSM COMMANDS

The following list of commands consists of “core” HSM commands which cannot be disabled.

Console	Host
CL – Configure the Alarms	BS – Erase the Key Change Storage
QL – Query the Alarm Configuration	RA – Cancel the Authorise Activity state
CC – Configure the Console Port	LG – Set HSM Response Delay
QC – Query the Console Port Configuration	NC – Perform Diagnostics and Obtain LMK
CS – Configure Security	Check Value
QS – Query the Security Configuration	NO – Return HSM Status Information
CP – Configure the Printer Port	
QC – Query the Printer Port Configuration	
CA – Configure the Auxiliary Port	
QA – Query the Auxiliary Port Configuration	
CM – Configure the Management Port	
QM – Query the Management Port Configuration	
CH – Configure the Host Port	
QH – Query the Host Port Configuration	
LK – Load new LMK	
LO – Load old LMK into Key Change Storage	
V – Verify LMK	
DC – Duplicate LMK Components	
A – Enter Authorise Activity state	
C – Cancel Authorise Activity state	
VR – Display Version Information	
GETCMDS – View the Available Host Commands	
ERRLOG – View Error Log	
CLEARERR – Clear Error Log	
AUDITLOG – View Audit Log	
CLEARAUDIT – Clear Audit Log	
AUDITOPTIONS – Select Audit Options	
SETTIME – Set the time & date	
GETTIME – Get the time & date	
SS – Save Settings to Smartcard	
RS – Retrieve Settings to Smartcard	
DT – Diagnostic Test	
N – Single-length DES calculator	
\$ – Double-length DES calculator	
T – Triple-length DES calculator	
FC – Format Smartcard	
CO – Copy Authorising Officer Smartcard	
VC – Verify Smartcard	
NP – Change Smartcard PIN	
RC – Read Unidentifiable Smartcard Details	
EJECT – Eject a Smartcard	
CONFIGCMDS – Configure Commands	
CONFIGPB – Configure PIN Blocks	

Appendix C PIN BLOCK FORMATS

The following table lists the PIN Block formats supported by the standard base software, together with their default state. Note that PIN Blocks implemented as part of a custom software development will always be enabled.

PIN Block Format	Description	Default
01	ISO 9564-I & ANSI X9.8 format 0	Enabled
02	Docutel ATM format	Disabled
03	Diebold & IBM ATM format	Disabled
04	PLUS Network format	Disabled
05	ISO 9564-I format 1	Enabled
34	Standard EMV 1996 format	Disabled
35	MasterCard Pay Now & Pay Later format	Enabled
41	Visa/Amex new PIN only format	Enabled
42	Visa/Amex new & old PIN format	Enabled
47	ISO 9564-I & ANSI X9.8 format 3	Enabled

Appendix D KEY SCHEME TABLE

Whenever a key is entered into the HSM, it must be prefixed by a Key Scheme Tag which allows the HSM to interpret the key correctly. The following values are supported:

Key Scheme Tag	Notes
None/Z	Encryption of a single length DES key using ANSI X9.17 methods. Used for encryption of keys under a variant LMK, and can also be used for the import or export of keys.
U	Encryption of a double length DES key using the variant method; used for encryption of keys under a variant LMK.
T	Encryption of a triple length DES key using the variant method; used for encryption of keys under a variant LMK.
X	Encryption of a double length key using ANSI X9.17 methods; only available for import and export of keys. This mode is enabled within the Configure Security command,
Y	Encryption of a triple length key using ANSI X9.17 methods; only available for import and export of keys. This mode is enabled within the Configure Security command.
V	Encryption of keys using Verifone/GISKE methods; used for export of DES keys.
R	Encryption of keys using ANSI TR-31 keyblock methods; only used for import or export of keys. The use of this scheme requires optional licence HSM8-LIC006.
S	Encryption of DES, RSA & HMAC keys using Thales keyblock methods; used for encryption of keys under a keyblock LMK.

Appendix E VARIANT LMKs

This section contains information relating to keys encrypted under Variant LMKs.

Key Type Table

The HSM provides a set of commands for key generation, key export and key import. An export command is one that translates a key from LMK encryption to encryption under a ZMK, for sending to another party. Import is the reverse, for receiving keys and translating to local storage. The Key Type Table controls ‘permitted actions’ for the console and host commands used to generate, import and export keys.

Errors are reported when an action breaks the rules imposed by the table. For example:

29 : Key function not permitted

The table below shows the actions that can be applied to each specific LMK pair.

Variant ⇨		0			1			2			3			4			5			6			7			8			9		
LMK ⇩		G	E	I	G	E	I	G	E	I	G	E	I	G	E	I	G	E	I	G	E	I	G	E	I	G	E	I	G	E	I
Pair	Code																														
04 – 05	00	ZMK			ZMK (Comp)			KML																							
		A			U	A	U	U	A	U																					
06 – 07	01	ZPK																													
		U	A	U																											
14 – 15	02	PVK TPK												CVK																	
		TMK												CSVK																	
16 – 17	03	TAK																													
		U	A	U																											
18 – 19	04				DTAB			IPB																							
20 – 21	05																														
22 – 23	06	WWK																													
		U	A	U																											
24 – 25	07																														
26 – 27	08	ZAK																													
		U	A	U																											
28 – 29	09	BDK			MK-AC			MK-SMI			MK-SMC			MK-DAK			MK-DN						MK-CVC3								
		U	A	U	U	A	U	U	A	U	U	A	U	U	A	U	U	A	U				U	A	U						
30 – 31	0A	ZEK																													
		U	A	U																											
32 – 33	0B	DEK																													
		U																													
34 – 35	0C	RSA-SK			HMAC																										
		A			U	A	U																								
36 – 37	0D	RSA-PK																													
		A	A																												
38 – 39	0E																														

Table of actions applied to each specific LMK pair and variant in generic HSM commands

G = Generate. E = Export. I = Import.

blank = Not allowed.

A = allowed in Authorised State.

U = allowed Unconditionally, i.e. without Authorised State.

Not all key type codes are available in all commands for security reasons.

The Key type code used within commands is formed by using the Variant code as the first character then the LMK pair code as the second character. For example the code for a ZPK is 001.

Appendix F KEYBLOCK LMKs

This section contains information relating to keys encrypted under Keyblock LMKs.

Key Usage Table

A keyblock's Key Usage field defines the primary usage of the key contained within the keyblock. The following values are supported:

Key Usage	Description
01*	WatchWord Key (WWK)
02*	RSA Public Key
03*	RSA Private Key
B0	Base Derivation Key (BDK)
C0	Card Verification Key
11*	Card Verification Key (American Express CSC)
12*	Card Verification Key (Mastercard CVC)
13*	Card Verification Key (Visa CVV)
D0	Data Encryption Key (Generic)
21*	Data Encryption Key (DEK)
22*	Data Encryption Key (ZEK)
E0	EMV/Chip card Master Key: Application Cryptogram (MK _{AC})
E1	EMV/Chip card Master Key: Secure Messaging for Confidentiality (MK _{SMC})
E2	EMV/Chip card Master Key: Secure Messaging for Integrity (MK _{SMI})
E3	EMV/Chip card Master Key: Data Authentication Code (MK _{DAC})
E4	EMV/Chip card Master Key: Dynamic Numbers (MK _{DN})
E5	EMV/Chip card Master Key: Card Personalisation
31*	Visa Cash Master Load Key (KML)
32*	Dynamic CVV Master Key (MK-CVC3)
K0	Key Encryption / Wrapping Key (Generic)
51*	Terminal Key Encryption (TMK)
52*	Zone Key Encryption (ZMK)
M0	ISO 16609 MAC algorithm 1 (using 3-DES)
M1	ISO 9797-1 MAC algorithm 1
M3	ISO 9797-1 MAC algorithm 3
61*	HMAC key (using SHA-1)
62*	HMAC key (using SHA-224)
63*	HMAC key (using SHA-256)
64*	HMAC key (using SHA-384)
65*	HMAC key (using SHA-512)
P0	PIN Encryption Key (Generic)
71*	Terminal PIN Encryption Key (TPK)
72*	Zone PIN Encryption Key (ZPK)
73*	Transaction Key Scheme Terminal Key Register (TKR)
V0	PIN Verification Key (Generic)
V1	PIN Verification Key (IBM 3624 algorithm)

V2	PIN Verification Key (Visa PVV algorithm)
----	---

* Thales Key Block proprietary value. All other values are common to both Thales Key Block and ANSI TR-31 format.

Mode of Use Table

A keyblock's Mode of Use field defines the operation that the key contained within the keyblock can perform. The following values are supported:

Mode of Use	Description
B	The key may be used to perform both encrypt and decrypt operations.
C	The key may be used to perform MAC calculation (both generate & verify) operations.
D	The key may only be used to perform decrypt operations.
E	The key may only be used to perform encrypt operations.
G	The key may only be used to perform MAC generate operations.
N	No special restrictions apply.
S	The key may only be used to perform digital signature generation operations.
V	The key may only be used to perform digital signature verification operations.

Note: These values are identical to those defined in TR-31.

Exportability Table

A keyblock's Exportability field defines the conditions under which the key contained within the keyblock can be exported outside the cryptographic domain in which the key is found. The following values are supported:

Exportability	Description
E	May only be exported in a keyblock format (i.e. schemes S, R, V), provided that the wrapping key itself is in a keyblock format (i.e. scheme S).
N	The key cannot be exported in any format.
S	May be exported to non-keyblock formats (e.g. schemes X, Y), provided such export is explicitly allowed via Configure Security.

Appendix G LIST OF AUTHORISABLE ACTIVITIES

The following table indicates, for each sensitive host or console command, which activity needs to be authorised. Note that the authorisation requirements may differ depending on whether a Variant or Keyblock LMK is used.

Examples:

- Authorising the activity `export.001.host` allows export of ZPKs using a host command (e.g. A8).
- Authorising the activity `export` allows export of any (valid) key using a host or console command.
- Authorising the activity `export..console` allows export of any (valid) key using a console command.

Note: When using a Variant LMK, the key type table determines whether the HSM needs to be authorised in order to generate, import or export a certain key. Where the key type table entry indicates 'U' (unconditional), it is not necessary to authorise the HSM for that activity. Moreover, authorised activities `genprint.*` and `component.*` do not examine the KTT and are always required.

The following table defines a number of 'groups' which, in turn, contain a set of either key types or key usages. These groups are subsequently referenced in the tables further below.

	Group	Contents
Variant LMK	<i>generate key types</i>	000 200 001 002 402 003 006 008 009 109 209 309 409 509 709 00a 00b
	<i>genprint key types</i>	000 200 001 002 402 003 006 008 009 109 209 309 409 509 709 00a 00b
	<i>component key types</i>	000 200 001 002 402 003 006 008 009 109 209 309 409 509 709 00a 00b
	<i>export key types</i>	200 001 002 402 003 006 008 009 109 209 309 409 509 709 00a
Keyblock LMK	<i>genprint key usages</i>	01 B0 C0 11 12 13 D0 21 22 E0 E1 E2 E3 E4 E5 31 32 K0 51 52 M0 M1 M3 P0 71 72 73 V0 V1 V2
	<i>component key usages</i>	01 B0 C0 11 12 13 D0 21 22 E0 E1 E2 E3 E4 E5 31 32 K0 51 52 M0 M1 M3 P0 71 72 73 V0 V1 V2
	<i>export key usages</i>	01 B0 C0 11 12 13 D0 21 22 E0 E1 E2 E3 E4 E5 31 32 K0 51 52 M0 M1 M3 61 62 63 64 65 P0 71 72 73 V0 V1 V2
	<i>import key usages</i>	01 B0 C0 11 12 13 D0 21 22 E0 E1 E2 E3 E4 E5 31 32 K0 51 52 M0 M1 M3 61 62 63 64 65 P0 71 72 73 V0 V1 V2

Command (H=Host, C=Console)		Description	Category	Sub-Category	Interface
SYMMETRIC KEY GENERATION					
Variant LMK	H – A0	Generate Key	generate	000	host
	C – KG	Generate Key			console
	C – GZ	Generate Zone Master Key and Write to Smartcards		generate key types	host
	H – BU	Generate Key Check Value (for Key Check Values > 6 digits)			console
	C – CK	Generate Key Check Value (for Key Check Values > 6 digits)			
	H – FG	Generate a Pair of PVKs		002	host
SYMMETRIC KEY GENERATION / PRINTING					
Keyblock LMK	H – A2	Generate and Print Component	genprint	genprint key usages	host
	H – NE	Generate and Print Key as Split Components			
Variant LMK	H – A2	Generate and Print Component	genprint	genprint key types	host
	H – NE	Generate and Print Key as Split Components			
	H – OC	Generate and Print ZMK component		000	
	H – OE	Generate and Print TMK, TPK or PVK		002	
SYMMETRIC KEY COMPONENT MANIPULATION					
Keyblock LMK	H – A4	Form a Key from Encrypted Components	component	component key usages	host
	C – EC	Encrypt Clear Component			console
	C – FK	Form Key from Components			
	C – GS	Generate Key Components and Write to a Smartcard			
	C – GC	Generate Key Component			
Variant LMK	H – GG	Form ZMK from Three ZMK Components	component	000	host
	H – GY	Form ZMK from 2 to 9 ZMK Components			
	H – A4	Form Key from Encrypted Components		component key types	console
	C – BK	Form Key from Components			
	C – EC	Encrypt Clear Component			
	C – FK	Form Key from Components			
	C – GS	Generate Key Components and Write to a Smartcard			

	C – GC	Generate Key Component			
	C – DE	Form ZMK from Clear Components			
	C – D	Form ZMK from Encrypted Components		000	
	C – Z	Encrypt clear ZMK Component			
	C – K	Encrypt Key under LMK Pair 14-15 (from components)		002	
SYMMETRIC KEY IMPORT					
Keyblock LMK	H – BY	Translate (Import) ZMK from ZMK to LMK Encryption (from non-keyblock format)	import	K0 52	host
	H – A6	Import Key (from non-keyblock format)		import key usages	host
	C – IK	Import Key (from non-keyblock format)		import key usages	console
	H – LU	Import HMAC Key (from non-keyblock format)		10C	host
Variant LMK	H – FC	Translate (Import) TMK, TPK or PVK from ZMK to LMK Encryption	import	002	host
	H – BY	Translate (Import) ZMK from ZMK to LMK Encryption		000	
	C – IV	Import CVK or PVK from ZMK to LMK		402 002	console
SYMMETRIC KEY EXPORT					
Keyblock LMK	H – A0	Generate Key (when requested to export generated key to non-keyblock format)	export	export key usages	host
	H – A8	Export Key (to non-keyblock format)		61 62 63 64 65	
	H – LW	Export HMAC Key (to non-keyblock format)		export key usages	console
	C – KG	Generate Key (when requested to export generated key to non-keyblock format)			
	C – KE	Export Key (to non-keyblock format)			
Variant LMK	H – A0	Generate Key (when requested to export generated key)	export	export key types	host
	H – A8	Export Key		002	
	H – FE	Translate (Export) TMK, TPK or PVK from LMK to ZMK Encryption		10c	
	H – LW	Export HMAC Key		export key types	console
	C – KG	Generate Key (when requested to export generated key)			
	C – KE	Export Key			
	C – WK	Translate (Export) Zone PIN Key		001	

ASYMMETRIC KEY MANAGEMENT					
Keyblock LMK	H – EI	Generate a Public/Private Key Pair	generate	rsa	host
	H – EO	Import a Public Key	import	rsa-pk	
Variant LMK	H – EI	Generate a Public/Private Key Pair	generate	rsa	host
	H – EO	Import a Public Key	import	rsa-pk	
CLEAR PIN					
H – BA	Encrypt a Clear PIN	pin	clear	host	
H – NG	Decrypt an Encrypted PIN				
PIN MAILER					
H – PE	Print PIN/PIN Solicitation Data	pin	mailer	host	
H – OA	Print a PIN Solicitation Mailer				
AUDIT					
H – Q6	Delete Audit Record	audit		host	
C – CLEARAUDIT	Clear the Audit Log			console	
C – AUDITOPTIONS	Audit Options				
C – A5	Configure Fraud Detection				
C – A7	Re-enable PIN Verification				
ADMINISTRATION					
C – DM	Delete LMK	admin		console	
C – SS	Save HSM Settings to a Smartcard				
C – RS	Retrieve HSM Settings from a Smartcard				
C – LO	Move ‘Old’ LMKs Into Key Change Storage				
C – SETTIME	Set the Time and Date				
DIAGNOSTICS					
H – KQ	ARQC (or TC/AAC) Verification and/or ARPC Generation	diag		host	
H – K2	Verify Truncated Application Cryptogram (CAP)				
H – KW	ARQC (or TC/ACC) Verification and/or ARPC Generation (EMV4.1 including CCD)				
H – KS	Data Authentication Code and Dynamic Number Verification				
H – PM	Verify a Dynamic CVV				

MISCELLANEOUS				
H – B0	Translate Key Scheme	misc		host
H – CS	Modify Key Block Header			
C – R	Load the Diebold Table			console
C – CV	Generate a VISA Card Verification Value			
C – PV	Generate a VISA PIN Verification Value			
C – TD	Translate Decimalisation Table			
C – ED	Encrypt Decimalisation Table			
C – MI	Generate a MAC on an IPB			
COMMAND				
H – GI	Import DES Key (Auth required if backward compatibility mode is enabled by console CS)	command	gi	host
H – TA	Print TMK Mailer		ta	
CUSTOM				
		custom	specific custom command code (s)	host console

Appendix H REDUCED ASCII CHARACTER SETS

Reduced ASCII Char Set	ASCII chars (hex)	ASCII chars
All printable characters	0x20 – 0x7E	<Space Character> !"#\$%&'()*+,-./0123456789:;<=>? @ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_ `abcdefghijklmnopqrstuvwxyz{ }~
Base94 characters	0x21 – 0x7E	!"#\$%&'()*+,-./0123456789:;<=>? @ABCDEFGHIJKLMNOPQRSTUVWXYZ[\]^_ `abcdefghijklmnopqrstuvwxyz{ }~
Base64 characters	0x30 – 0x39 0x41 – 0x5A 0x61 – 0x7A 0x2B 0x2F	0123456789 ABCDEFGHIJKLMNOPQRSTUVWXYZ abcdefghijklmnopqrstuvwxyz + /
Hex-only chars	0x30 – 0x39 0x41 – 0x46	0123456789 ABCDEF

Appendix I CONFIGURE SECURITY SETTINGS

Parameter	Default value
PIN length: 4..12 <i>This is the length of the PIN to be stored as an encrypted PIN under the LMK.</i> <i>Note: The “encrypted PIN” is one character longer than the length specified for the PIN by this parameter.</i>	4
Echo: On or Off <i>If the answer to this question is ‘On’ then passwords and other secret values are displayed on the console as entered. Characters can be hidden by using ‘^’ prior to entering the component or key.</i> <i>Note: Enabling Echo is a security hazard and should not be used in a live system.</i>	Off
Atalla ZMK variant support: On or Off <i>For interoperation with Atalla systems. This enables the optional Atalla variants within commands. Any console command providing key support will prompt for an Atalla variant.</i> <i>Note: Selection has no effect on host commands - Atalla variants can be supplied with any appropriate command regardless of this setting.</i>	Off
Transaction key scheme: Racal, Australian or None <i>Transaction key schemes are techniques whereby data-encrypting keys change with each transaction in a manner that cannot be followed by a third party. The HSM supports three variants of transaction key schemes: Racal, Australian (AS2805) and DUKPT. There are command conflicts between the Racal and Australian scheme so only one can be selected. The use of the DUKPT commands is not affected by this setting.</i> <i>Note: The default value has changed to ‘None’. In this case, none of the Racal or Australian transaction key scheme commands are available to the host.</i>	None
User storage key length: Single, Double or Triple <i>This is the length of the keys stored in user storage; it can be single, double or triple length. The number of keys that can be stored depends upon this setting.</i>	Single
Select clear PINs: Yes or No <i>This enables the clear PIN support via host commands ‘NG’ and ‘BA’. Authorised state is a requirement for these commands to be processed by a host application.</i> <i>Note: This is a security risk unless precautions are taken at the host.</i>	No
Enable ZMK translate command: Yes or No <i>This enables the ‘BY’ command that allows the translation of Zone Master Keys from under another Zone Master Key. Authorised state is required for this command to process within a host application.</i> <i>Note: The availability of this command is a significant security risk.</i>	No
Enable X9.17 for import: Yes or No <i>This enables support for the ANSI X9.17 mechanism for key import. When being imported, each key of double or triple length is encrypted separately using the Electronic Code Book (ECB) mode of encryption. This is a lower security option. It is strongly recommended that the ANSI TR-31 keyblock is used instead of X9.17.</i>	No

Parameter	Default value
Enable X9.17 for export: Yes or No <i>Similar to the previous item, but used when exporting keys.</i>	No
Solicitation batch size: 1..1024 <i>A method supported by the HSM to enable customers to self-select their own PINs is to use Solicitation mailers. This is a turnaround form that is sent to the cardholder. The cardholder records the PIN selection on the form and returns it to the issuer. The mailer data consists of the cardholder name and address and a reference number (an encrypted account number). As a security measure, the form returned to the issuer contains only the reference number and the PIN selection. A batch process is used to process these requests when returned.</i>	1024
Enable single-DES: Yes or No <i>This parameter is only valid if ZMK length is double. If enabled, it permits the use of single-length DES keys.</i> <i>Note: The default value has been changed to 'No'.</i>	No
Prevent Single-DES keys masquerading as double or triple-length key: Yes or No <i>When set to Yes, all HSM commands that import double or triple-length DES keys will ensure that the imported key is not simply a single-length key masquerading as a double or triple-length key.</i>	Yes
ZMK length: Single or Double <i>The length of the Zone Master Key: single or double. This is a backwards-compatible mode to enable the switching between 16H and 32H for ZMKs.</i> <i>Note: The default value of this parameter has been changed to 'Double'.</i>	Double
Decimalisation table Encrypted/Plaintext: Encrypted or Plaintext <i>This option determines if the decimalisation table will be encrypted or in plain text. The default setting is encrypted, however, to allow for backward compatibility plaintext decimalisation tables can be selected. It is recommended that encrypted decimalisation tables are used to protect against decimalisation table manipulation attacks.</i>	Encrypted
Enable decimalisation table checks: Yes or No <i>The values in the decimalisation tables, used for deriving and verifying PIN offset values, are normally restricted to provide additional security by rejecting values which are potentially insecure. This can cause problems where existing tables fail the checks, so for backward compatibility this parameter allows the restrictions to be disabled.</i>	Yes
PIN encryption algorithm: A (Visa method) or B (Racal method) <i>This selects the PIN encryption algorithm to be used when encrypted PINs are stored by the card issuer. The Racal algorithm is the best choice for a new installation; it is the stronger of the two methods. The Visa algorithm is offered for compatibility with older HSMs and for customers who already have a database of encrypted PINs.</i> <i>When the Racal method is used, the output of the encryption is hex characters whereas the Visa method produces numeric digits. Commands that use encrypted PINs describe them as 'LN or LH'.</i>	A (Visa method)

Parameter	Default value
Card/password authorisation: Card or Password <i>This option selects the method of authenticating security officers requesting a security state change. The Authorised State is a mode that the HSM can be placed in for sensitive data processing. This authorised mode is required when input commands at the console or host use clear text data such as key components or unencrypted pins. Authorised mode can be used in both Online and Offline host states and requires the authorising officers to invoke the higher security level. Before the authorised state can be set the authorising officers need to be verified by the HSM. Officer verification is done by checking either a smartcard and PIN or a password (16 alphanumeric characters.) If the Password option is not set when the LMK is created, the password option will not be available as no password is created and stored with the LMK components.</i>	Card
Card issuer password: 8 characters <i>This option enables Thales to change the card issuer password for LMK and authorising officer Smartcards. If the password has been changed you will be advised when new cards are delivered. Special care must be taken that no key is inadvertently struck, placing a character in this field. If this setting is changed, it will not be possible to format the Smartcards using the 'FC' command.</i>	GUARDATA
Authorised state required when importing DES key under RSA Key: Yes or No <i>This setting determines whether Authorised State is mandatory for the import of DES keys using RSA keys (host command GI). When set to Yes, the GI command always requires Authorised State (and the use of the signature field is optional). When set to 'No', the GI command does not require Authorised State.</i>	Yes
Minimum HMAC verification length in bytes: Yes or No <i>This setting determines the minimum and maximum lengths of HMAC keys that the HSM can generate. HMAC keys must satisfy the equation $L/2 \leq \text{key length}$, where L = is the size of the hash function output. For SHA-1 HMAC keys, L=20, and therefore the key length must be at least 10.</i>	10
Enable PKCS#11 import and export for HMAC keys: Yes or No <i>This setting determines whether the host commands LU and LW can import or export HMAC keys in PKCS#11 format.</i>	No
Enable ANSI X9.17 import and export for HMAC keys: Yes or No <i>This setting determines whether the host commands LU and LW can import or export HMAC keys in ANSI X9.17 format.</i>	No
Enable ZEK encryption of all printable ASCII chars²: Yes or No <i>This setting determines whether the 'Message Encryption' host commands M0, M2 and M4 can encrypt/decrypt/translate messages (using a ZEK) where the plaintext contains only printable ASCII characters.</i>	No
Enable ZEK encryption of "Base94" ASCII chars: Yes or No <i>This setting determines whether the 'Message Encryption' host commands M0, M2 and M4 can encrypt/decrypt/translate messages (using a ZEK) where the plaintext contains only Base94 characters.</i>	No

² Refer to Appendix H for details of ASCII reduced character sets.

Parameter	Default value
Enable ZEK encryption of “Base64” ASCII chars: Yes or No <i>This setting determines whether the ‘Message Encryption’ host commands M0, M2 and M4 can encrypt/decrypt/translate messages (using a ZEK) where the plaintext contains only Base64 characters.</i>	No
Enable ZEK encryption of “Hex-only” ASCII chars: Yes or No <i>This setting determines whether the ‘Message Encryption’ host commands M0, M2 and M4 can encrypt/decrypt/translate messages (using a ZEK) where the plaintext contains only hexadecimal ASCII characters.</i>	No
Restrict Key Check Value to 6 hex chars: Yes or No <i>This setting determines whether Key Check Values (KCVs) should be restricted to consist of only 6 hex characters. The overall length of the KCV field will remain the same, regardless of this setting. However, when set to ‘Yes’, only the first 6 characters will contain the KCV: any remaining characters will be ignored (when input to the HSM) or set to ‘0’ (when returned from the HSM).</i>	Yes
Enable Multiple Authorised Activities: Yes or No <i>If enabled, will allow precise selection of authorised activities (including timeout period if required). If disabled HSM reverts to global Authorised state.</i>	Yes
Enable Persistent Authorised Activities: Yes or No <i>If enabled, will allow “persistent” authorised activities to be automatically restored when the HSM restarts following a power failure.</i>	Yes
Enable variable length PIN offset: Yes or No <i>If enabled, this will allow the IBM 3624 PIN Offset commands to return an Offset whose length matches the PIN, rather than being restricted to the Check Length parameter.</i>	No
Enable weak PIN checking: Yes or No <i>If enabled, the HSM's PIN generation/derivation host commands will check to ensure that the PIN does not match one of the entries in the appropriate global 'Excluded PIN Table'. If a match is found in the list, then the command fails, returning error code 86.</i>	No
Enable PIN Block format 34 as output format for PIN translations to ZPK: Yes or No <i>If enabled, the HSM will permit PIN block format 34 to be used as the output format of PIN translation commands. In previous versions of HSM software, translations to this format were not possible.</i>	No
Default LMK identifier: 00..99 <i>Identifies the Default LMK, which the HSM will use if it receives a command that does not explicitly state which LMK is to be used. The use of the Default LMK provides a “backward-compatible” mode, even when multiple LMKs are loaded in the HSM.</i>	00
Management LMK identifier: 00..99 <i>Identifies the Management LMK, which will be used for authorising certain management functions (e.g. setting the HSM's date/time), and for encrypting the audit MAC key.</i>	00
Use HSM clock for date/time validation: Yes or No <i>If enabled, the HSM uses its integral real-time clock to validate check the start/end date/time optional header blocks of keyblocks (when present).</i>	Yes

Parameter	Default value
Additional padding to disguise key length: Yes or No <i>If enabled, the HSM disguises the length of single or double length keys within a keyblock by adding 8 or 16 extra padding bytes, such that single, double and triple length DES keys all appear to be triple length keys.</i>	No
Key export and import in trusted format only: Yes or No <i>If enabled, the HSM will only import/export keys using a keyblock format. In this case, any export/import process using keys in variant format (including X9.17 format) will be prohibited.</i>	Yes
Secure Link password: 8 characters <i>This setting allows the user to configure the password which is to be used to protect the HSM Manager – HSM communications. The HSM Manager will be made available at some future date.</i>	password

Appendix J FRAUD DETECTION FUNCTIONS

The HSM has fraud detection functions designed to detect and prevent 'brute force' attacks, where (for example) large numbers of PINs are submitted until the correct PIN is discovered.

The detection works by counting how many failed PIN verifies are detected in one minute and in one hour. The console command **A5** is used to configure these limits. If either of these counts exceeds the specified limits then the HSM increments the PIN Attack Counter by one and forces all PIN verification commands to return an error 39 in their response. The HSM will continue to return error 39 until the console command **A7** is used to re-enable PIN verification.

If the PIN Attack Counter reaches the PIN attack limit then the HSM will clear the LMKs from its memory. Installing the LMKs will set the PIN attack counter to 0.

The following list specifies the PIN verification host commands to which the limits apply:

- DA** – Verify a Terminal PIN Using the IBM Method
- EA** – Verify an Interchange PIN Using the IBM Method
- CG** – Verify a Terminal PIN Using the Diebold Method
- EG** – Verify an Interchange PIN Using the Diebold Method
- DC** – Verify a Terminal PIN Using the Visa Method
- EC** – Verify an Interchange PIN Using the Visa Method
- BC** – Verify a Terminal PIN Using the Comparison Method
- BE** – Verify an Interchange PIN Using the Comparison Method

Appendix K THALES KEYBLOCK/TR31 KEY USAGE CONVERSION

When exporting from Thales Keyblock format to ANSI TR-31 format, the following table is used to convert the keyblocks' Key Usage field:

Internal (Thales) Key Block	Key Usage	Exported (TR-31) Key Block
'11'	Card verification (American Express CSC)	'C0'
'12'	Card verification (Mastercard CVC)	'C0'
'13'	Card verification (Visa CVV)	'C0'
'21'	Data encryption using a DEK (DEK)	'D0'
'22'	Data encryption using a ZEK (ZEK)	'D0'
'31'	Visa Cash Master Load Key (KML)	'E6'
'32'	Dynamic CVV Master Key (MK-CVC3)	'E6'
'51'	Terminal key encryption (TMK)	'K0'
'52'	Zone key encryption (ZMK)	'K0'
'71'	Terminal PIN encryption (TPK)	'P0'
'72'	Zone PIN encryption (ZPK)	'P0'
'73'	Terminal Key Register (TKR)	'P0'